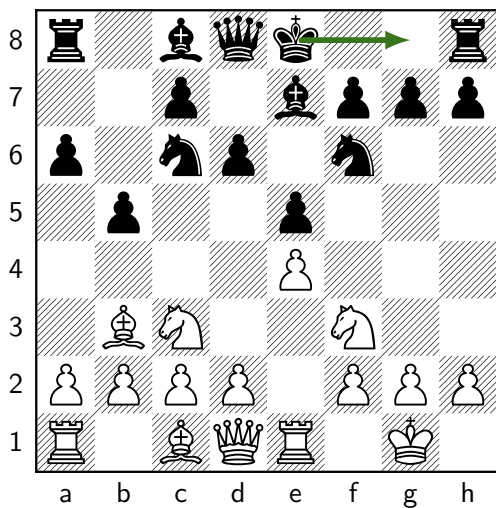


Melanie-MASTERS (human masters) — Ruy Lopez, Closed

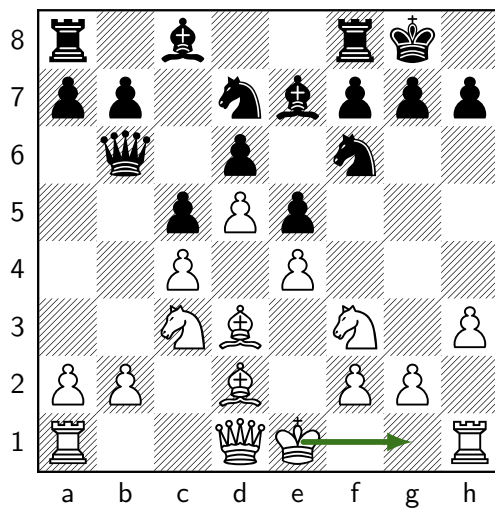
Melanie (Deep_Vision)

Abstract

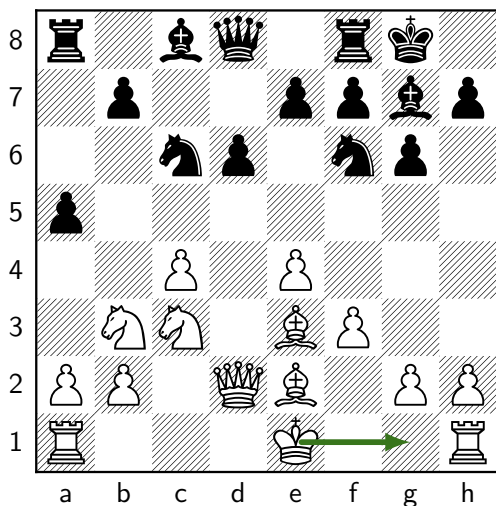
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



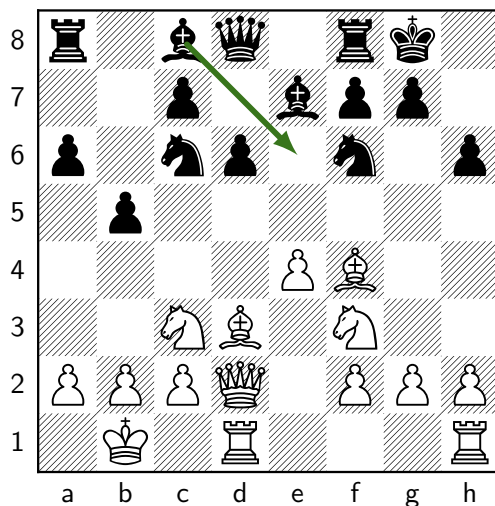
(a)



(b)

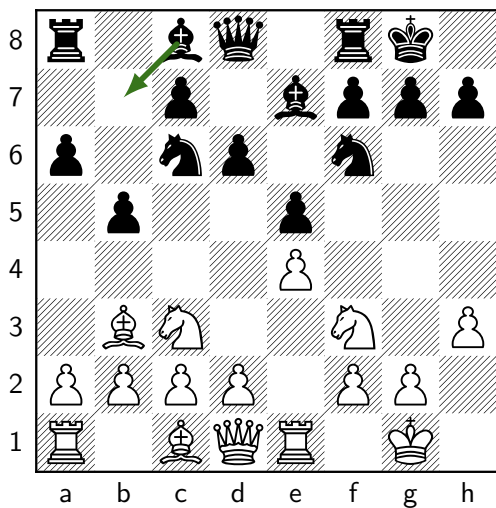


(c)

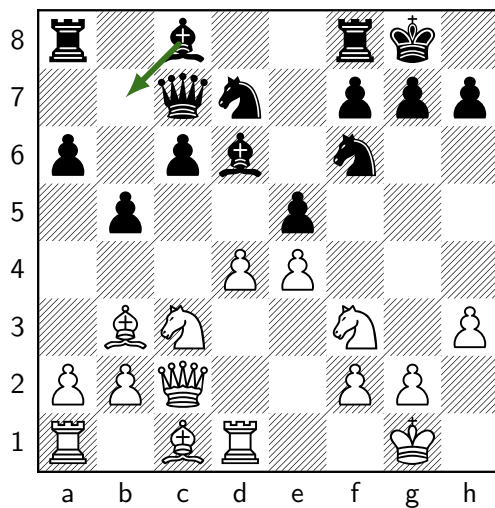


(d)

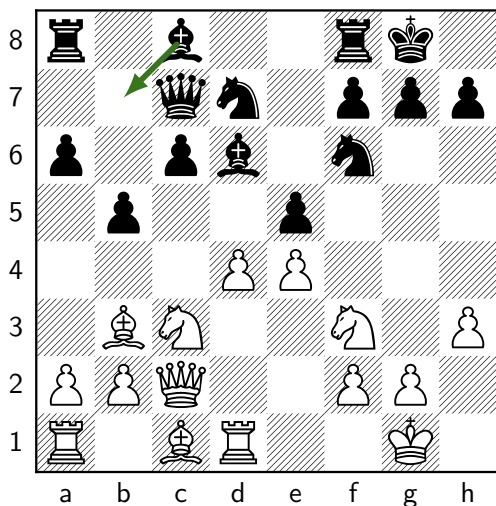
Figure 1: **Move 8, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2025-11_p0000#2263` (ply 20) [structural distance $d=95$, QAP obj=79]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa6* as their *Ra1*, our *Ph7* as their *Bd2*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#606` (ply 22) [structural distance $d=82$, QAP obj=62]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pa2* as their *Ra1*, our *Nf6* as their *Qd2*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#2472` (ply 21) [structural distance $d=93$, QAP obj=110]: the master played Be6; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Pd6* as their *Qd8*, our *Qd8* as their *Rf8*; transferred move c8e6. Synthesis: 2 retrieved analogs transferred to 0-0 (summed QAP objective 142, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



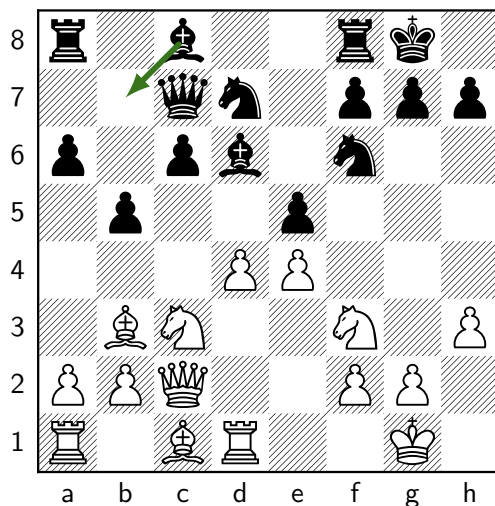
(a)



(b)

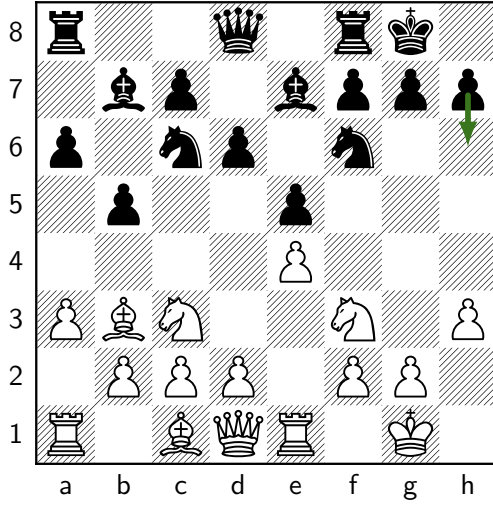


(c)

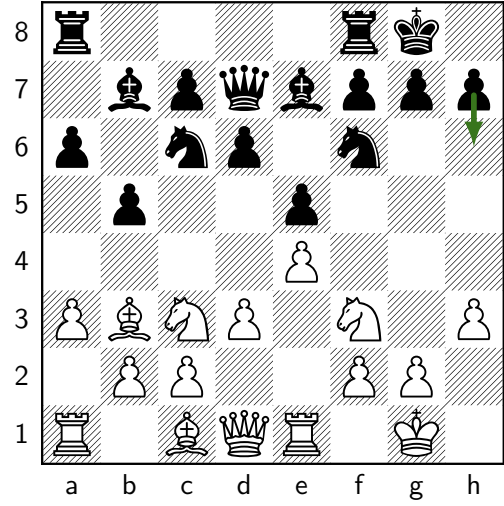


(d)

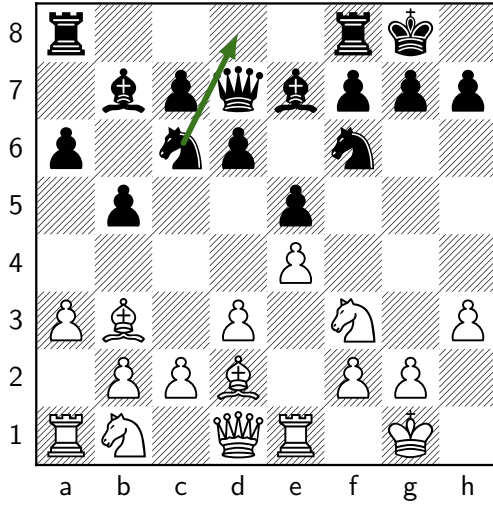
Figure 2: **Move 9, Black: Bb7.** (a) Our position; Melanie plays Bb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0003#179` (ply 25) [structural distance $d=80$, QAP obj=100]: the master played Bb7; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Be7* as their *Nd7*, our *Nc6* as their *Bd6*; transferred move c8b7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0003#1613` (ply 25) [structural distance $d=80$, QAP obj=100]: the master played Bb7; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Be7* as their *Nd7*, our *Nc6* as their *Bd6*; transferred move c8b7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#3002` (ply 25) [structural distance $d=80$, QAP obj=100]: the master played Bb7; under μ our *Bc1* plays the role of that game’s *Bc1*; cross-type roles: our *Be7* as their *Nd7*, our *Nc6* as their *Bd6*; transferred move c8b7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bb7 (summed QAP objective 598, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



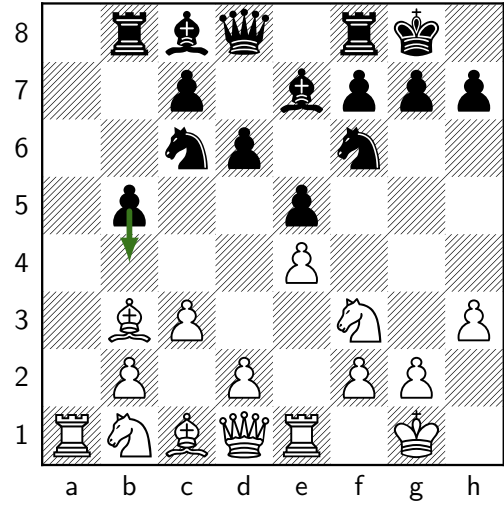
(a)



(b)

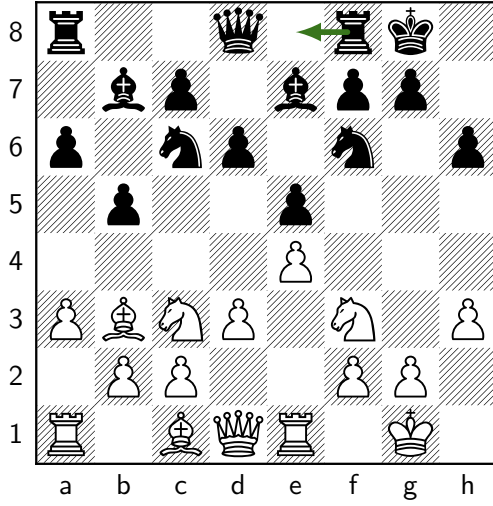


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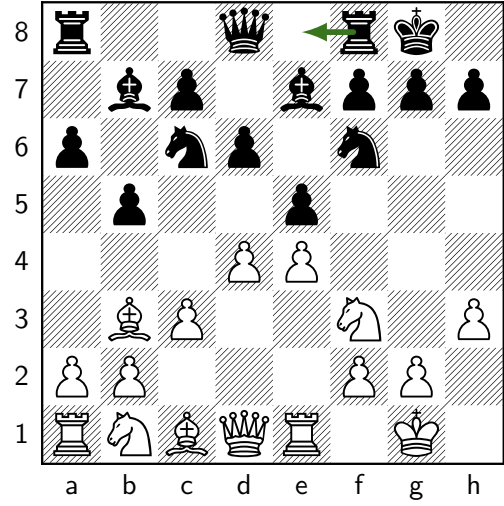


(d)

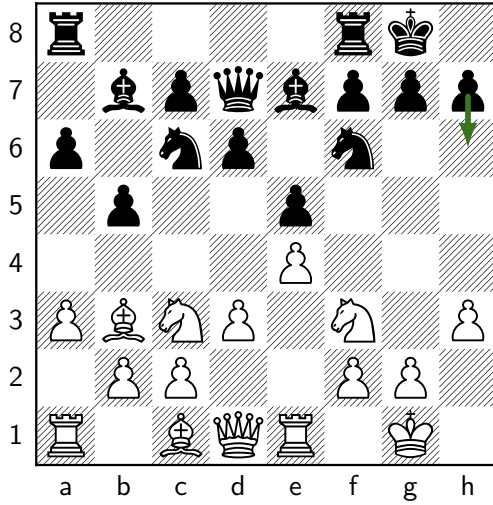
Figure 3: **Move 10, Black: h6**. (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3170` (ply 21) [structural distance $d=55$, QAP obj=138]: the master played h6; under μ our *Ph2* plays the role of that game’s *Ph2*; cross-type roles: our *nf3* as their *Qd7*; transferred move h7h6 (**endorses this move**). (c) Retrieved from `twic1500#328` (ply 21) [structural distance $d=86$, QAP obj=113]: the master played Nd8; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *Nf6* as their *Qd7*, our *bc1* as their *Ph7*; transferred move c6b8. (d) Retrieved from `lichess_elite_2024-09_p0003#86` (ply 21) [structural distance $d=80$, QAP obj=112]: the master played b4; under μ our *Ra1* plays the role of that game’s *Pb4* (a **cross-type** correspondence); cross-type roles: our *pc2* as their *Rb8*, our *nc3* as their *Bc8*; transferred move a8a7. Synthesis: 1 retrieved analog transferred to h6 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



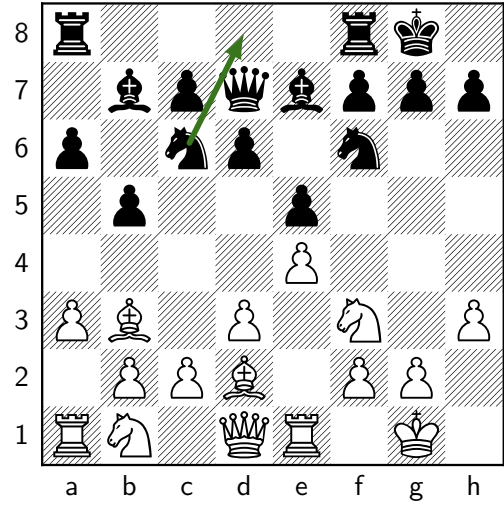
(a)



(b)

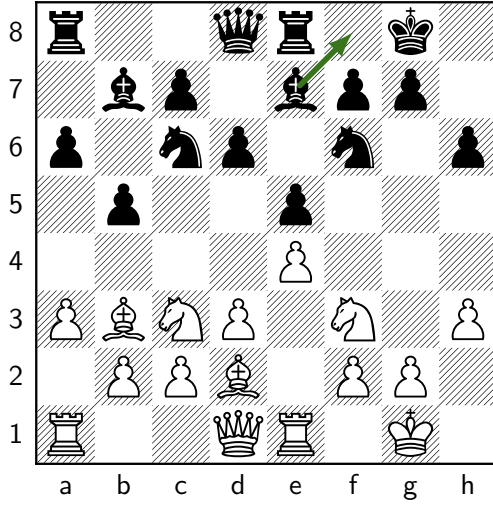


(c)

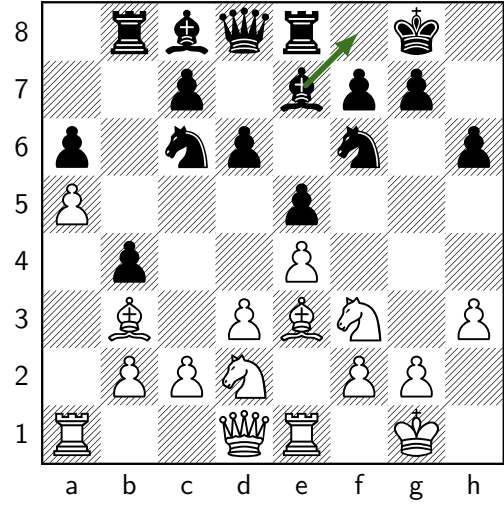


(d)

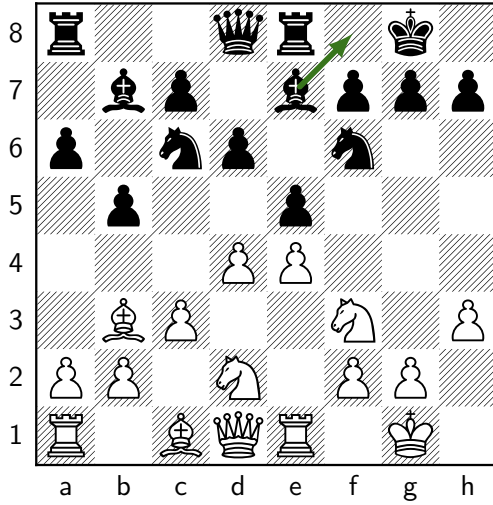
Figure 4: **Move 11, Black: Re8.** (a) Our position; Melanie plays **Re8** (**f8e8**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3963` (ply 23) [structural distance $d=41$, QAP obj=84]: the master played **Re8**; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Nc6* as their *Ra8*, our *Ph6* as their *Qd8*; transferred move **f8e8** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#3170` (ply 21) [structural distance $d=42$, QAP obj=140]: the master played **h6**; under μ our *Ph3* plays the role of that game’s *Ph2*; cross-type roles: our *nc3* as their *Qd7*; transferred move **h6h5**. (d) Retrieved from `twic1500#328` (ply 21) [structural distance $d=80$, QAP obj=112]: the master played **Nd8**; under μ our *Nc3* plays the role of that game’s *Nc3*; cross-type roles: our *Nf6* as their *Qd7*, our *Qd8* as their *Ph7*; transferred move **c6b8**. Synthesis: 2 retrieved analogs transferred to **Re8** (summed QAP objective 160, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



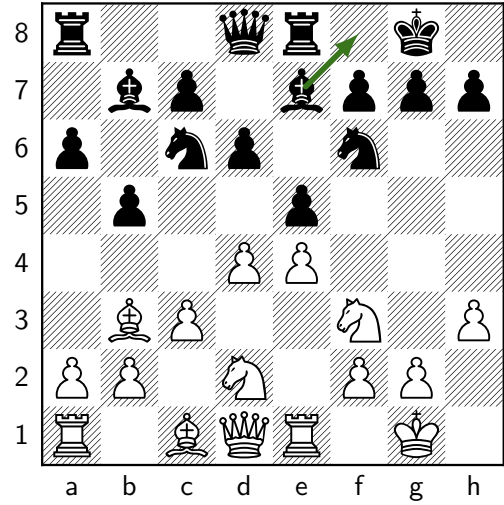
(a)



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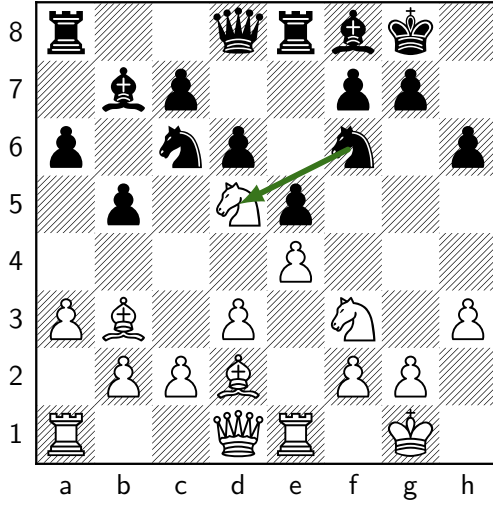


(c)

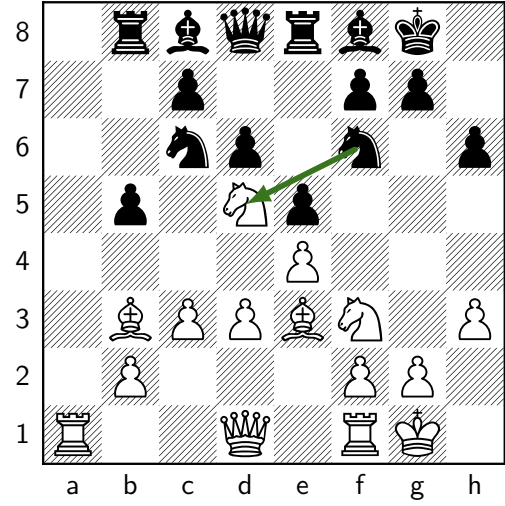


(d)

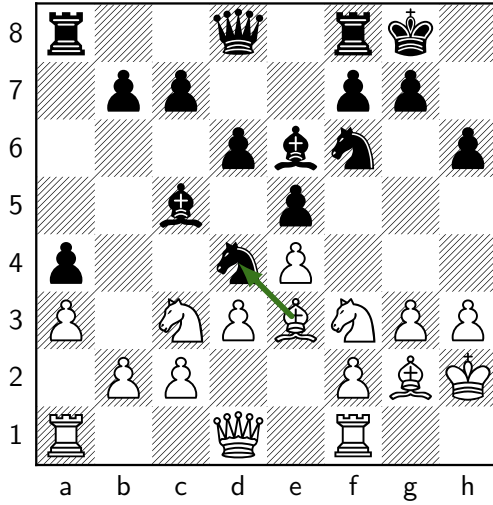
Figure 5: **Move 12, Black: Bf8.** (a) Our position; Melanie plays Bf8 (e7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `twic1500#137` (ply 25) [structural distance $d=77$, QAP obj=129]: the master played Bf8; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *pa3* as their *Rb8*, our *Nf6* as their *Bc8*; transferred move e7f8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#614` (ply 21) [structural distance $d=74$, QAP obj=90]: the master played Bf8; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Ph6* as their *Ra8*, our *Ra8* as their *Qd8*; transferred move e7f8 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#518` (ply 21) [structural distance $d=74$, QAP obj=90]: the master played Bf8; under μ our *Be2* plays the role of that game’s *Be2*; cross-type roles: our *Ph6* as their *Ra8*, our *Ra8* as their *Qd8*; transferred move e7f8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bf8 (summed QAP objective 755, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



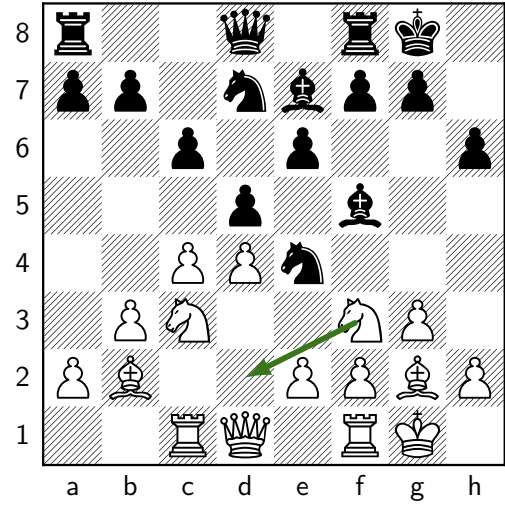
(a)



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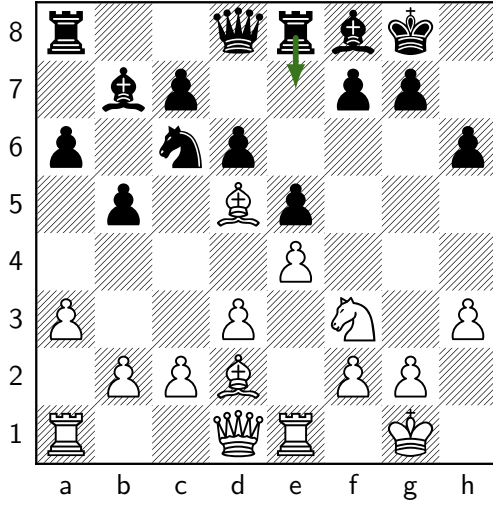


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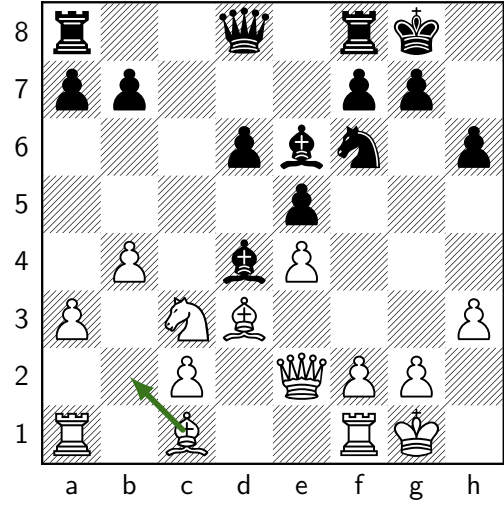


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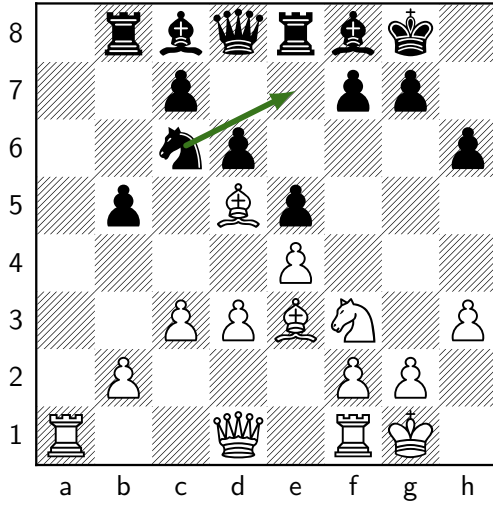
Figure 6: **Move 13, Black: Nxd5.** (a) Our position; Melanie plays Nxd5 (f6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#3414` (ply 27) [structural distance $d=65$, QAP obj=113]: the master played Nxd5; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pd6* as their *Bc8*, our *Nc6* as their *Re8*; transferred move f6d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#1604` (ply 22) [structural distance $d=71$, QAP obj=65]: the master played Bxd4; under μ our *Nf3* plays the role of that game’s *Be3* (a **cross-type** correspondence); cross-type roles: our *Pd6* as their *Rf1*, our *Pb5* as their *Bg2*; transferred move f6d5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0010#3980` (ply 20) [structural distance $d=57$, QAP obj=93]: the master played Nd2; under μ our *Nc3* plays the role of that game’s *Nf3*; cross-type roles: our *Pb5* as their *Rc1*, our *Bf8* as their *Rf1*; transferred move c6a7. Synthesis: 2 retrieved analogs transferred to Nxd5 (summed QAP objective 178, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



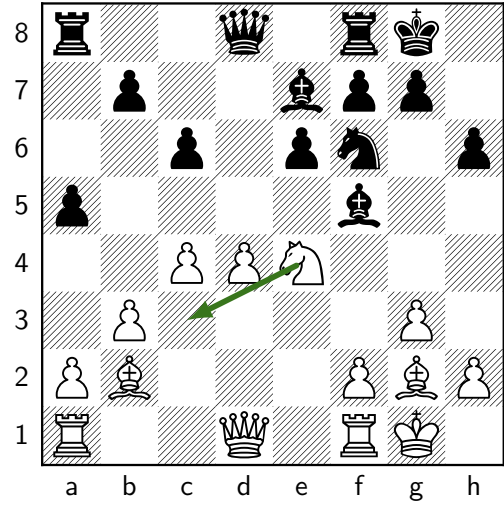
(a)



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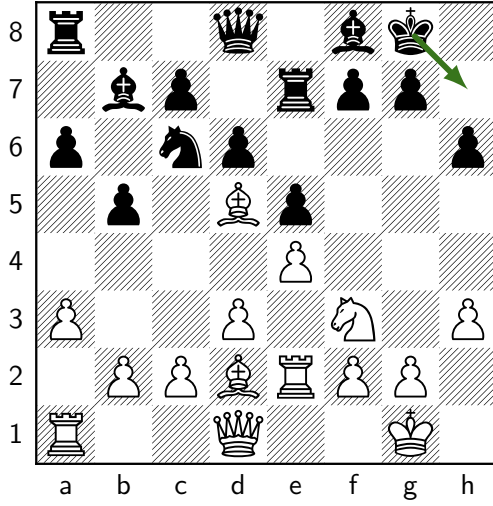


(c)

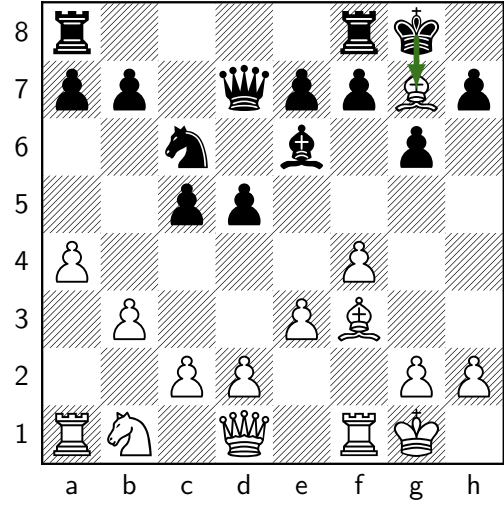


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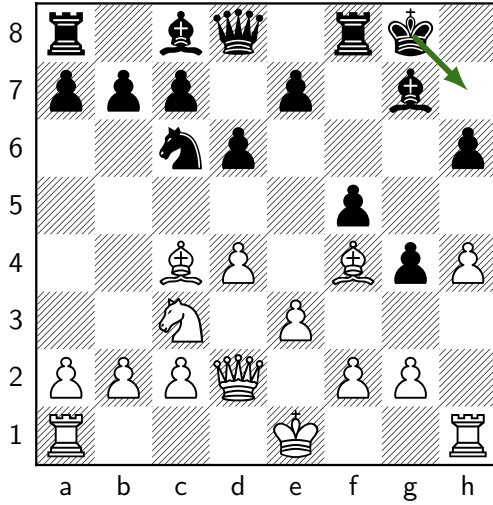
Figure 7: **Move 14, Black: Re7.** (a) Our position; Melanie plays Re7 (e8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3141` (ply 26) [structural distance $d=78$, QAP obj=44]: the master played Bb2; under μ our Re1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our bd2 as their Ra1, our Bb7 as their Rf1; transferred move e8e7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#3414` (ply 29) [structural distance $d=78$, QAP obj=84]: the master played Ne7; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our pg2 as their Bc8, our Bb7 as their pg2; transferred move c6e7. (d) Retrieved from `lichess_elite_2024-09_p0026#337` (ply 26) [structural distance $d=78$, QAP obj=40]: the master played Nc3; under μ our Nc3 plays the role of that game’s Ne4; cross-type roles: our ph3 as their Ra1, our Re8 as their Qd1; transferred move c6a7. Synthesis: 1 retrieved analog transferred to Re7 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



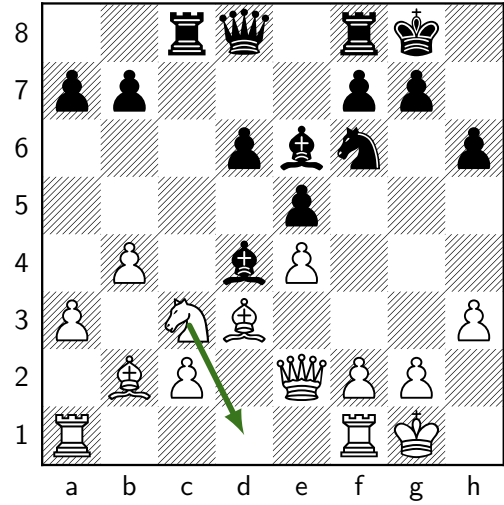
(a)



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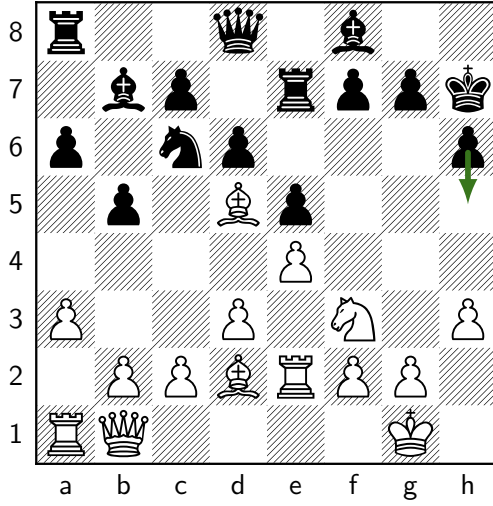


(c)

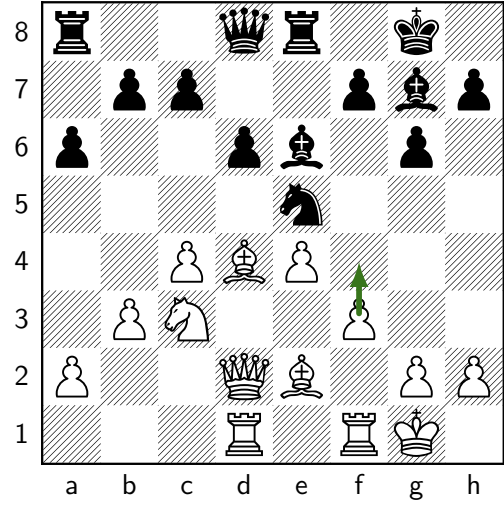


(d)

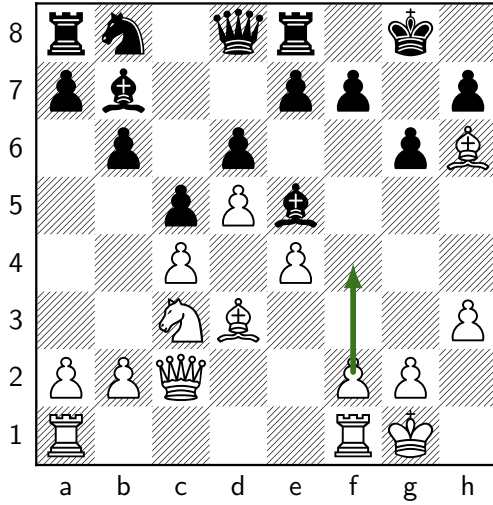
Figure 8: **Move 15, Black: Kh7.** (a) Our position; Melanie plays Kh7 (g8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#2544` (ply 23) [structural distance $d=97$, QAP obj=33]: the master played Kxg7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Nc6 as their Rf8, our Qd8 as their Pa7; transferred move g8h7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#847` (ply 23) [structural distance $d=89$, QAP obj=18]: the master played Kh7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our Pg7 as their Ra8, our Pc7 as their Bc8; transferred move g8h7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#3141` (ply 28) [structural distance $d=93$, QAP obj=82]: the master played Nd1; under μ our Nc3 plays the role of that game’s Nc3; cross-type roles: our Bb7 as their Rf1, our pa3 as their Bb2; transferred move c6b8. Synthesis: 2 retrieved analogs transferred to Kh7 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



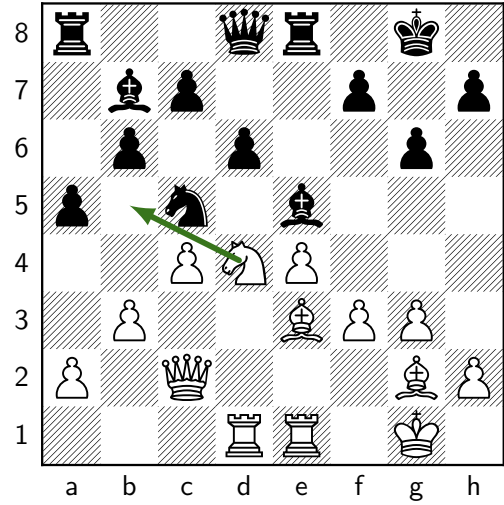
(a)



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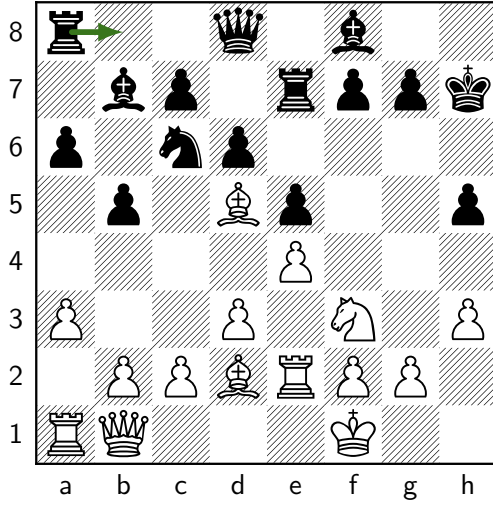


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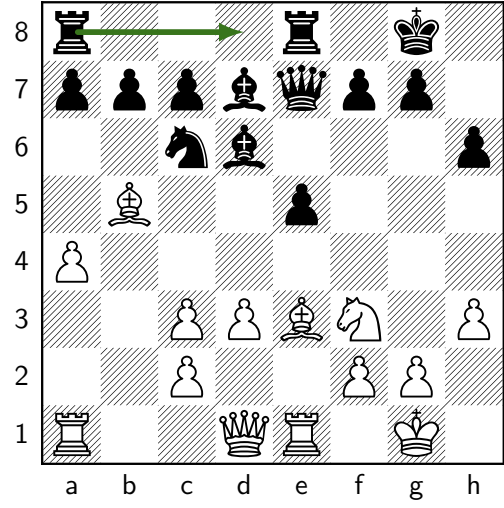


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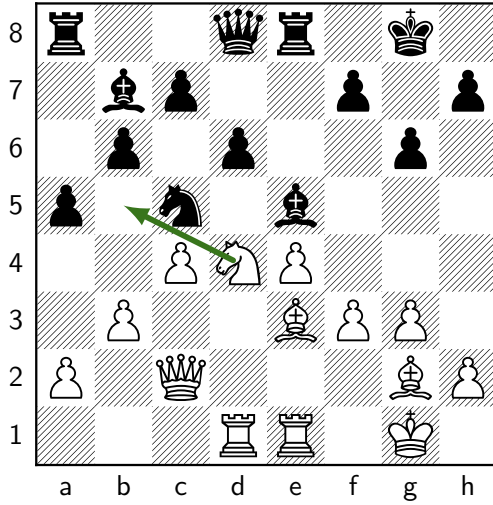
Figure 9: **Move 16, Black: h5**. (a) Our position; Melanie plays h5 (h6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#823` (ply 28) [structural distance $d=92$, QAP obj=25]: the master played f4; under μ our *Ph3* plays the role of that game’s *Pf3*; cross-type roles: our *Pc7* as their *Rd1*, our *Bf8* as their *Rf1*; transferred move h6h5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#449` (ply 24) [structural distance $d=78$, QAP obj=17]: the master played f4; under μ our *Ph3* plays the role of that game’s *Pf2*; cross-type roles: our *pf2* as their *Ra1*, our *Pg7* as their *Rf1*; transferred move h6h5 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#952` (ply 34) [structural distance $d=97$, QAP obj=35]: the master played Nb5; under μ our *Nc3* plays the role of that game’s *Nd4*; cross-type roles: our *Bb7* as their *Rd1*, our *Ph6* as their *Re1*; transferred move c6a7. Synthesis: 2 retrieved analogs transferred to h5 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



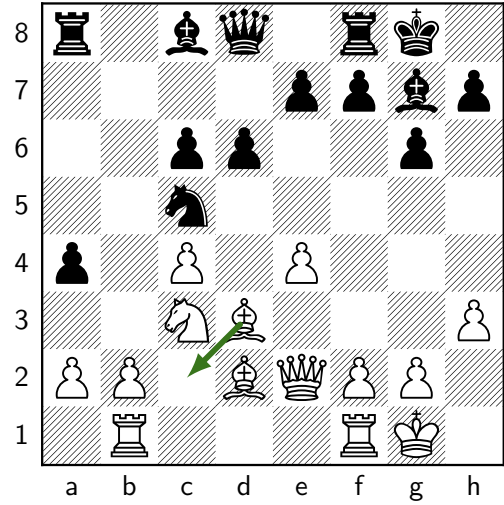
(a)



(b)

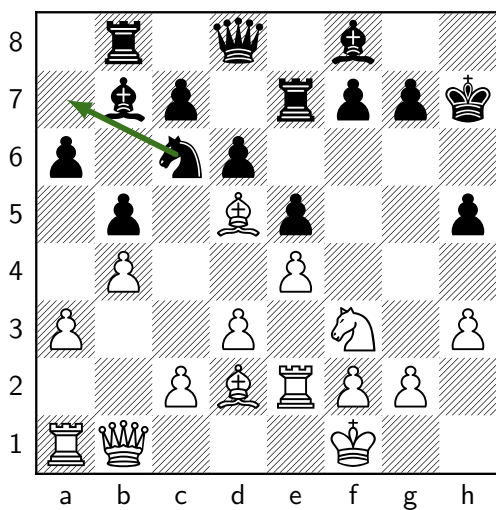


(c)

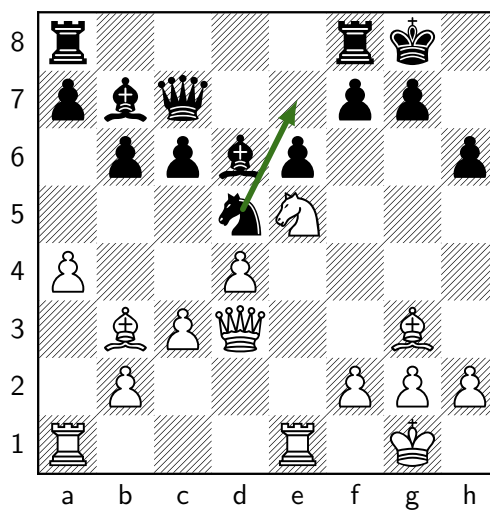


(d)

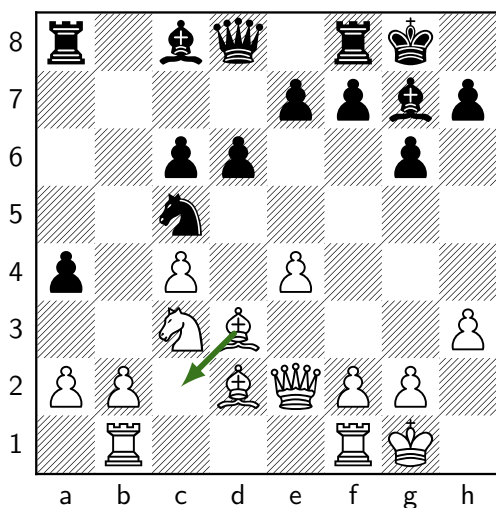
Figure 10: **Move 17, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#1274` (ply 25) [structural distance $d=89$, QAP obj=31]: the master played Rad8; under μ our Ra1 plays the role of that game’s Ra1; cross-type roles: our Bf8 as their Re8, our Bb7 as their Pa7; transferred move a8b8 (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#952` (ply 34) [structural distance $d=88$, QAP obj=35]: the master played Nb5; under μ our Nc3 plays the role of that game’s Nd4; cross-type roles: our Pe5 as their Rd1, our Pd6 as their Re1; transferred move c6a7. (d) Retrieved from `lichess_elite_2024-09_p0010#491` (ply 26) [structural distance $d=65$, QAP obj=25]: the master played Bc2; under μ our Nc3 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rb1, our Pg7 as their Rf1; transferred move c6a7. Synthesis: 1 retrieved analog transferred to Rb8 (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



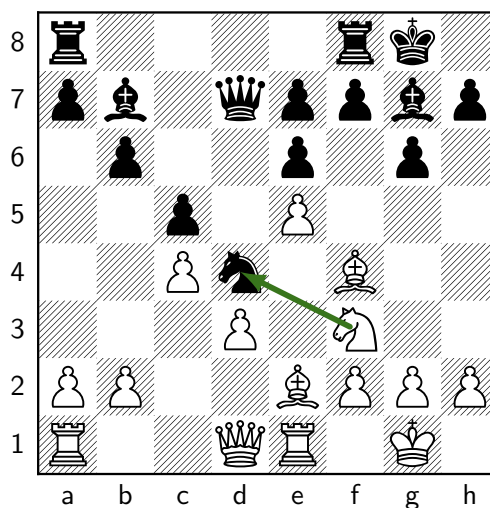
(a)



(b)

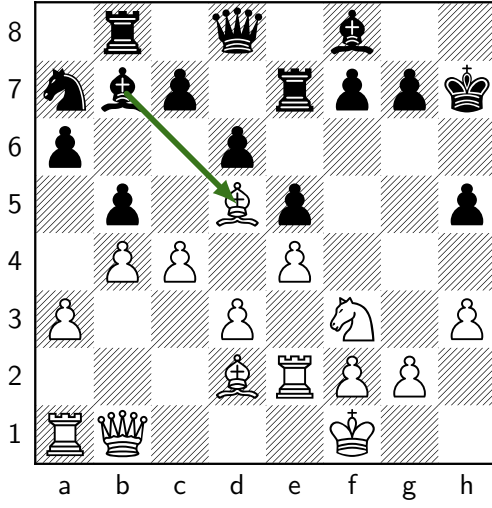


(c)

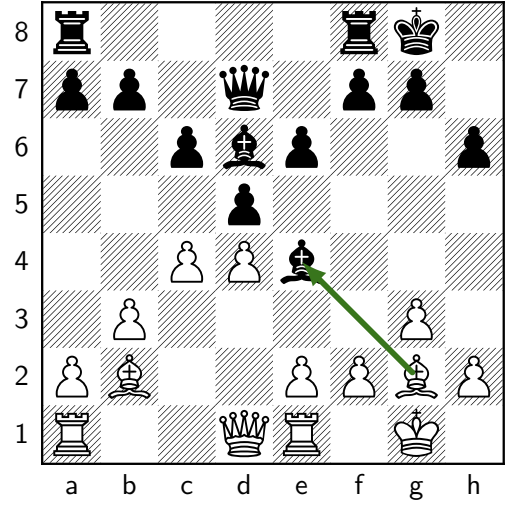


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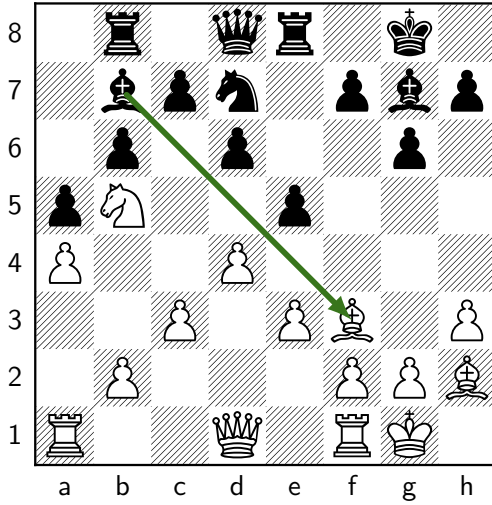
Figure 11: **Move 18, Black: Na7.** (a) Our position; Melanie plays Na7 (c6a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3310` (ply 33) [structural distance $d=85$, QAP obj=45]: the master played Ne7; under μ our Nc3 plays the role of that game’s Nd4; cross-type roles: our Pd6 as their Ra8, our bd2 as their Rf8; transferred move c6a7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#491` (ply 26) [structural distance $d=74$, QAP obj=28]: the master played Bc2; under μ our Nc3 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Bf8 as their Rf1, our bd5 as their nc5; transferred move c6a7 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0005#1010` (ply 28) [structural distance $d=83$, QAP obj=28]: the master played Nxd4; under μ our Nc3 plays the role of that game’s Nf3; cross-type roles: our ph3 as their Ra1, our Pa6 as their Qd1; transferred move c6a5. Synthesis: 2 retrieved analogs transferred to Na7 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



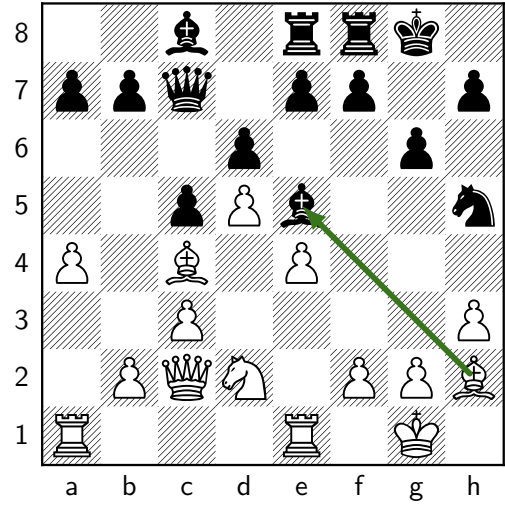
(a)



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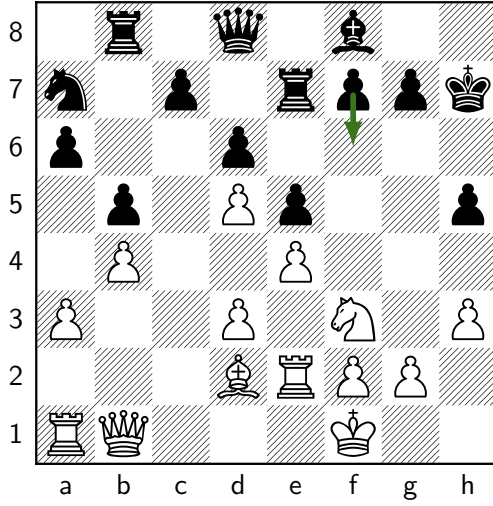


(c)

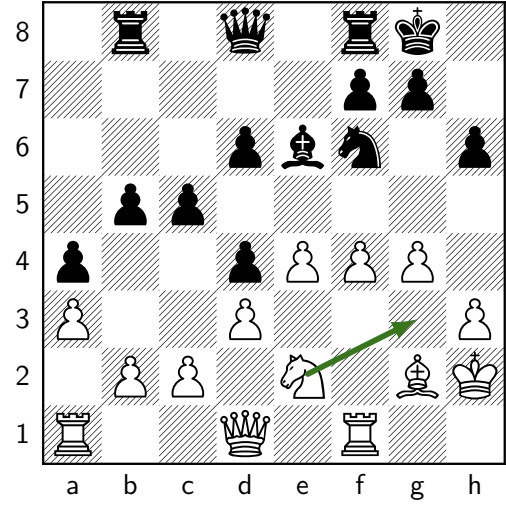


(d)

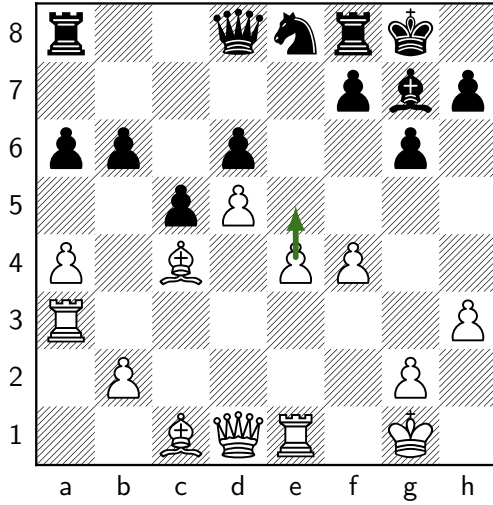
Figure 12: **Move 19, Black: Bxd5.** (a) Our position; Melanie plays Bxd5 (b7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#1717` (ply 26) [structural distance $d=87$, QAP obj=81]: the master played Bxe4; under μ our Bb2 plays the role of that game’s Bg2; cross-type roles: our Bf8 as their Ra1, our Pc7 as their Qd1; transferred move b7d5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0013#3061` (ply 31) [structural distance $d=83$, QAP obj=62]: the master played Bxf3; under μ our Bb2 plays the role of that game’s Bb2; cross-type roles: our Re7 as their Qd8, our pb4 as their Re8; transferred move b7d5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#2275` (ply 36) [structural distance $d=79$, QAP obj=38]: the master played Bxe5; under μ our Bb2 plays the role of that game’s Bh2; cross-type roles: our Pg7 as their Re1, our Pd6 as their Qc2; transferred move b7d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bxd5 (summed QAP objective 181, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



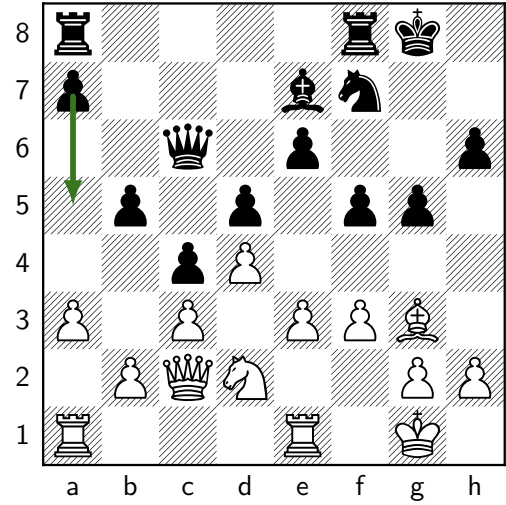
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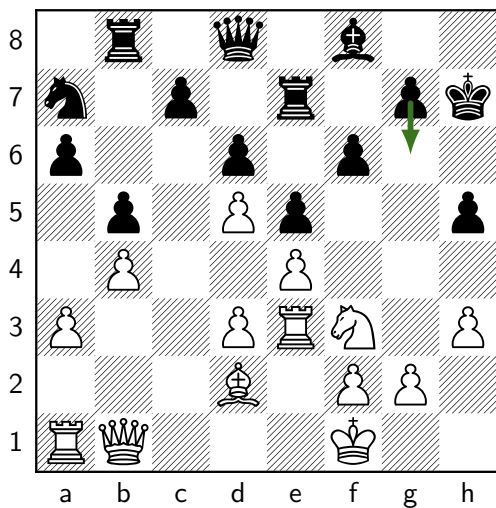


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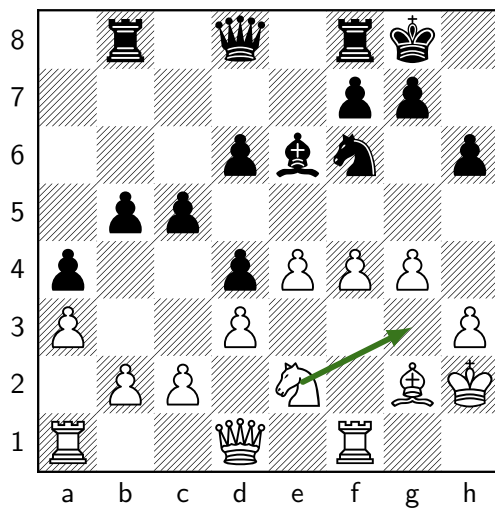


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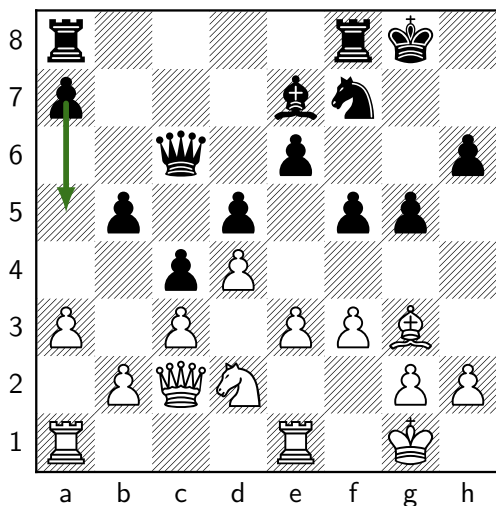
Figure 13: **Move 20, Black: f6**. (a) Our position; Melanie plays f6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1604` (ply 32) [structural distance $d=69$, QAP obj=58]: the master played Ng3; under μ our Pf2 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Rf1, our Na7 as their Pb2; transferred move f7f6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#715` (ply 34) [structural distance $d=68$, QAP obj=27]: the master played e5; under μ our Pa3 plays the role of that game’s Pe4; cross-type roles: our Qd8 as their Bc1, our Pd6 as their Qd1; transferred move a6a5. (d) Retrieved from `lichess_elite_2024-09_p0031#2423` (ply 41) [structural distance $d=82$, QAP obj=24]: the master played a5; under μ our Rb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our qb1 as their Ra8, our bd2 as their Rf8; transferred move b8a8. Synthesis: 1 retrieved analog transferred to f6 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



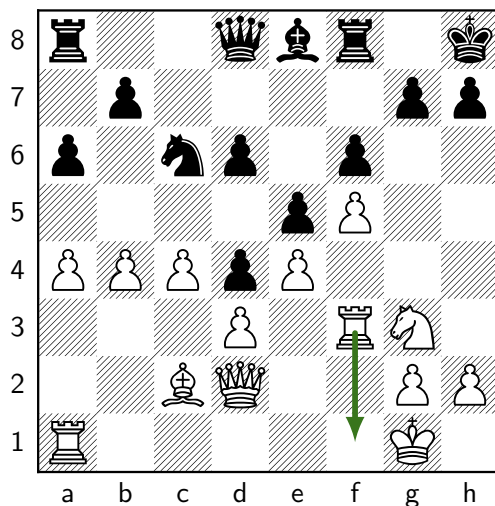
(a)



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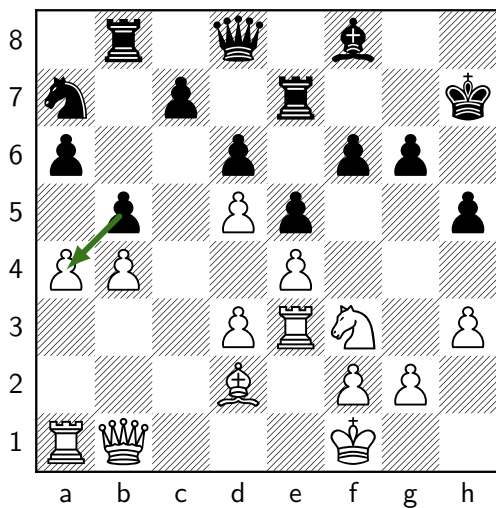


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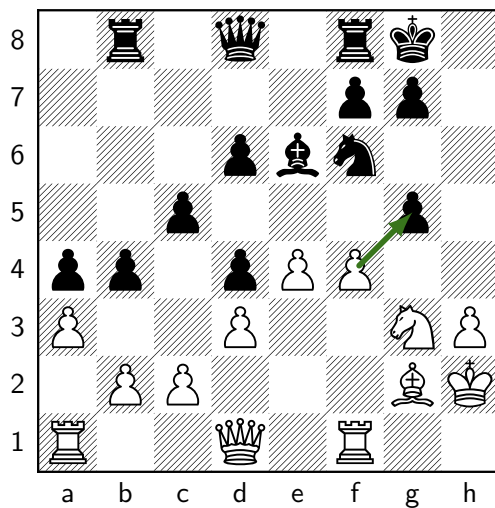


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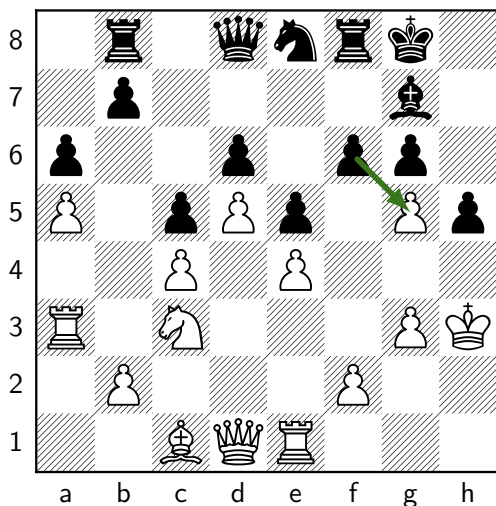
Figure 14: **Move 21, Black: g6**. (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1604` (ply 32) [structural distance $d=67$, QAP obj=59]: the master played Ng3; under μ our *Pg2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *pa3* as their *Ra1*, our *Bf8* as their *Rf1*; transferred move g7g6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#2423` (ply 41) [structural distance $d=69$, QAP obj=24]: the master played a5; under μ our *Rb1* plays the role of that game’s *Pa2* (a **cross-type** correspondence); cross-type roles: our *Bf8* as their *Ra8*, our *Pg7* as their *Rf8*; transferred move b8a8. (d) Retrieved from `lichess_elite_2024-09_p0019#1688` (ply 36) [structural distance $d=84$, QAP obj=23]: the master played Rff1; under μ our *Pf3* plays the role of that game’s *Rf3* (a **cross-type** correspondence); cross-type roles: our *pe4* as their *Ra1*, our *pd5* as their *Bc2*; transferred move f6f5. Synthesis: 1 retrieved analog transferred to g6 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



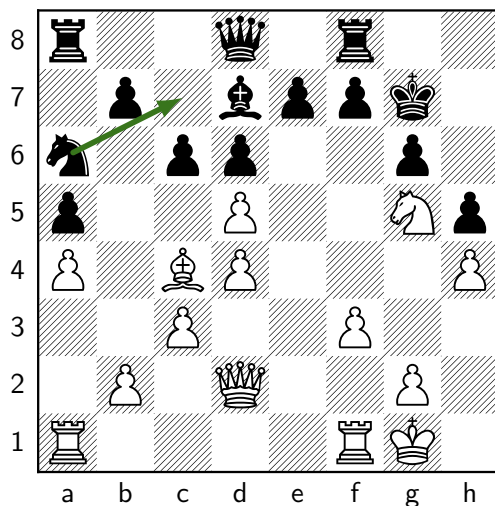
(a)



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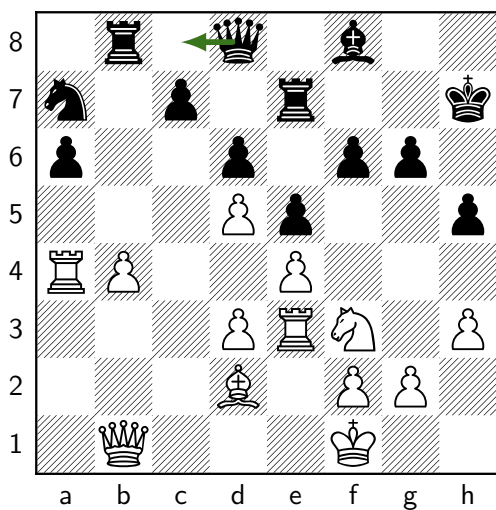


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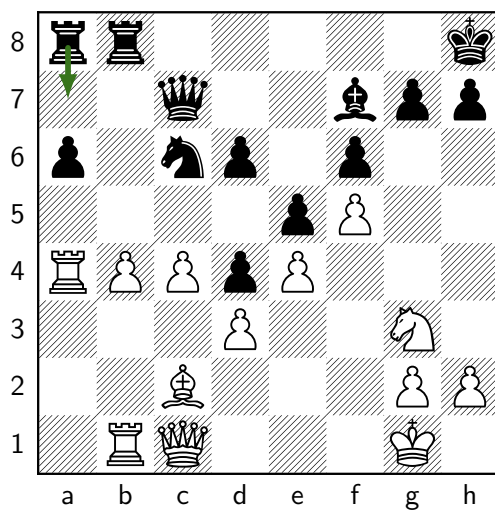


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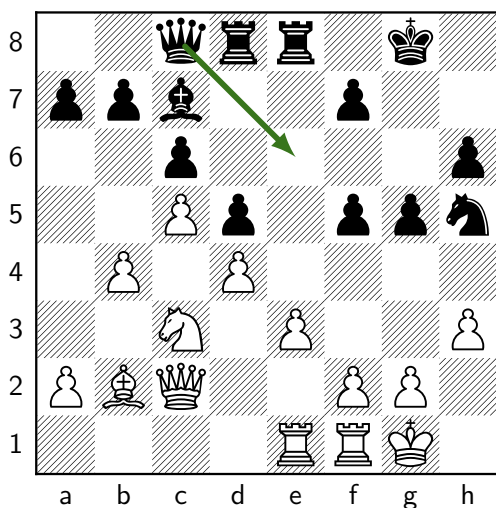
Figure 15: **Move 22, Black: bxa4.** (a) Our position; Melanie plays **bxa4** (**b5a4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#1604` (ply 36) [structural distance $d=80$, QAP obj=70]: the master played **fxg5**; under μ our $Pb4$ plays the role of that game’s $Pf4$; cross-type roles: our $Qd8$ as their $Ra1$, our $Na7$ as their $Qd1$; transferred move **b5a4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3460` (ply 45) [structural distance $d=68$, QAP obj=48]: the master played **fxg5**; under μ our $Pb4$ plays the role of that game’s $Pf3$; cross-type roles: our $pg2$ as their $Rb8$, our $nf3$ as their $Qd8$; transferred move **b5a4** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0013#840` (ply 37) [structural distance $d=56$, QAP obj=41]: the master played **Nc7**; under μ our $Pb4$ plays the role of that game’s $Na3$ (a **cross-type** correspondence); cross-type roles: our $Na7$ as their $Ra8$, our $Pg6$ as their $Qd8$; transferred move **b5a4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **bxa4** (summed QAP objective 158, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



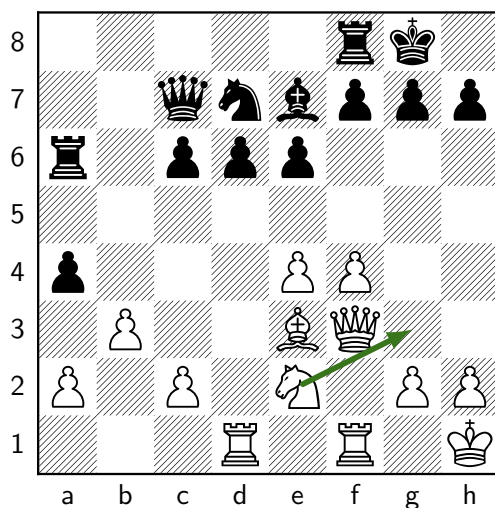
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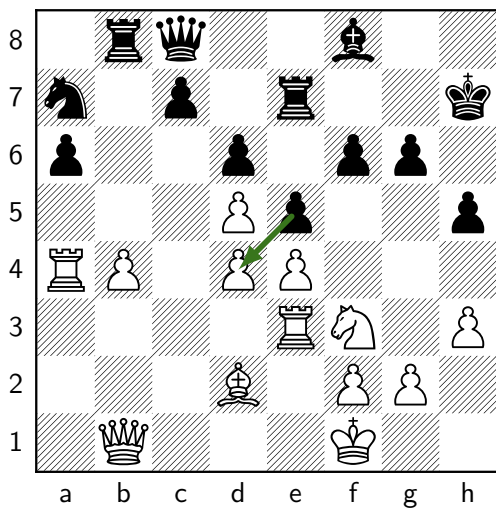


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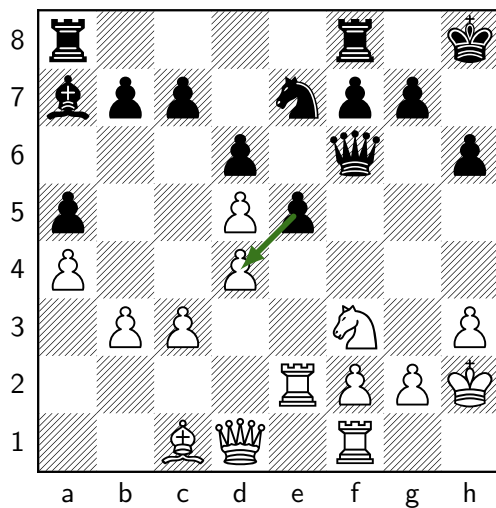


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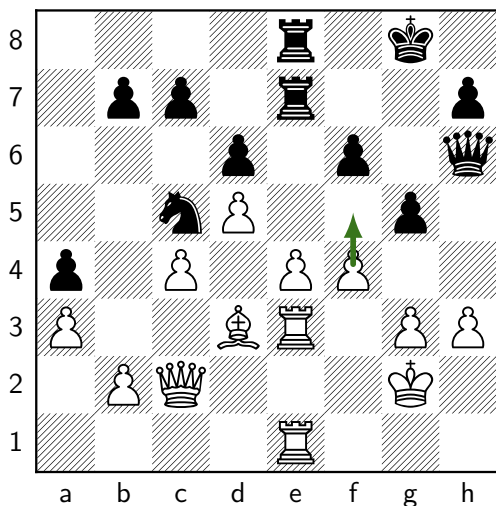
Figure 16: **Move 23, Black: Qc8**. (a) Our position; Melanie plays Qc8 (d8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0019#1688 (ply 47) [structural distance $d=77$, QAP obj=92]: the master played Ra7; under μ our Qd1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Na7 as their Qc7, our nf3 as their Bf7; transferred move d8c8 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0023#1619 (ply 45) [structural distance $d=68$, QAP obj=50]: the master played Qe6; under μ our Qd1 plays the role of that game’s Qc1; cross-type roles: our Bf8 as their Rd8, our Na7 as their Bc7; transferred move d8c8 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0020#3782 (ply 32) [structural distance $d=77$, QAP obj=26]: the master played Ng3; under μ our Qd1 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our Bf8 as their Ph2, our pe4 as their Be3; transferred move d8c8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qc8 (summed QAP objective 169, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



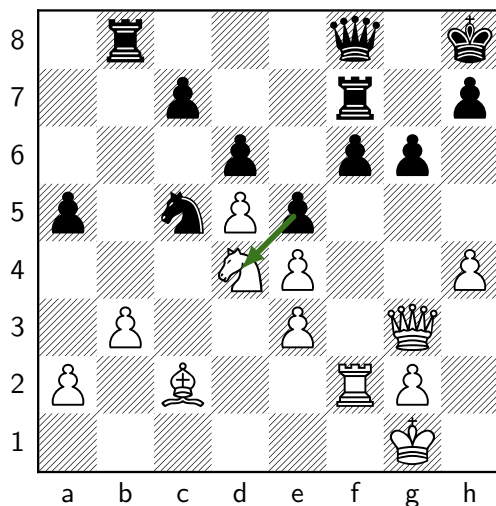
(a)



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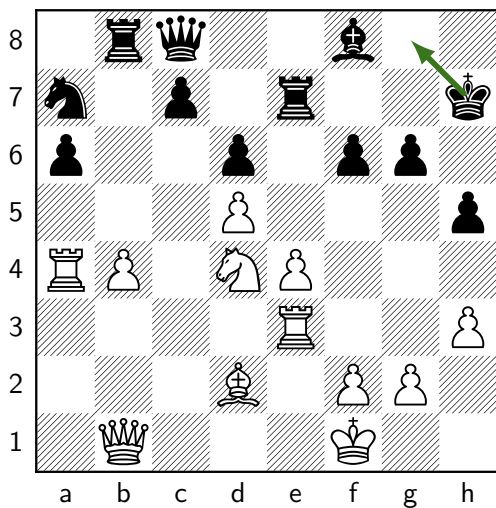


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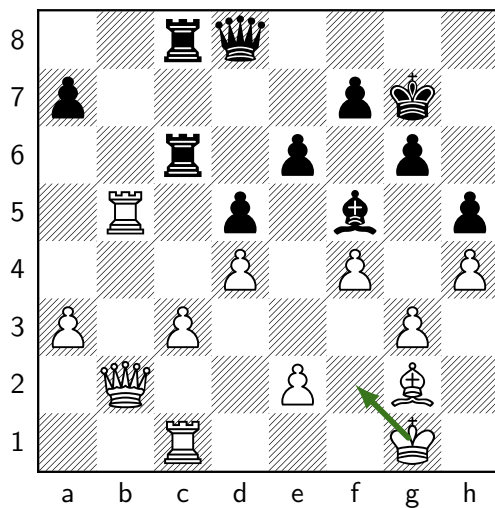


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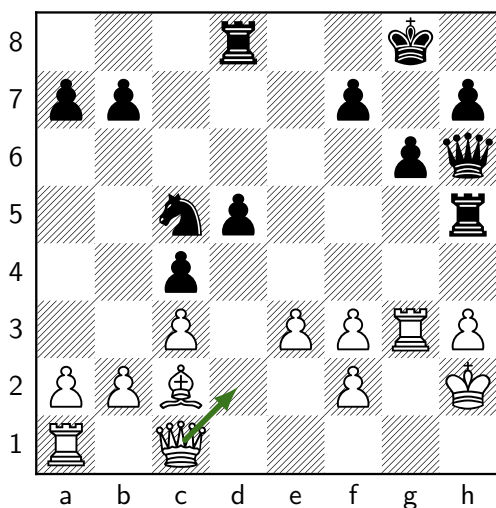
Figure 17: **Move 24, Black: exd4**. (a) Our position; Melanie plays **exd4** (e5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0020#961` (ply 33) [structural distance $d=76$, QAP obj=87]: the master played **exd4**; under μ our *Pe4* plays the role of that game’s *Pe4*; cross-type roles: our *pe4* as their *Rf8*, our *Pe5* as their *Ba7*; transferred move **e5d4** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#2630` (ply 52) [structural distance $d=84$, QAP obj=59]: the master played **f5**; under μ our *Pe4* plays the role of that game’s *Pf4*; cross-type roles: our *Na7* as their *Pb2*, our *Re7* as their *Qc2*; transferred move **e5d4** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#2084` (ply 61) [structural distance $d=69$, QAP obj=36]: the master played **exd4**; under μ our *Pe4* plays the role of that game’s *Pe4*; cross-type roles: our *Re7* as their *Qf8*, our *qb1* as their *Rf7*; transferred move **e5d4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **exd4** (summed QAP objective 181, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



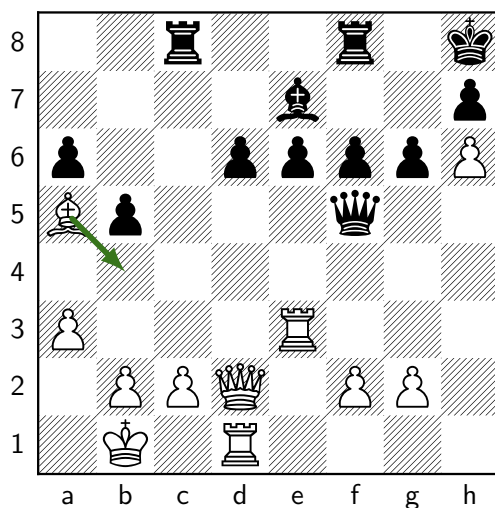
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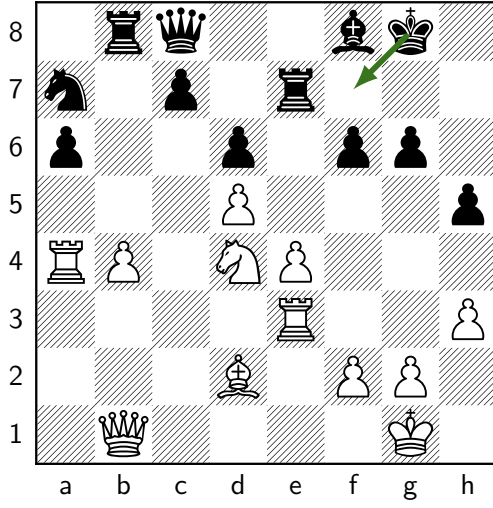


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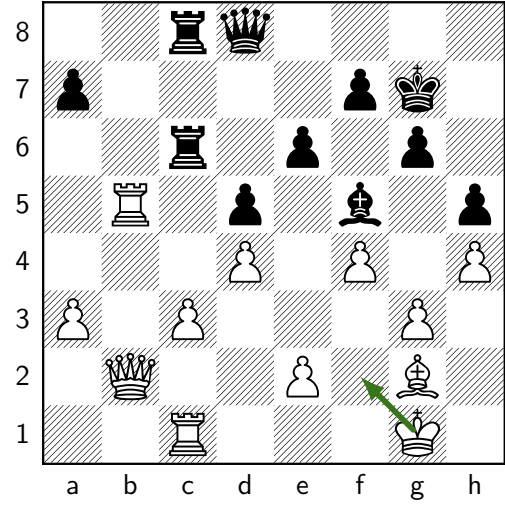


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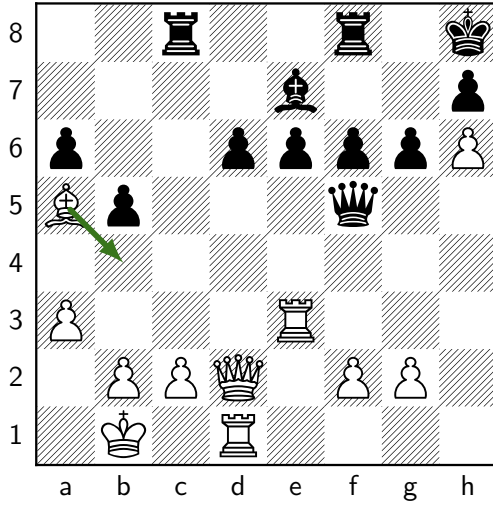
Figure 18: **Move 25, Black: Kg8.** (a) Our position; Melanie plays Kg8 (h7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#1239` (ply 50) [structural distance $d=79$, QAP obj=108]: the master played Kf2; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our ra4 as their Bg2, our Na7 as their Pg3; transferred move h7g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#820` (ply 52) [structural distance $d=76$, QAP obj=92]: the master played Qd2; under μ our Ph4 plays the role of that game’s Qc1 (a **cross-type** correspondence); cross-type roles: our Na7 as their Pb2, our Re7 as their Bc2; transferred move h5h4. (d) Retrieved from `lichess_elite_2024-09_p0019#3093` (ply 52) [structural distance $d=79$, QAP obj=77]: the master played Bb4; under μ our Qc1 plays the role of that game’s Ba5 (a **cross-type** correspondence); cross-type roles: our nd4 as their Rd1, our Na7 as their Pb2; transferred move c8e8. Synthesis: 1 retrieved analog transferred to Kg8 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



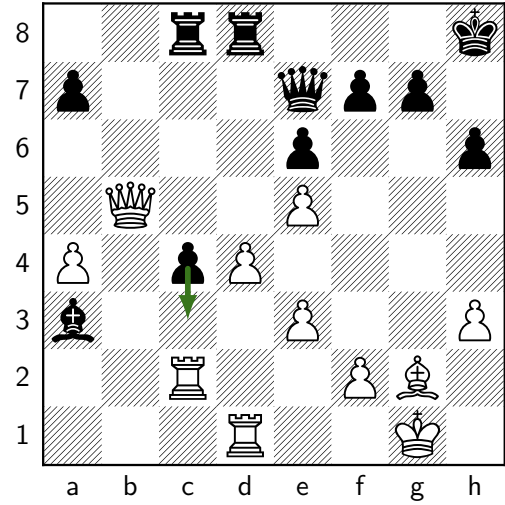
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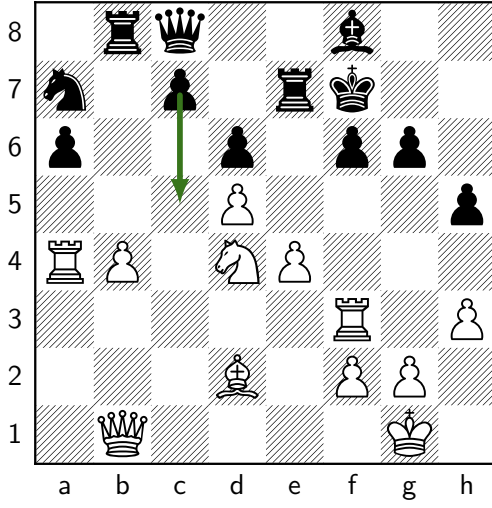


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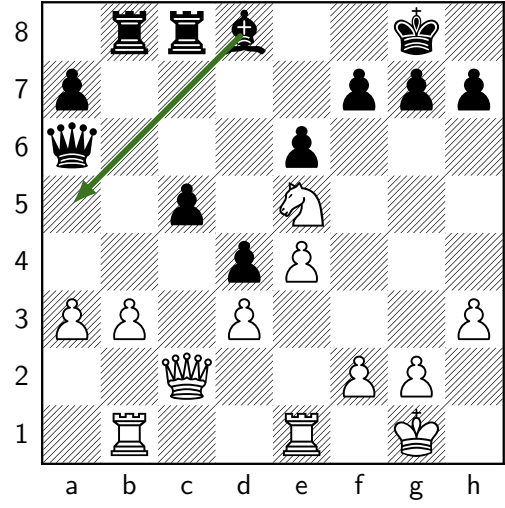


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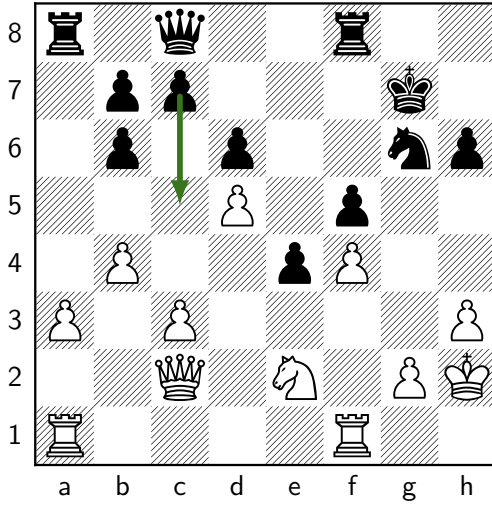
Figure 19: **Move 26, Black: Kf7**. (a) Our position; Melanie plays Kf7 (g8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#1239` (ply 50) [structural distance $d=89$, QAP obj=108]: the master played Kf2; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *ra4* as their *Bg2*, our *Rb8* as their *Pa3*; transferred move g8f7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#3093` (ply 52) [structural distance $d=86$, QAP obj=76]: the master played Bb4; under μ our *Na2* plays the role of that game’s *Ba5* (a **cross-type** correspondence); cross-type roles: our *re3* as their *Pg2*, our *Re7* as their *qf5*; transferred move a7b5. (d) Retrieved from `lichess_elite_2024-09_p0021#405` (ply 57) [structural distance $d=93$, QAP obj=60]: the master played c3; under μ our *Pa3* plays the role of that game’s *Pc5*; cross-type roles: our *Qc8* as their *Rc8*, our *Bf8* as their *Pa7*; transferred move a6a5. Synthesis: 1 retrieved analog transferred to Kf7 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



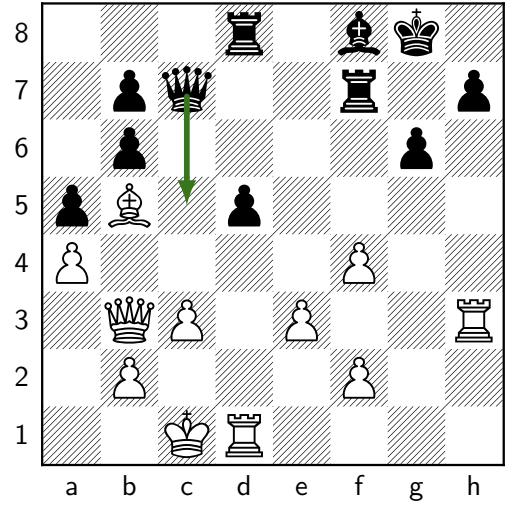
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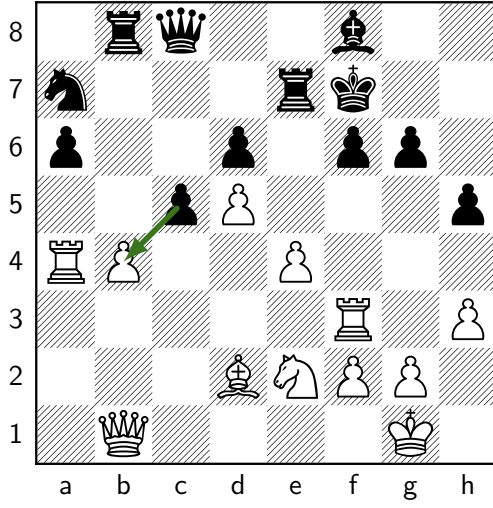


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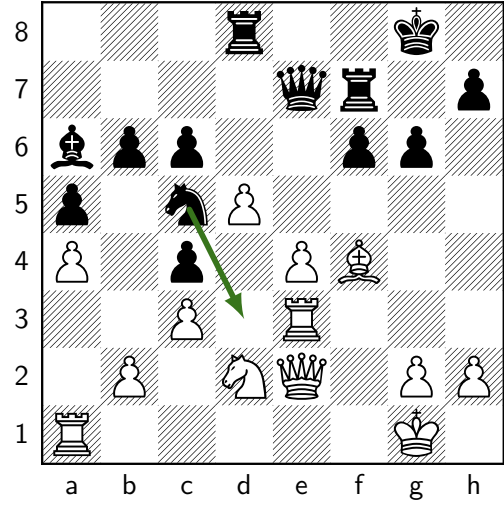


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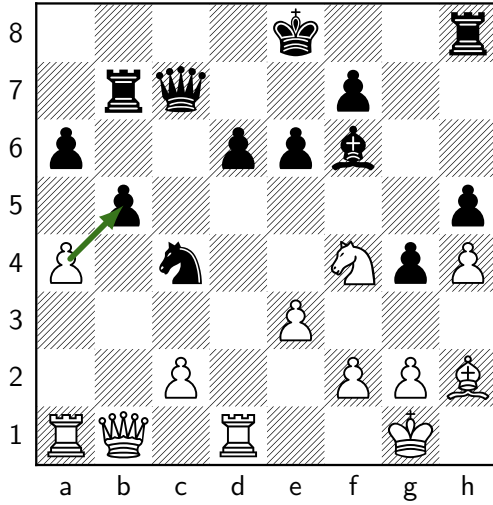
Figure 20: **Move 27, Black: c5.** (a) Our position; Melanie plays c5 (c7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#137` (ply 51) [structural distance $d=72$, QAP obj=86]: the master played Bb5; under μ our *Pc2* plays the role of that game’s *Bd1* (a **cross-type** correspondence); cross-type roles: our *Qc8* as their *Rc8*, our *nd4* as their *Pa7*; transferred move c7c5 (**endorses this move**). (c) Retrieved from `twic1500#180` (ply 43) [structural distance $d=83$, QAP obj=52]: the master played c5; under μ our *Pc2* plays the role of that game’s *Pc2*; cross-type roles: our *Pd6* as their *Qc8*, our *Pg6* as their *Rf8*; transferred move c7c5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#2952` (ply 47) [structural distance $d=84$, QAP obj=106]: the master played Qc5; under μ our *Rb1* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Qc8* as their *Rd8*, our *Na7* as their *Bf8*; transferred move b8b6. Synthesis: 2 retrieved analogs transferred to c5 (summed QAP objective 137, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



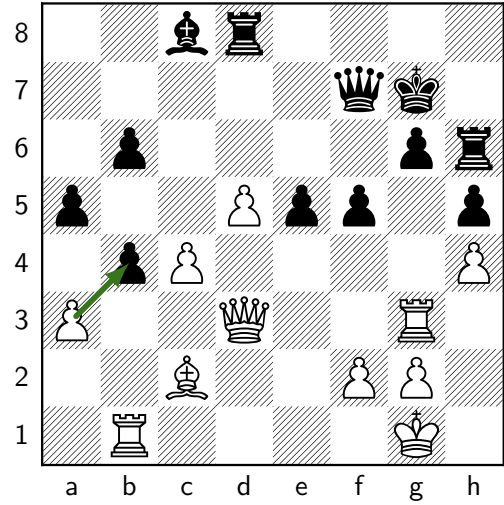
(a)



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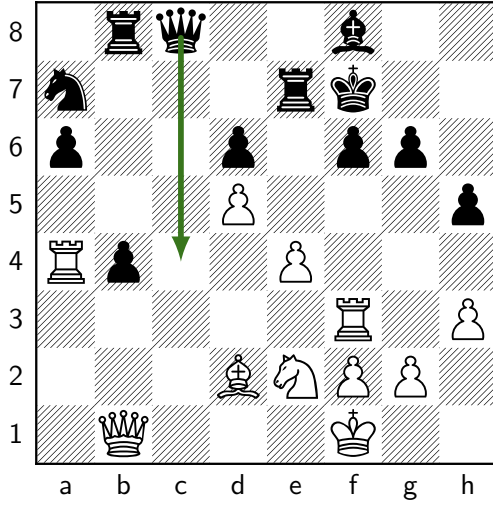


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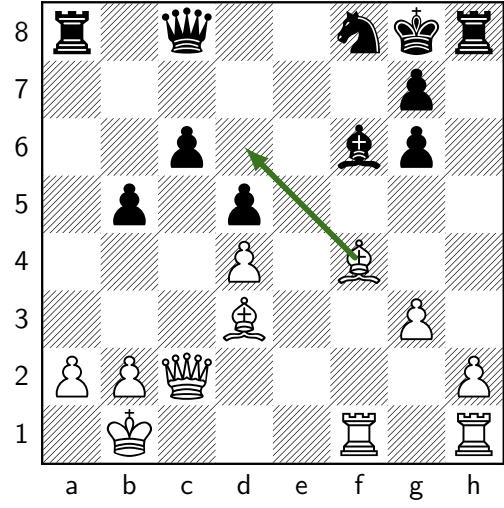


(d)

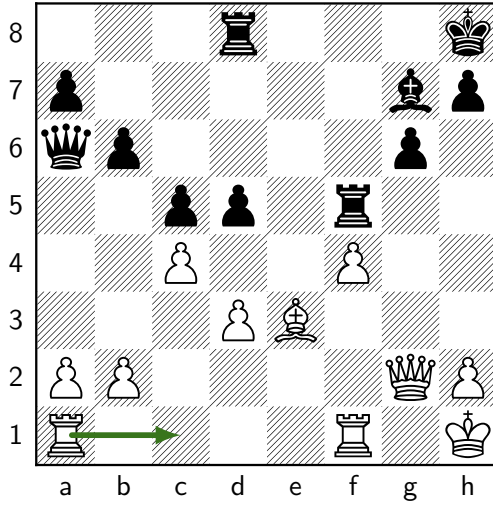
Figure 21: **Move 28, Black: cxb4.** (a) Our position; Melanie plays cxb4 (c5b4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#20` (ply 43) [structural distance $d=81$, QAP obj=126]: the master played Nd3; under μ our $Pc4$ plays the role of that game’s $Nc4$ (a **cross-type** correspondence); cross-type roles: our $Rb8$ as their $Qe7$, our $Bf8$ as their $Rf7$; transferred move c5b4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#2298` (ply 48) [structural distance $d=89$, QAP obj=115]: the master played axb5; under μ our $Pc4$ plays the role of that game’s $Pa4$; cross-type roles: our $Pc5$ as their $Ra1$, our $Rb8$ as their $Qb1$; transferred move c5b4 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0020#1920` (ply 68) [structural distance $d=67$, QAP obj=102]: the master played axb4; under μ our $Pc4$ plays the role of that game’s $Pa3$; cross-type roles: our $Qc8$ as their $Bc2$, our $ra4$ as their $Qd3$; transferred move c5b4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to cxb4 (summed QAP objective 440, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



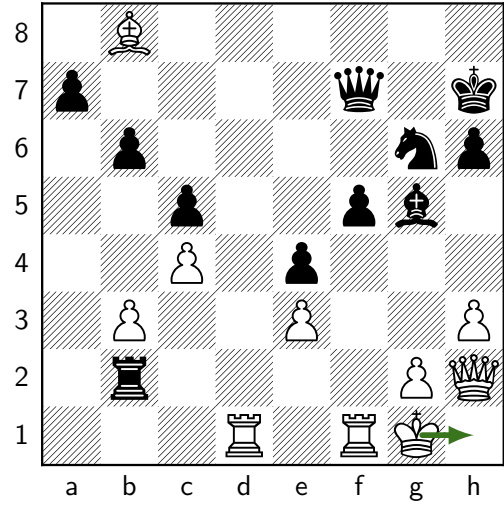
(a)



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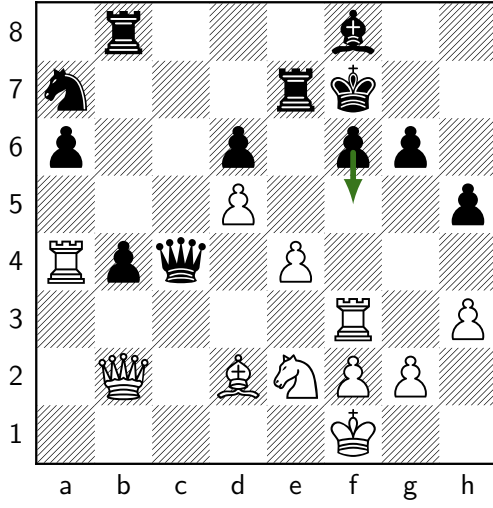


(c)

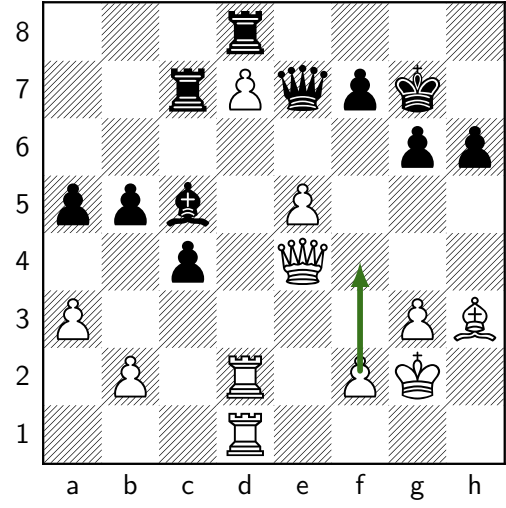


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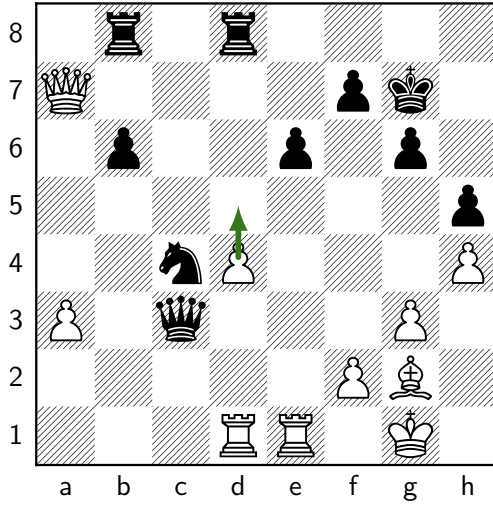
Figure 22: **Move 29, Black: Qc4.** (a) Our position; Melanie plays Qc4 (c8c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#314` (ply 46) [structural distance $d=94$, QAP obj=53]: the master played Bd6; under μ our Qc1 plays the role of that game’s Bf4 (a **cross-type** correspondence); cross-type roles: our ph3 as their Rf1, our pf2 as their Rh1; transferred move c8c4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0018#3280` (ply 52) [structural distance $d=91$, QAP obj=85]: the master played Rac1; under μ our Rb1 plays the role of that game’s Ra1; cross-type roles: our Pg6 as their Rf1, our Re7 as their Qg2; transferred move b8b7. (d) Retrieved from `lichess_elite_2024-09_p0031#2365` (ply 68) [structural distance $d=83$, QAP obj=79]: the master played Kh1; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Bf8 as their Rd1, our bd2 as their pc5; transferred move f7g7. Synthesis: 1 retrieved analog transferred to Qc4 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



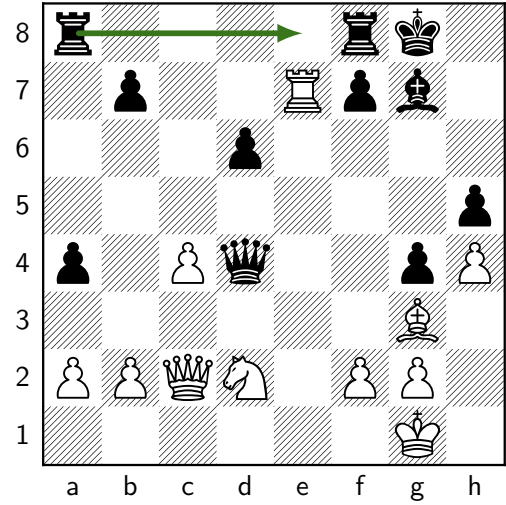
(a)



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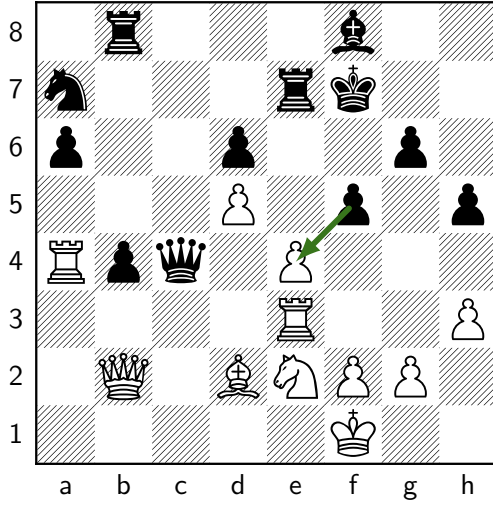


(c)

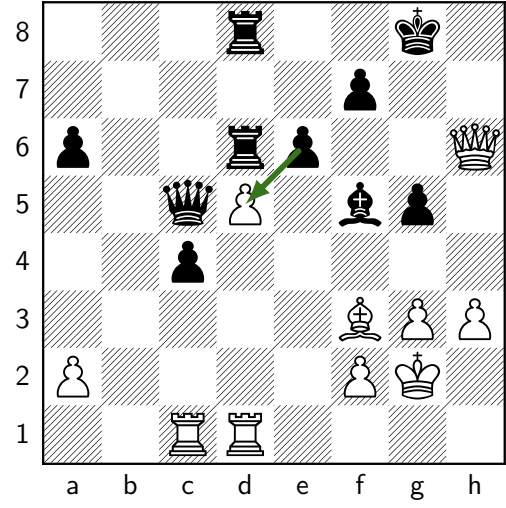


(d)

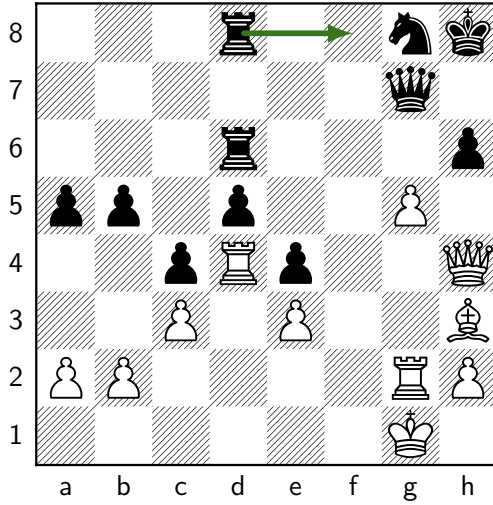
Figure 23: **Move 30, Black: f5**. (a) Our position; Melanie plays f5 (f6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#1427` (ply 60) [structural distance $d=79$, QAP obj=97]: the master played f4; under μ our *Pf3* plays the role of that game’s *Pf2*; cross-type roles: our *Bf8* as their *Rd1*, our *Pd6* as their *Rd2*; transferred move f6f5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#1695` (ply 66) [structural distance $d=95$, QAP obj=21]: the master played d5; under μ our *Pf3* plays the role of that game’s *Pd4*; cross-type roles: our *Bf8* as their *Rd1*, our *Pd6* as their *Bg2*; transferred move f6f5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0018#1915` (ply 47) [structural distance $d=106$, QAP obj=95]: the master played *Rae8*; under μ our *Rb1* plays the role of that game’s *Ra1*; cross-type roles: our *rf3* as their *bg3*, our *bd2* as their *pa2*; transferred move b8c8. Synthesis: 2 retrieved analogs transferred to f5 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



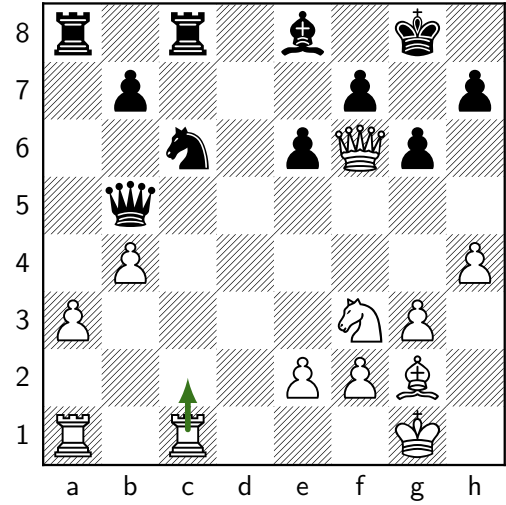
(a)



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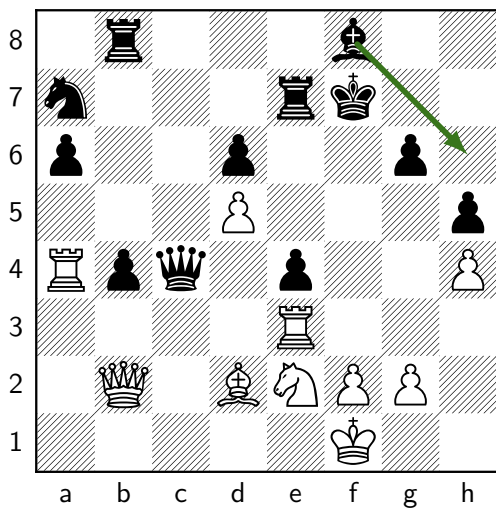


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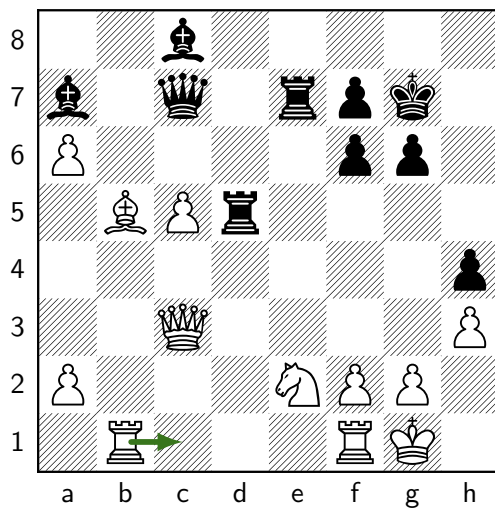


(d)

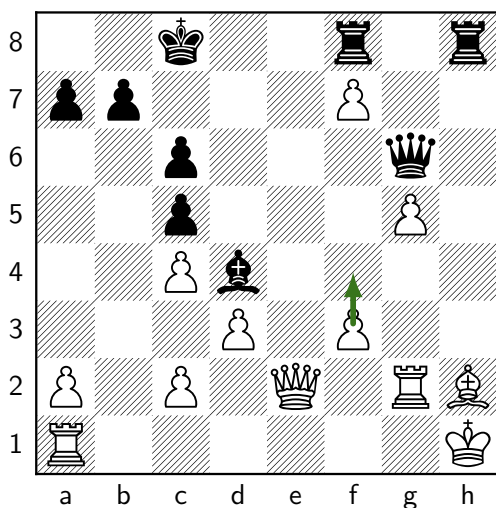
Figure 24: **Move 31, Black: fxe4.** (a) Our position; Melanie plays fxe4 (f5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0012#1850 (ply 63) [structural distance $d=88$, QAP obj=110]: the master played exd5; under μ our Pf4 plays the role of that game’s Pe3; cross-type roles: our Bf8 as their Rd8, our Na7 as their Pa6; transferred move f5e4 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0010#135 (ply 75) [structural distance $d=90$, QAP obj=109]: the master played Rf8; under μ our Bf1 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Pg6 as their Ng8, our Re7 as their Qg7; transferred move f8h6. (d) Retrieved from lichess_elite_2024-09_p0020#3663 (ply 44) [structural distance $d=92$, QAP obj=101]: the master played Rc2; under μ our Re2 plays the role of that game’s Rc1; cross-type roles: our Pg6 as their Ra1, our Bf8 as their Pf2; transferred move e7d7. Synthesis: 1 retrieved analog transferred to fxe4 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



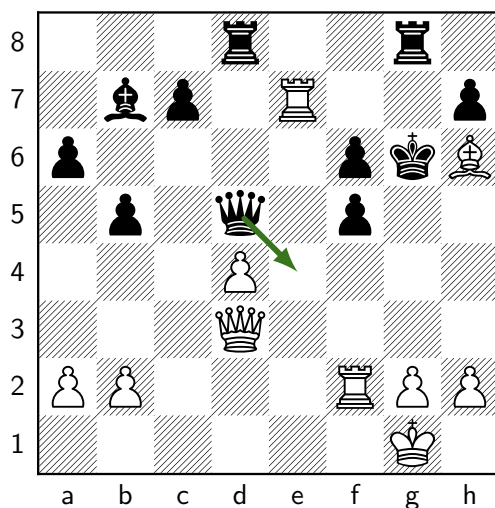
(a)



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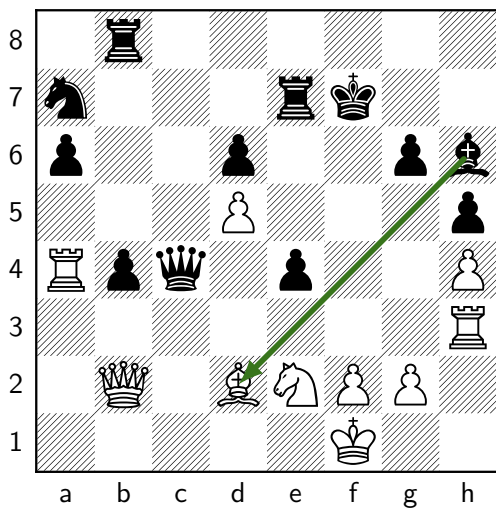


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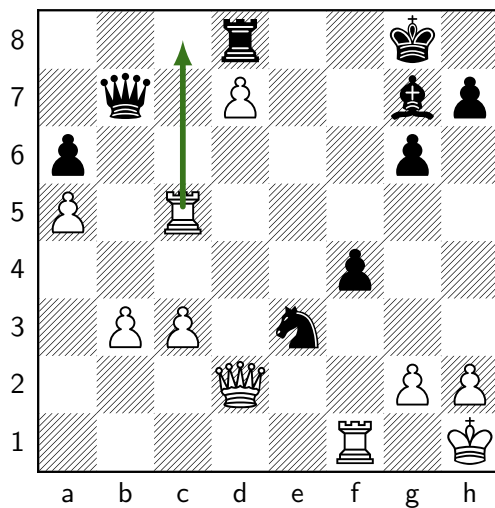


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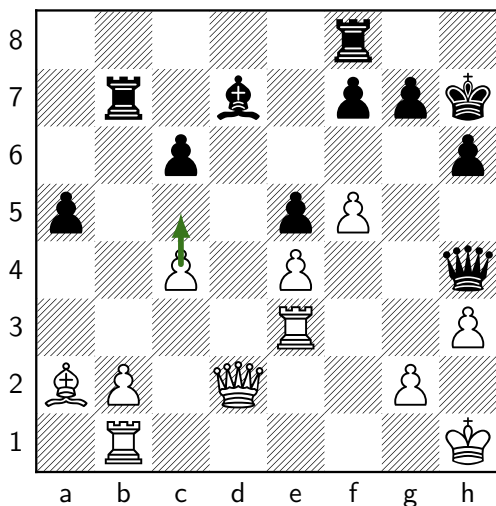
Figure 25: **Move 32, Black: Bh6**. (a) Our position; Melanie plays Bh6 (f8h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0030#462` (ply 70) [structural distance $d=96$, QAP obj=112]: the master played Rbc1; under μ our Bf1 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pa2, our pd5 as their Ne2; transferred move f8h6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3393` (ply 60) [structural distance $d=104$, QAP obj=71]: the master played f4; under μ our Bf1 plays the role of that game’s Pf3 (a **cross-type** correspondence); cross-type roles: our ne2 as their Ra1, our Rb8 as their Pa2; transferred move f8h6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0003#921` (ply 55) [structural distance $d=109$, QAP obj=48]: the master played Qe4; under μ our Qc5 plays the role of that game’s Qd4; cross-type roles: our Pd6 as their Rd8, our Ph5 as their Bb7; transferred move c4d3. Synthesis: 2 retrieved analogs transferred to Bh6 (summed QAP objective 183, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



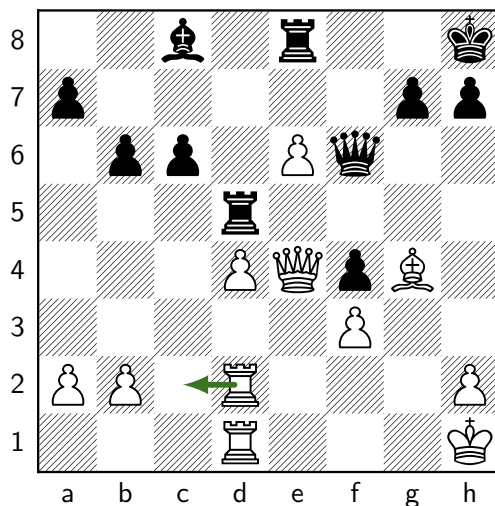
(a)



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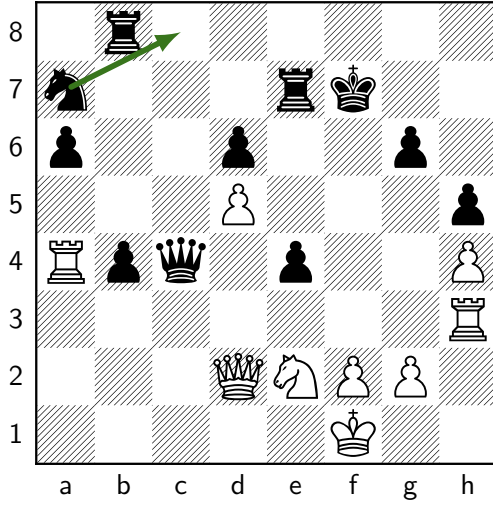


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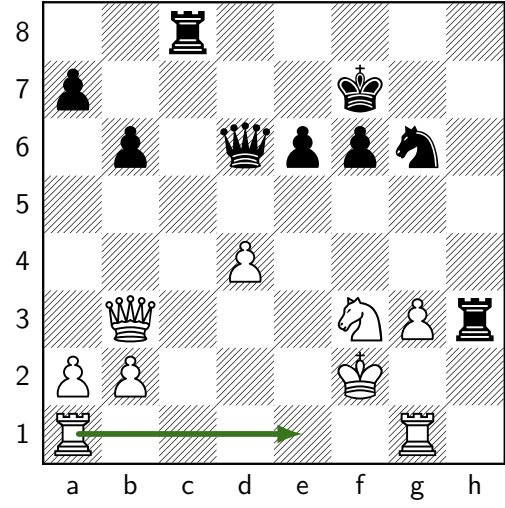


(d)

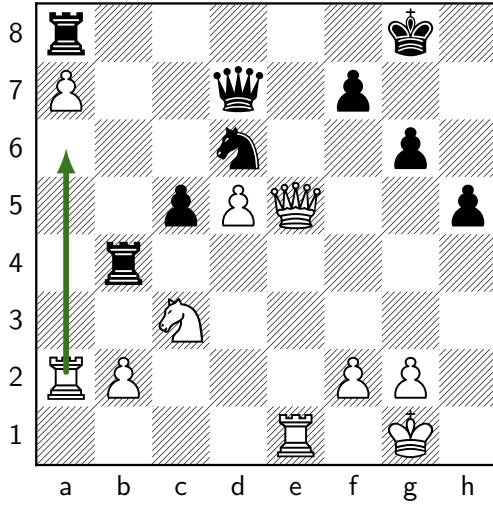
Figure 26: **Move 33, Black: Bxd2.** (a) Our position; Melanie plays Bxd2 (h6d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1637` (ply 58) [structural distance $d=88$, QAP obj=48]: the master played Rc8; under μ our *Bh3* plays the role of that game’s *Rc5* (a **cross-type** correspondence); cross-type roles: our *Na7* as their *Ph2*, our *bd2* as their *pg6*; transferred move h6d2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#3553` (ply 56) [structural distance $d=78$, QAP obj=45]: the master played c5; under μ our *Bh3* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *ph4* as their *Rb1*, our *pg2* as their *Ba2*; transferred move h6d2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0019#2353` (ply 52) [structural distance $d=91$, QAP obj=58]: the master played Rc2; under μ our *Pe5* plays the role of that game’s *Rd2* (a **cross-type** correspondence); cross-type roles: our *Na7* as their *Pa2*, our *Bh6* as their *Pf3*; transferred move e4e3. Synthesis: 2 retrieved analogs transferred to Bxd2 (summed QAP objective 94, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



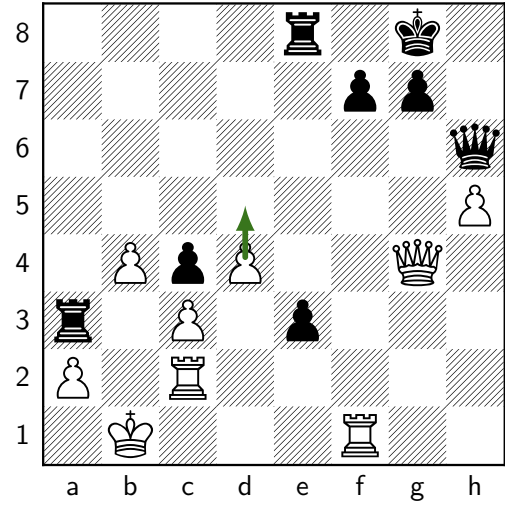
(a)



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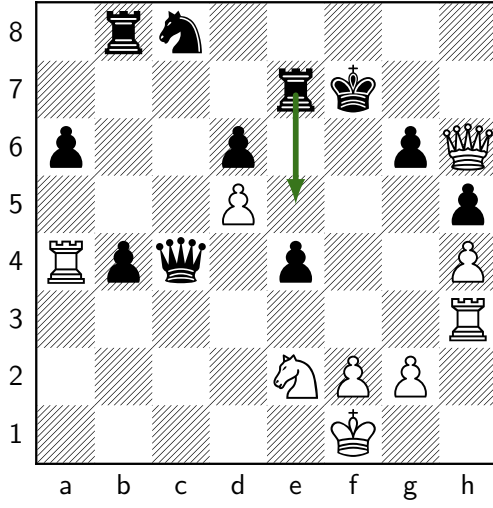


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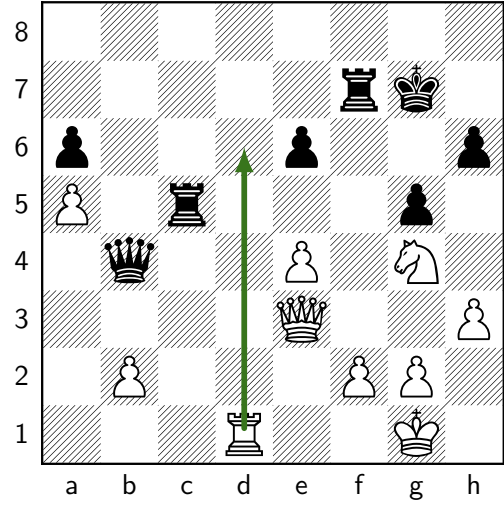


(d)

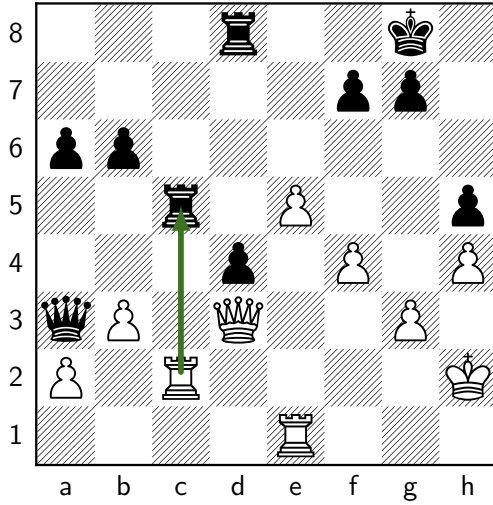
Figure 27: **Move 34, Black: Nc8.** (a) Our position; Melanie plays Nc8 (a7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#3981` (ply 60) [structural distance $d=71$, QAP obj=59]: the master played Rae1; under μ our Na2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pe4 as their Nf3, our ne2 as their pe6; transferred move a7c8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#2263` (ply 66) [structural distance $d=59$, QAP obj=42]: the master played Ra6; under μ our Rb1 plays the role of that game’s Ra2; cross-type roles: our Na7 as their Pf2, our Pa6 as their Nc3; transferred move b8a8. (d) Retrieved from `lichess_elite_2024-09_p0030#857` (ply 68) [structural distance $d=77$, QAP obj=39]: the master played d5; under μ our Pe5 plays the role of that game’s Pd4; cross-type roles: our rh3 as their pf7, our ne2 as their pg7; transferred move e4e3. Synthesis: 1 retrieved analog transferred to Nc8 (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



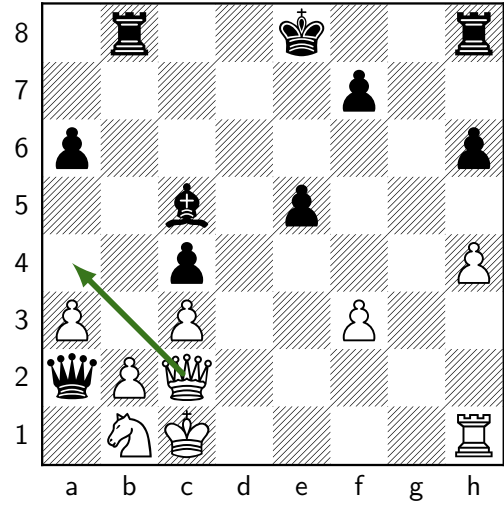
(a)



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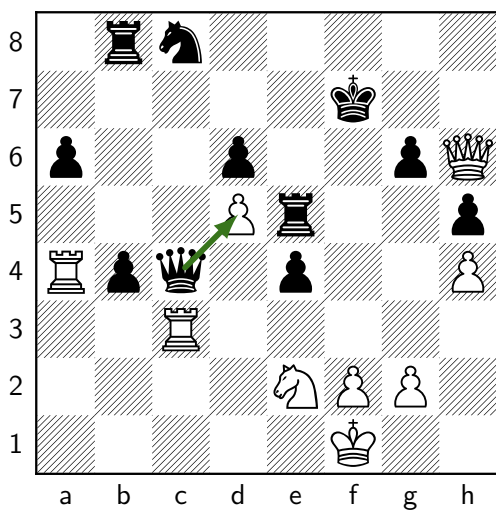


(c)

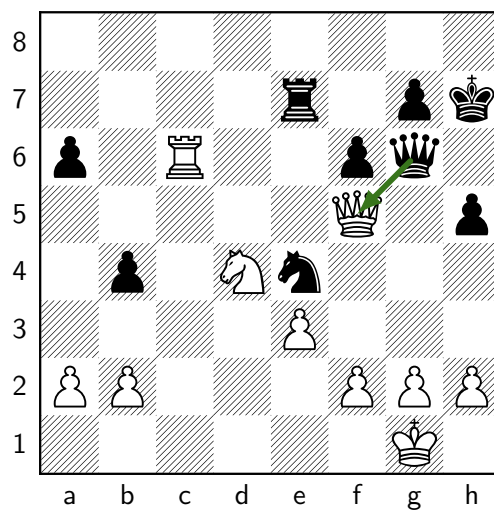


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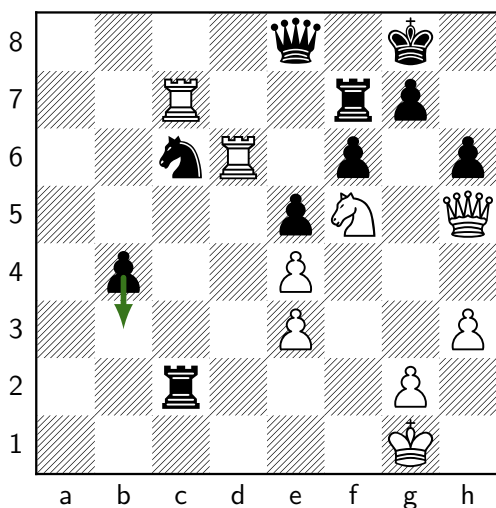
Figure 28: **Move 35, Black: Re5.** (a) Our position; Melanie plays Re5 (e7e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0016#2439` (ply 68) [structural distance $d=85$, QAP obj=24]: the master played Rd6; under μ our Re2 plays the role of that game’s Rd1; cross-type roles: our Nc8 as their Pg2, our ra4 as their qb4; transferred move e7e5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#3741` (ply 54) [structural distance $d=84$, QAP obj=81]: the master played Rxc5; under μ our Pg3 plays the role of that game’s Rc2 (a **cross-type** correspondence); cross-type roles: our Nc8 as their Re1, our ne2 as their Pb3; transferred move g6g5. (d) Retrieved from `lichess_elite_2024-09_p0021#671` (ply 56) [structural distance $d=86$, QAP obj=44]: the master played Qa4+; under μ our Re2 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our pf2 as their Rh1, our ne2 as their Pb2; transferred move e7d7. Synthesis: 1 retrieved analog transferred to Re5 (summed QAP objective 24, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



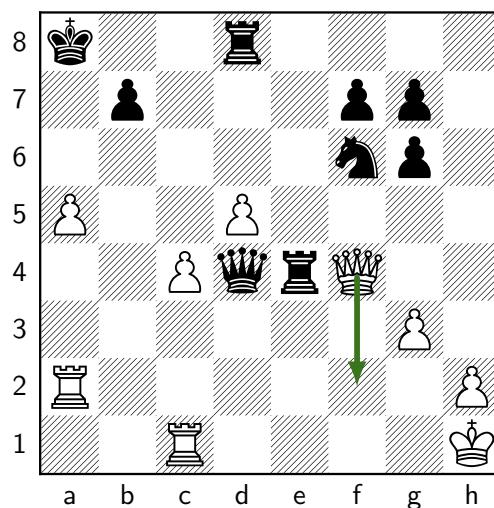
(a)



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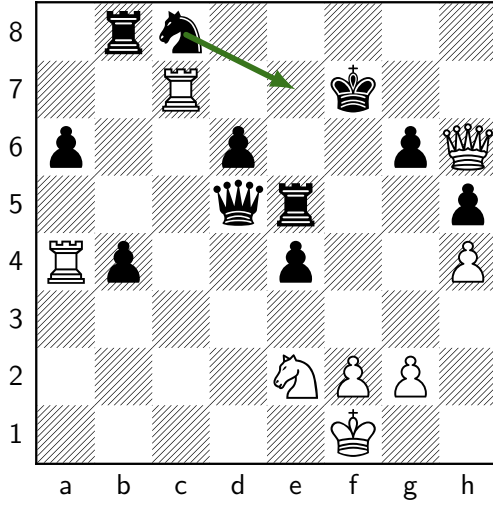


(c)

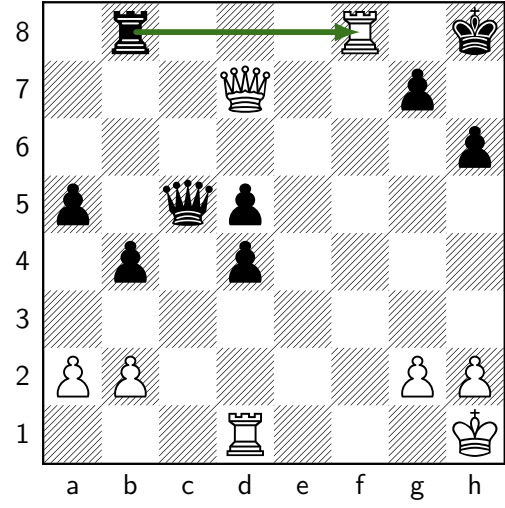


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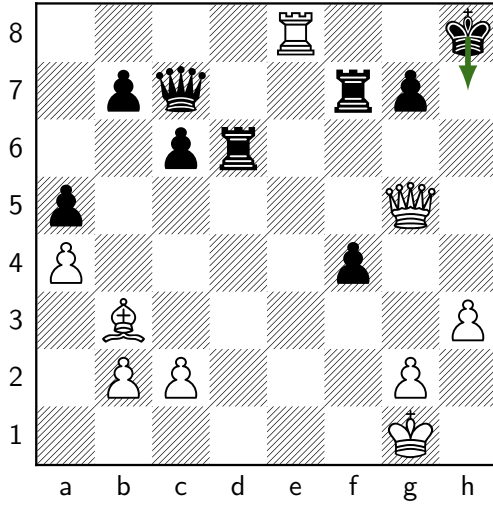
Figure 29: **Move 36, Black: Qxd5.** (a) Our position; Melanie plays Qxd5 (c4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0017#1385` (ply 63) [structural distance $d=97$, QAP obj=42]: the master played Qxf5; under μ our Qc5 plays the role of that game’s Qg3; cross-type roles: our Rb8 as their Pa6, our rc3 as their qf5; transferred move c4c3. (c) Retrieved from `twic1500#270` (ply 61) [structural distance $d=89$, QAP obj=40]: the master played b3; under μ our Rb1 plays the role of that game’s Pb5 (a **cross-type** correspondence); cross-type roles: our Qc4 as their rc7, our ra4 as their Pg7; transferred move b8b7. (d) Retrieved from `lichess_elite_2024-09_p0003#3510` (ply 62) [structural distance $d=112$, QAP obj=39]: the master played Qf2; under μ our Qc5 plays the role of that game’s Qf4; cross-type roles: our ra4 as their pg6, our ph4 as their rd8; transferred move c4c7. Synthesis: no retrieved analog transferred to Qxd5 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



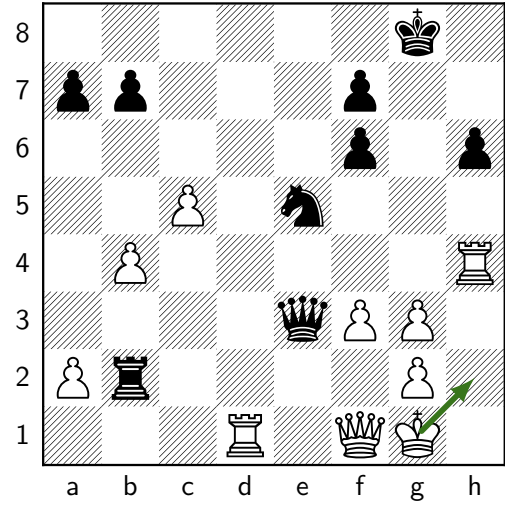
(a)



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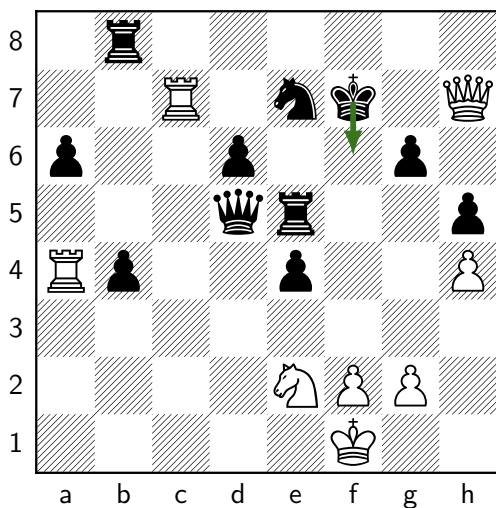


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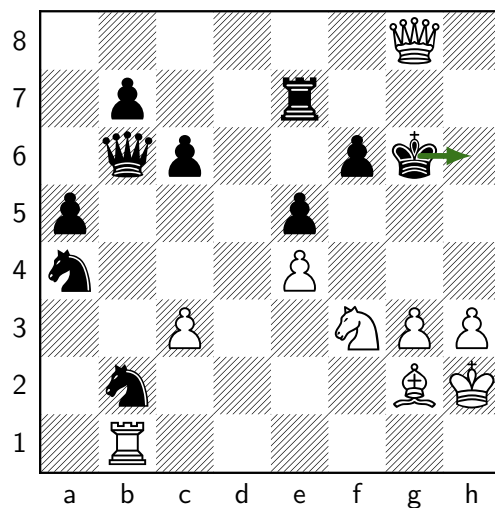


(d)

Figure 30: **Move 37, Black: Ne7**. (a) Our position; Melanie plays Ne7 (c8e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#3565` (ply 61) [structural distance $d=83$, QAP obj=47]: the master played Rxf8; under μ our Nc1 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our Re5 as their Qc5, our ra4 as their Pd4; transferred move c8e7 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0017#3648` (ply 57) [structural distance $d=95$, QAP obj=68]: the master played Kh7; under μ our Kf2 plays the role of that game’s Kh1; cross-type roles: our Re5 as their Qc7, our Nc8 as their Rf7; transferred move f7g8. (d) Retrieved from `lichess_elite_2024-09_p0005#2900` (ply 56) [structural distance $d=95$, QAP obj=49]: the master played Kh2; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our qh6 as their rb2, our rc7 as their qe3; transferred move f7g8. Synthesis: 1 retrieved analog transferred to Ne7 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

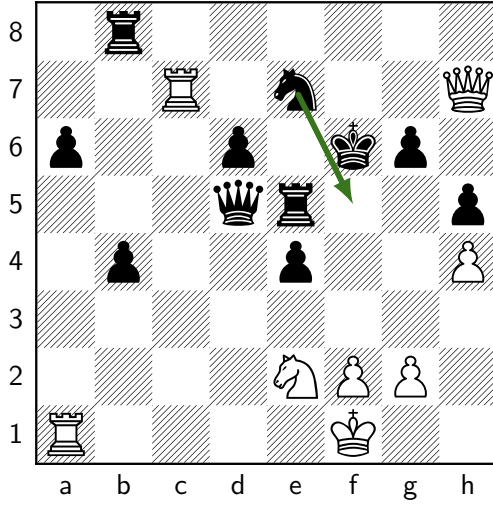


(a)

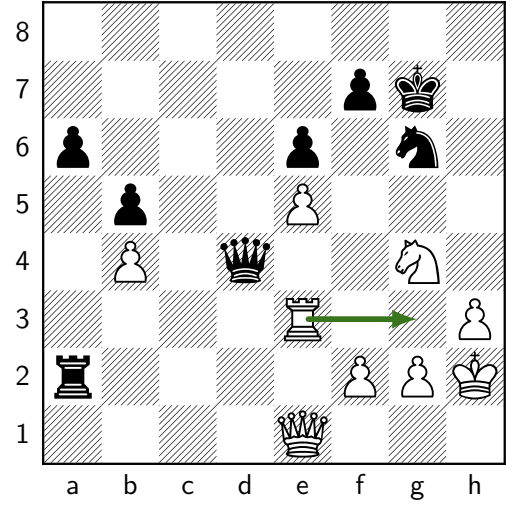


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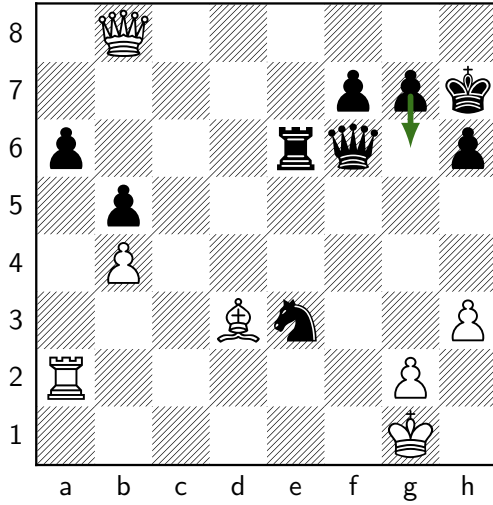
Figure 31: **Move 38, Black: Kf6.** (a) Our position; Melanie plays Kf6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#146` (ply 65) [structural distance $d=104$, QAP obj=79]: the master played Kh6; under μ our *Kf2* plays the role of that game’s *Kg3*; cross-type roles: our *ne2* as their *Pb7*, our *Pa6* as their *Re7*; transferred move f7f8. Synthesis: no retrieved analog transferred to Kf6 – the `search` selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



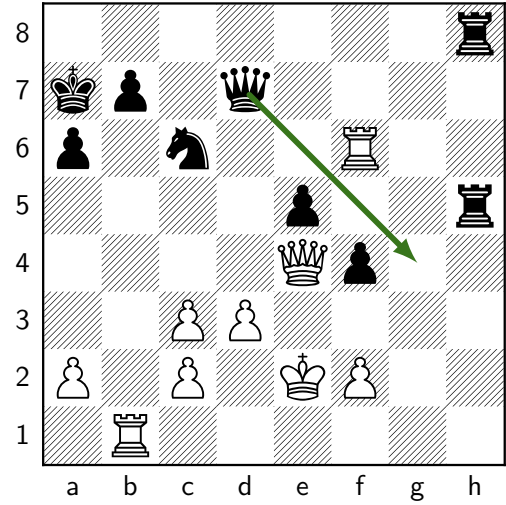
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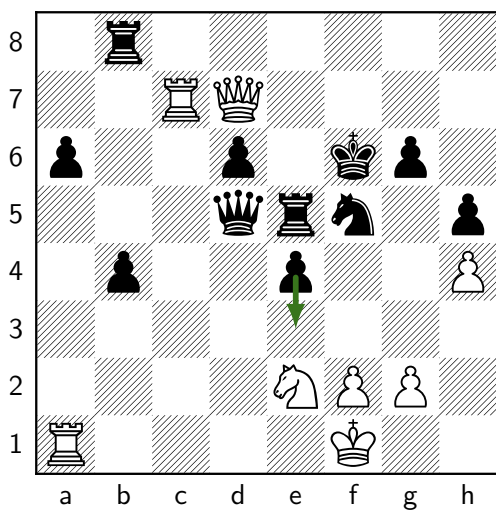


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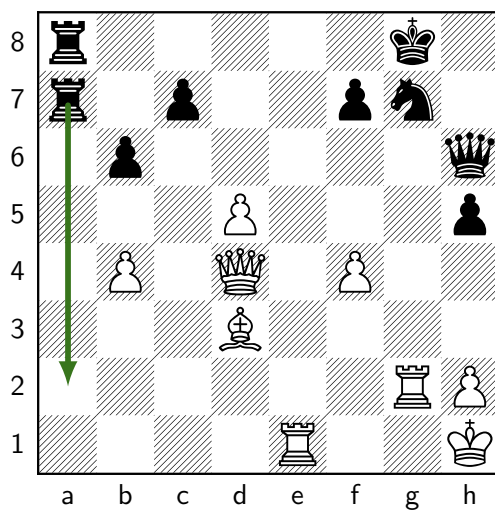


(d)

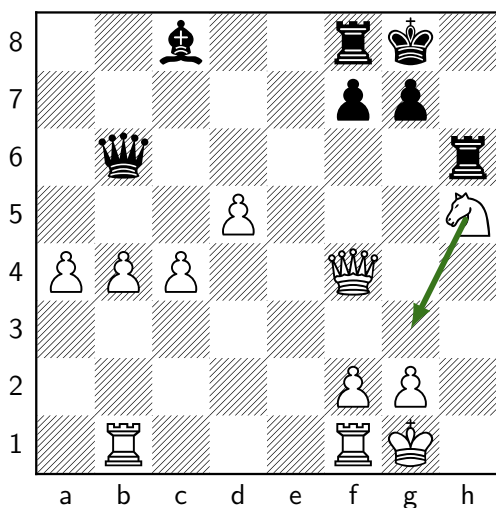
Figure 32: **Move 39, Black: Nf5.** (a) Our position; Melanie plays Nf5 (e7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0018#3016 (ply 68) [structural distance $d=77$, QAP obj=11]: the master played Rg3; under μ our Ne2 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our Re5 as their Qe1, our qh7 as their ra2; transferred move e7f5 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0025#3899 (ply 71) [structural distance $d=97$, QAP obj=5]: the master played g6; under μ our Ne2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Pg6 as their qb8, our qh7 as their Ph6; transferred move e7f5 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0024#1684 (ply 75) [structural distance $d=93$, QAP obj=60]: the master played Qg4+; under μ our Re4 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Pa6 as their Rh5, our Ne7 as their Pf4; transferred move e5g5. Synthesis: 2 retrieved analogs transferred to Nf5 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



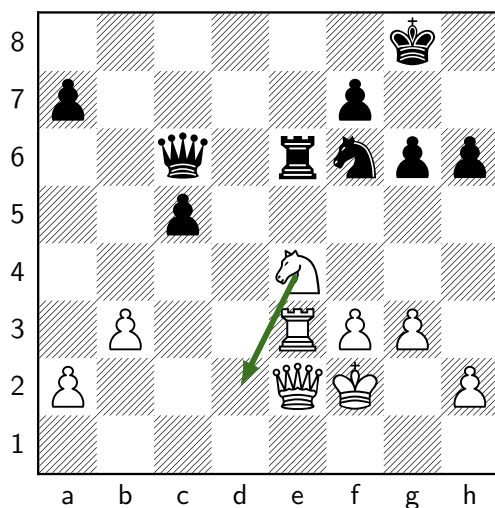
(a)



(b)

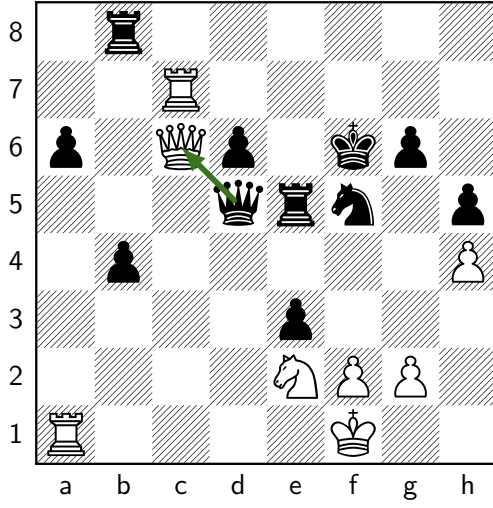


(c)

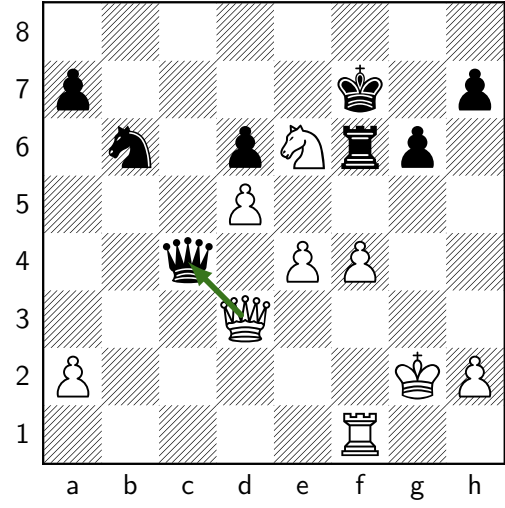


(d)

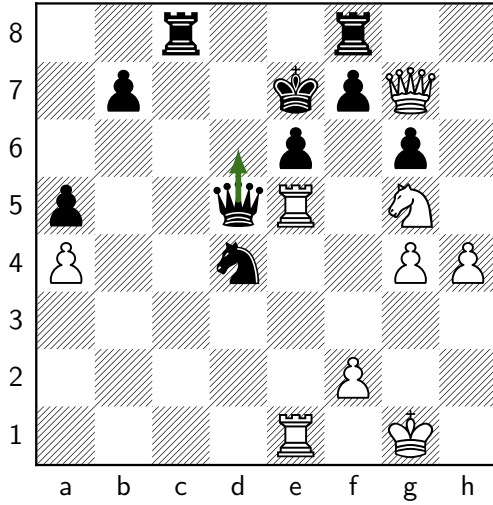
Figure 33: **Move 40, Black: e3 (e4e3).** (a) Our position; Melanie plays e3 (e4e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0021#1492` (ply 73) [structural distance $d=86$, QAP obj=52]: the master played Ra2; under μ our *Pe5* plays the role of that game’s *Ra2* (a **cross-type** correspondence); cross-type roles: our *Qd5* as their *Pc7*, our *Re5* as their *Pf7*; transferred move e4e3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0019#336` (ply 70) [structural distance $d=80$, QAP obj=34]: the master played Ng3; under μ our *Pe5* plays the role of that game’s *Nh5* (a **cross-type** correspondence); cross-type roles: our *Pg6* as their *Rf1*, our *rc7* as their *Pc4*; transferred move e4e3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#1498` (ply 68) [structural distance $d=82$, QAP obj=61]: the master played Nd2; under μ our *Nf4* plays the role of that game’s *Ne4*; cross-type roles: our *Rb8* as their *Pa2*, our *Pa6* as their *Qe2*; transferred move f5h6. Synthesis: 2 retrieved analogs transferred to e3 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



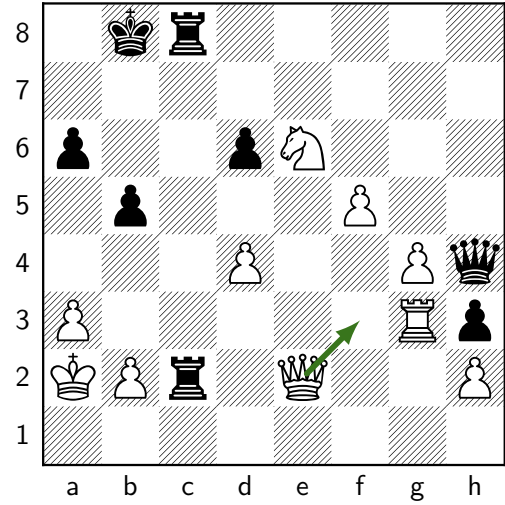
(a)



(b)

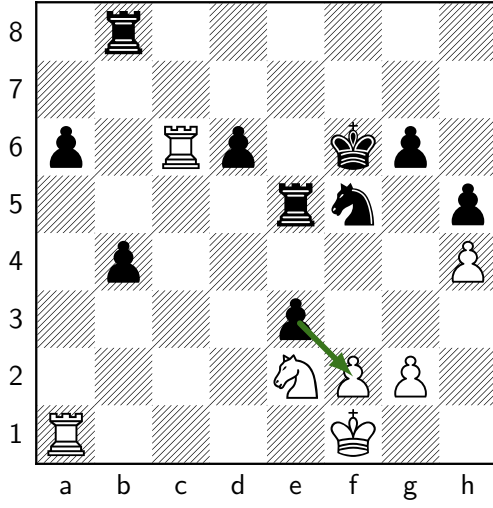


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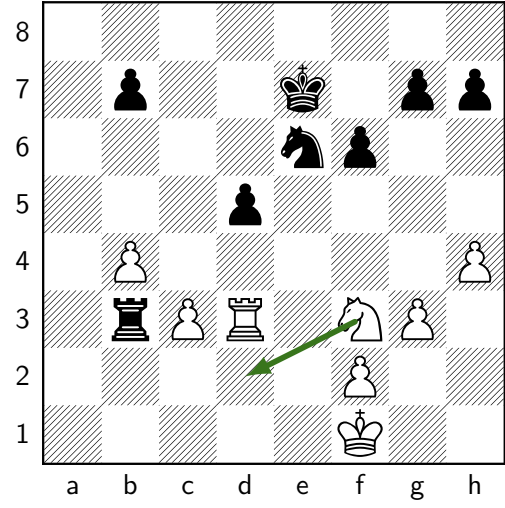


(d)

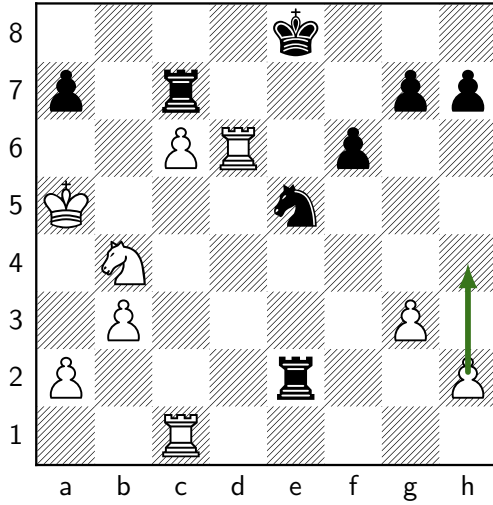
Figure 34: **Move 41, Black: Qxc6.** (a) Our position; Melanie plays Qxc6 (d5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#3074` (ply 64) [structural distance $d=96$, QAP obj=108]: the master played Qxc4; under μ our Qd4 plays the role of that game’s Qd3; cross-type roles: our Rb8 as their Ph2, our Qd5 as their Pe4; transferred move d5c6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2860` (ply 67) [structural distance $d=105$, QAP obj=142]: the master played Qd6; under μ our Qd4 plays the role of that game’s Qd4; cross-type roles: our Nf5 as their Rc8, our Qd5 as their Rf8; transferred move d5f7. (d) Retrieved from `lichess_elite_2024-09_p0018#1143` (ply 70) [structural distance $d=90$, QAP obj=57]: the master played Qf3; under μ our Qd4 plays the role of that game’s Qe2; cross-type roles: our qc6 as their rc2, our Re5 as their Ph2; transferred move d5c5. Synthesis: 1 retrieved analog transferred to Qxc6 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



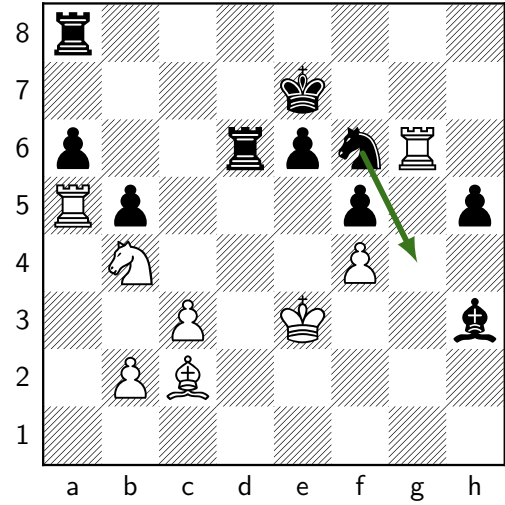
(a)



(b)

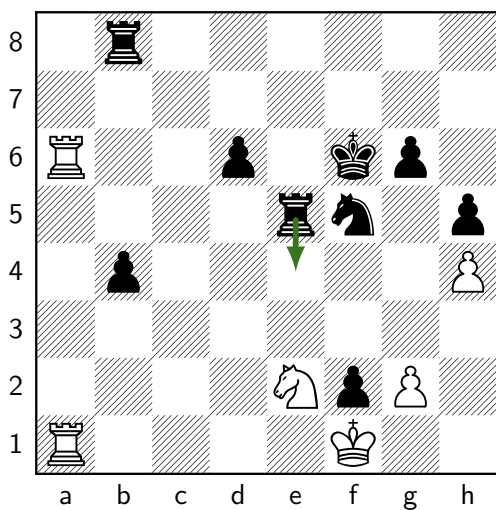


(c)

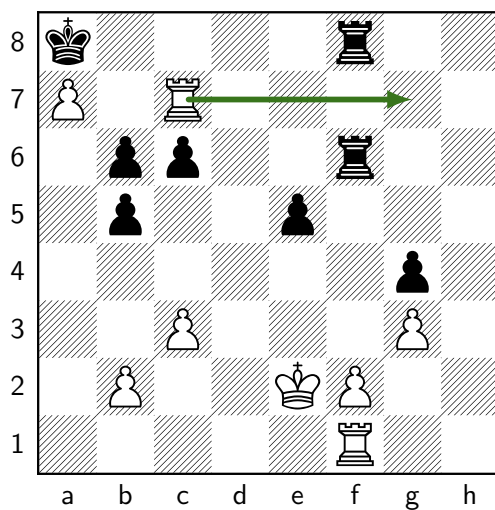


(d)

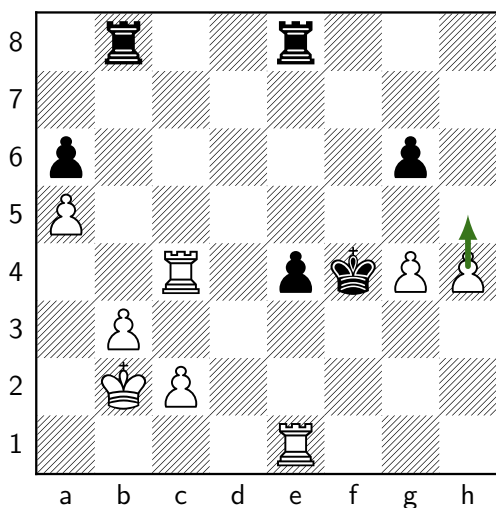
Figure 35: **Move 42, Black: exf2**. (a) Our position; Melanie plays **exf2** (**e3f2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#3571` (ply 74) [structural distance $d=70$, QAP obj=71]: the master played **Nd2**; under μ our *Pe6* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Nf5* as their *Rd3*, our *ra1* as their *Ph4*; transferred move **e3f2** (**endorses this move**). (c) Retrieved from `lichess_elite_2025-11_p0000#1560` (ply 80) [structural distance $d=68$, QAP obj=59]: the master played **h4**; under μ our *Pe6* plays the role of that game’s *Ph2*; cross-type roles: our *ne2* as their *Rc1*, our *pf2* as their *re2*; transferred move **e3f2** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0017#3712` (ply 75) [structural distance $d=72$, QAP obj=42]: the master played **Ng4+**; under μ our *Pe6* plays the role of that game’s *Nf3* (a **cross-type** correspondence); cross-type roles: our *Nf5* as their *Ra8*, our *Rb8* as their *Pe6*; transferred move **e3f2** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **exf2** (summed QAP objective 219, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



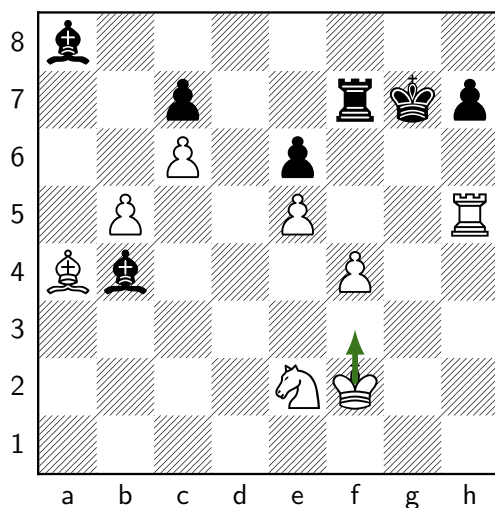
(a)



(b)

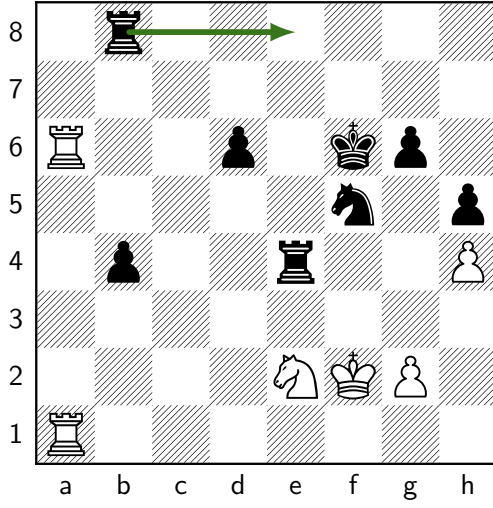


(c)

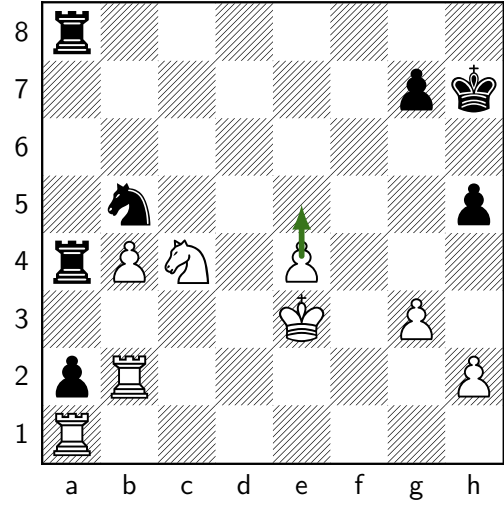


(d)

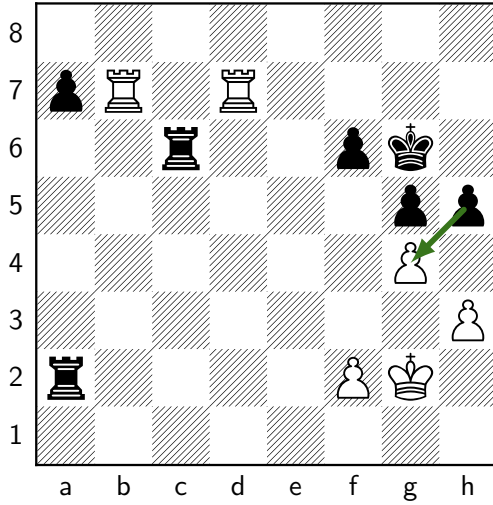
Figure 36: **Move 43, Black: Re4**. (a) Our position; Melanie plays Re4 (e5e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#3283` (ply 76) [structural distance $d=61$, QAP obj=65]: the master played Rg7; under μ our Re4 plays the role of that game’s Rc7; cross-type roles: our Nf5 as their Rf1, our Rb8 as their pe5; transferred move e5e4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0028#11` (ply 78) [structural distance $d=54$, QAP obj=58]: the master played h5; under μ our Rb1 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pc2, our ne2 as their pg6; transferred move b8h8. (d) Retrieved from `lichess_elite_2024-09_p0020#121` (ply 70) [structural distance $d=56$, QAP obj=46]: the master played Kf3; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our Pg6 as their Ne2, our Rb8 as their Ba4; transferred move f6f7. Synthesis: 1 retrieved analog transferred to Re4 (summed QAP objective 65, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



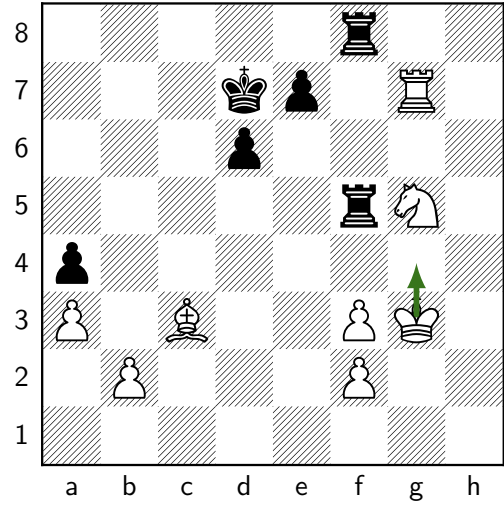
(a)



(b)

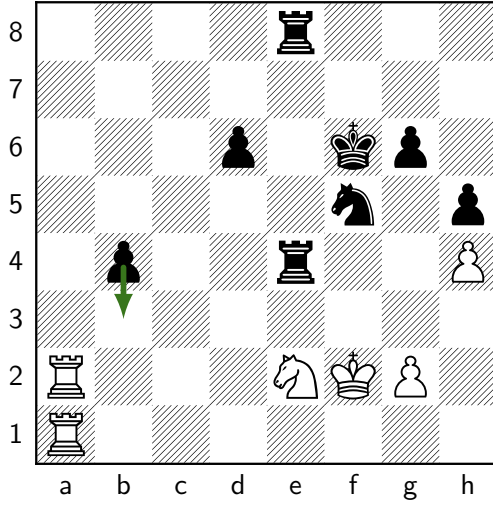


(c)

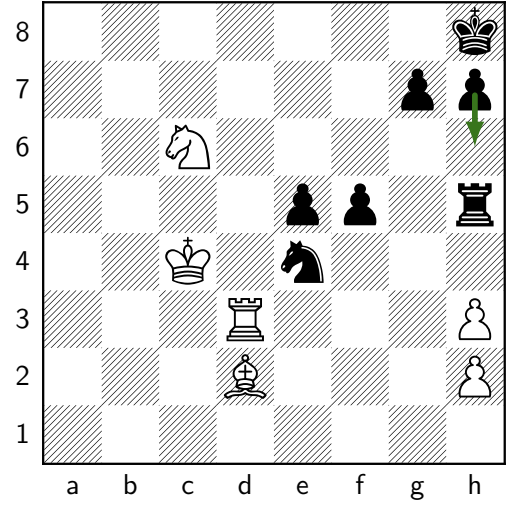


(d)

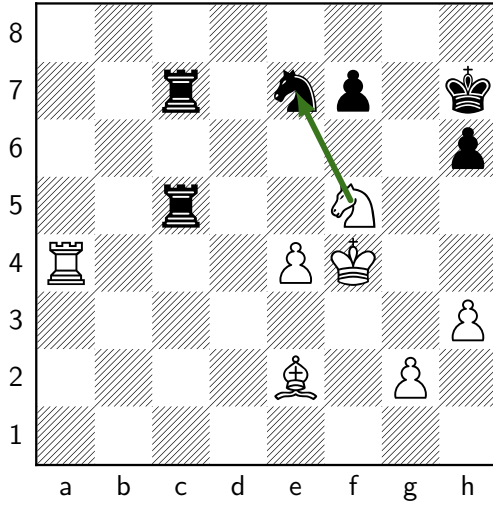
Figure 37: **Move 44, Black: Rbe8.** (a) Our position; Melanie plays Rbe8 (b8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3890` (ply 94) [structural distance $d=34$, QAP obj=41]: the master played e5; under μ our Rb1 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our ne2 as their pa2, our Nf5 as their Rb2; transferred move b8b7. (c) Retrieved from `lichess_elite_2024-09_p0021#683` (ply 71) [structural distance $d=43$, QAP obj=34]: the master played hxg4; under μ our Nf4 plays the role of that game’s Ph4 (a **cross-type** correspondence); cross-type roles: our ne2 as their Pa7, our Pb4 as their rd7; transferred move f5h6. (d) Retrieved from `lichess_elite_2024-09_p0024#3619` (ply 76) [structural distance $d=46$, QAP obj=31]: the master played Kg4; under μ our Kf3 plays the role of that game’s Kg3; cross-type roles: our Nf5 as their Pf2, our ra1 as their Bc3; transferred move f6e7. Synthesis: no retrieved analog transferred to Rbe8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



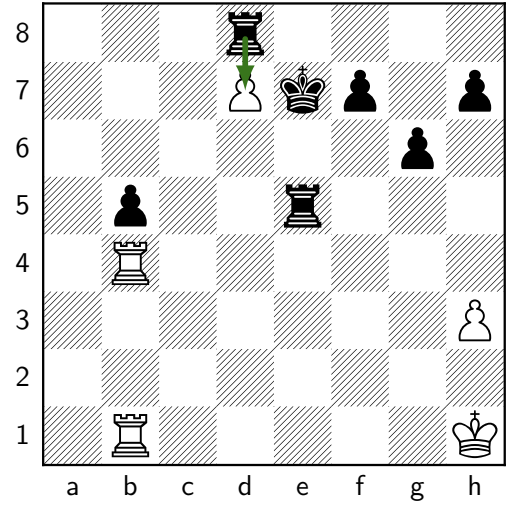
(a)



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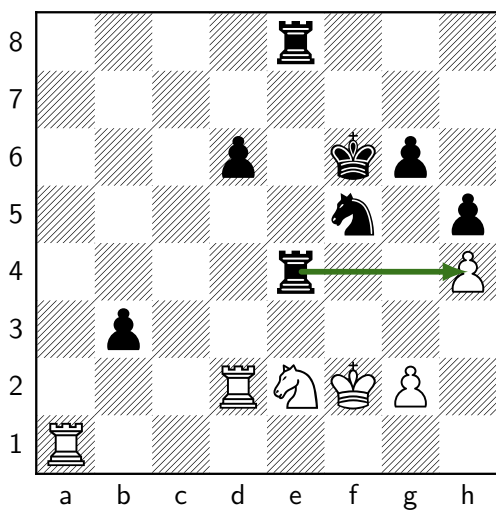


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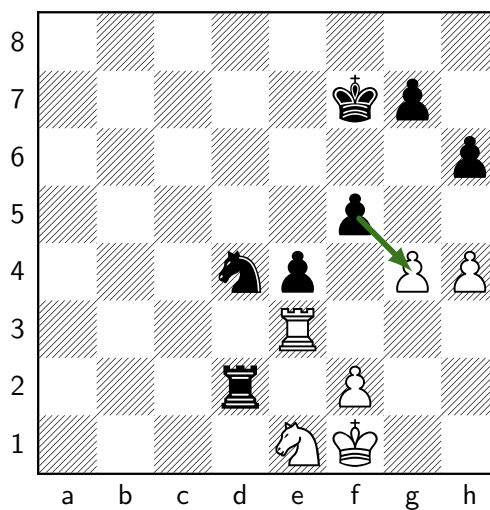


(d)

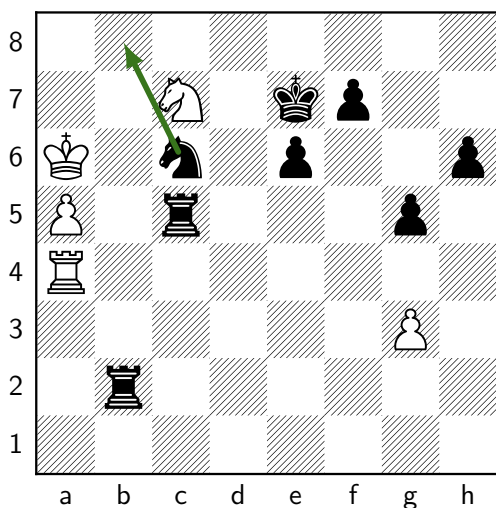
Figure 38: **Move 45, Black: b3**. (a) Our position; Melanie plays b3 (b4b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0028#3612` (ply 73) [structural distance $d=49$, QAP obj=30]: the master played h6; under μ our *Pb5* plays the role of that game’s *Ph2*; cross-type roles: our *pg2* as their *nc6*, our *Re8* as their *Pe5*; transferred move b4b3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0012#3261` (ply 82) [structural distance $d=50$, QAP obj=31]: the master played Nxe7; under μ our *Re5* plays the role of that game’s *Nf5* (a **cross-type** correspondence); cross-type roles: our *Pg6* as their *Be2*, our *Nf5* as their *Ph3*; transferred move e4c4. (d) Retrieved from `lichess_elite_2024-09_p0028#3842` (ply 75) [structural distance $d=48$, QAP obj=21]: the master played Rxd7; under μ our *Nf4* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *ne2* as their *Pb5*; transferred move f5g7. Synthesis: 1 retrieved analog transferred to b3 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



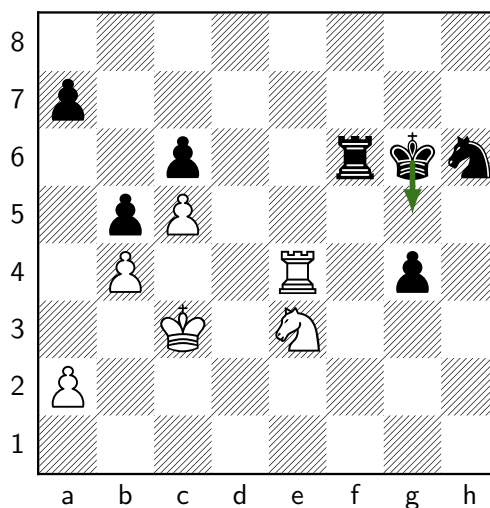
(a)



(b)

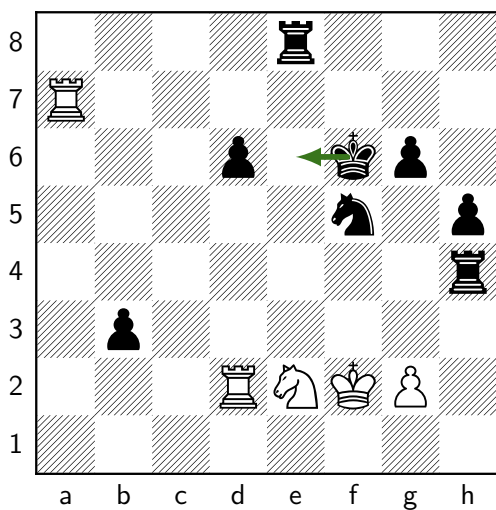


(c)

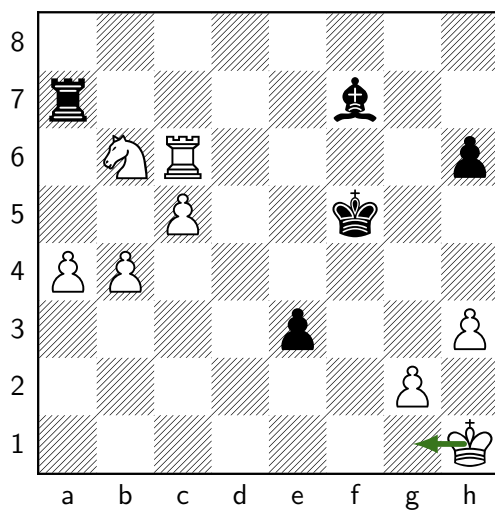


(d)

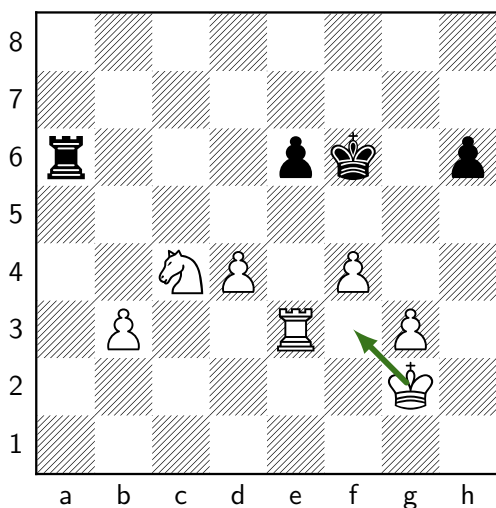
Figure 39: **Move 46, Black: Rxh4.** (a) Our position; Melanie plays Rxh4 (e4h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `twic1500#314` (ply 97) [structural distance $d=34$, QAP obj=25]: the master played fxf4; under μ our Re5 plays the role of that game’s Pf4 (a **cross-type** correspondence); cross-type roles: our Re8 as their Nd4, our ne2 as their Pe4; transferred move e4h4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#113` (ply 87) [structural distance $d=44$, QAP obj=28]: the master played Nb8+; under μ our Nf4 plays the role of that game’s Nc3; cross-type roles: our Re4 as their nc7, our ne2 as their Pe6; transferred move f5h6. (d) Retrieved from `lichess_elite_2024-09_p0005#1414` (ply 105) [structural distance $d=43$, QAP obj=25]: the master played Kg5; under μ our Kf3 plays the role of that game’s Kg3; cross-type roles: our ne2 as their Rf6, our rd2 as their Nh6; transferred move f6f7. Synthesis: 1 retrieved analog transferred to Rxh4 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



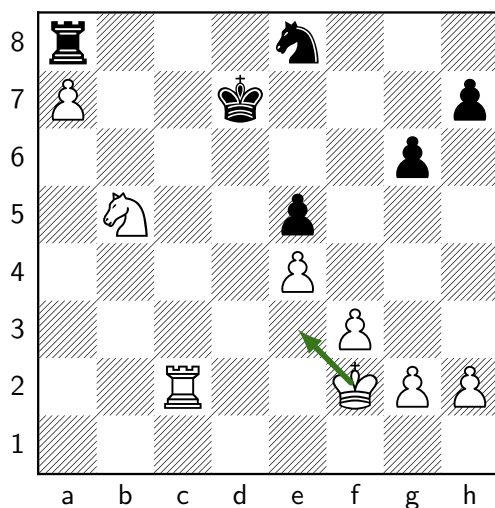
(a)



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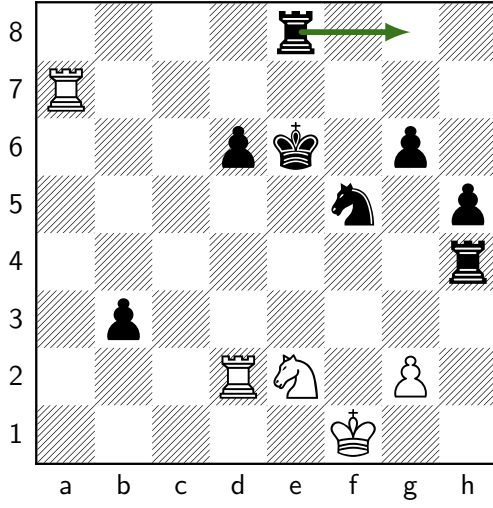


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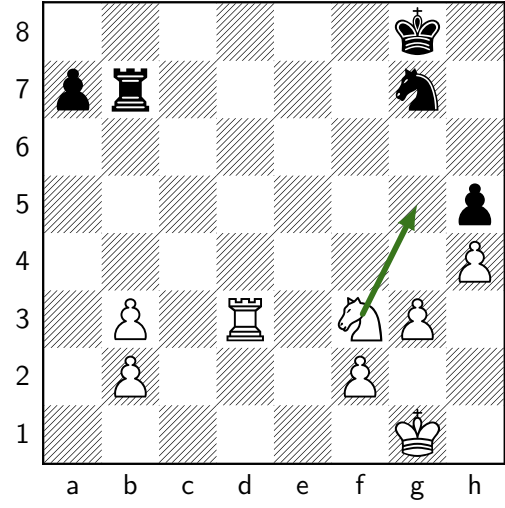


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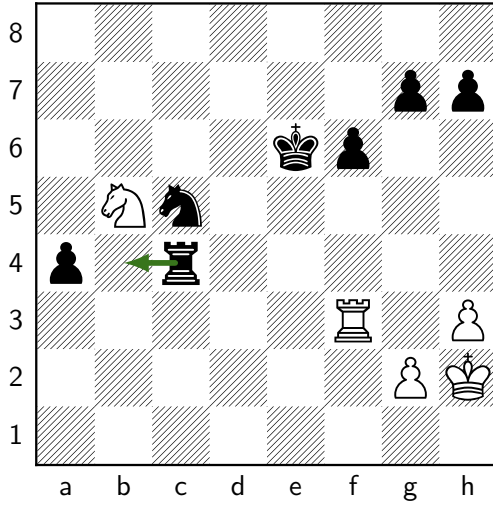
Figure 40: **Move 47, Black: Ke6 (f6e6)**. (a) Our position; Melanie plays Ke6 (f6e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#2540` (ply 84) [structural distance $d=40$, QAP obj=21]: the master played Kg1; under μ our Kf3 plays the role of that game’s Kh1; cross-type roles: our ra7 as their Ph3, our ne2 as their Pa4; transferred move f6e6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#3491` (ply 104) [structural distance $d=45$, QAP obj=16]: the master played Kf3; under μ our Kf3 plays the role of that game’s Kg2; cross-type roles: our Rh4 as their Nc4, our ra7 as their Pd4; transferred move f6e6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#3715` (ply 88) [structural distance $d=43$, QAP obj=11]: the master played Ke3; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our Nf5 as their Rc2, our ra7 as their Ph2; transferred move f6e6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ke6 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



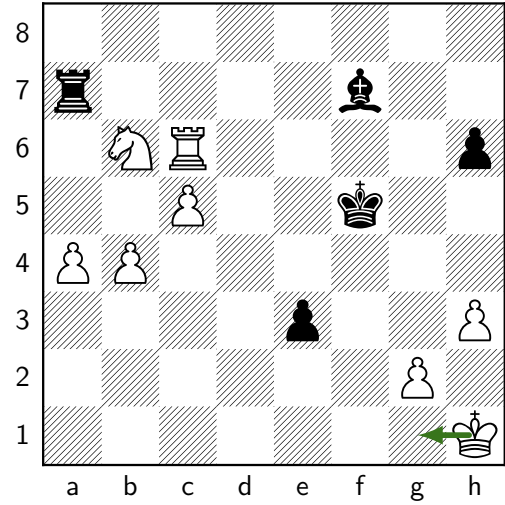
(a)



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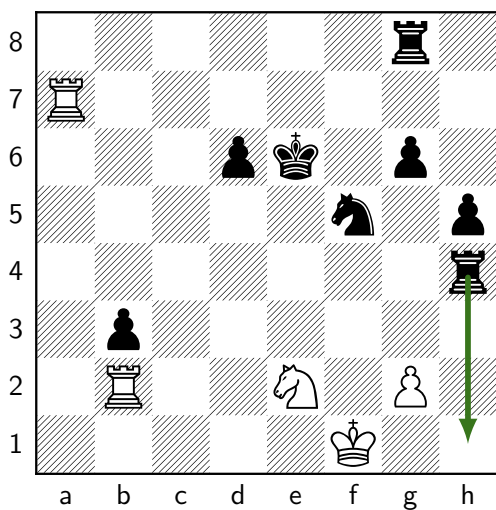


(c)

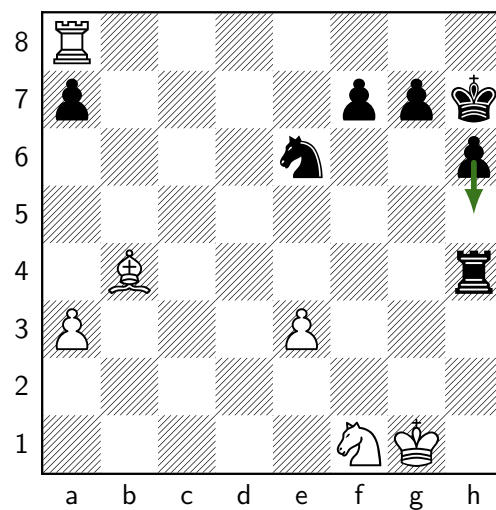


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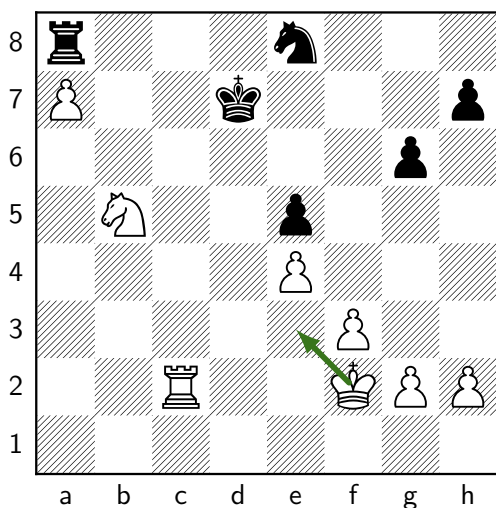
Figure 41: **Move 48, Black: Rg8**. (a) Our position; Melanie plays Rg8 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#2212` (ply 78) [structural distance $d=42$, QAP obj=23]: the master played Ng5; under μ our Re1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Nf5 as their Rd3, our Rh4 as their Ph4; transferred move e8g8 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0022#3710` (ply 79) [structural distance $d=44$, QAP obj=20]: the master played Rb4; under μ our Rh5 plays the role of that game’s Rc5; cross-type roles: our Nf5 as their Pg7, our ra7 as their Ph7; transferred move h4b4. (d) Retrieved from `lichess_elite_2024-09_p0025#2540` (ply 84) [structural distance $d=47$, QAP obj=18]: the master played Kg1; under μ our Ke3 plays the role of that game’s Kh1; cross-type roles: our ra7 as their Pg2, our Re8 as their pe3; transferred move e6f6. Synthesis: 1 retrieved analog transferred to Rg8 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



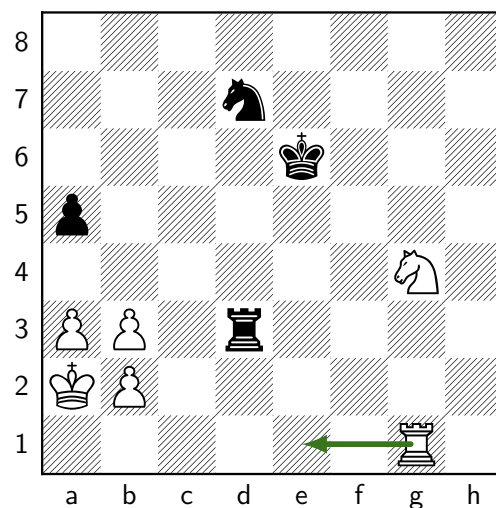
(a)



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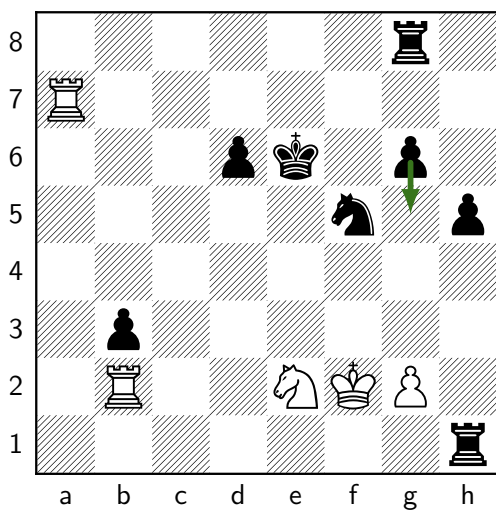


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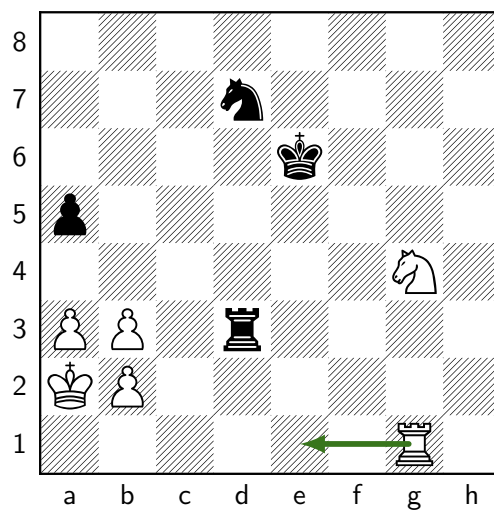


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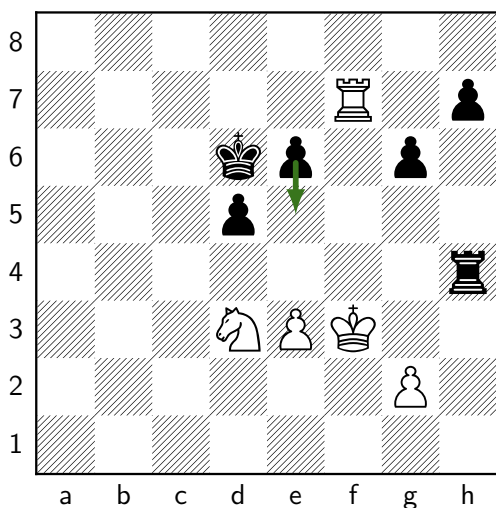
Figure 42: **Move 49, Black: Rh1+.** (a) Our position; Melanie plays Rh1+ (h4h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#747` (ply 91) [structural distance $d=45$, QAP obj=20]: the master played h5; under μ our *Pd3* plays the role of that game’s *Ph3*; cross-type roles: our *ra7* as their *Pf7*, our *Nf5* as their *Pg7*; transferred move d6d5. (c) Retrieved from `lichess_elite_2024-09_p0030#3715` (ply 88) [structural distance $d=38$, QAP obj=20]: the master played Ke3; under μ our *Ke3* plays the role of that game’s *Kf2*; cross-type roles: our *Nf5* as their *Pg2*, our *ra7* as their *Ph2*; transferred move e6d5. (d) Retrieved from `lichess_elite_2024-09_p0024#2834` (ply 82) [structural distance $d=44$, QAP obj=13]: the master played Re1+; under μ our *Rg1* plays the role of that game’s *Rg1*; cross-type roles: our *Nf5* as their *Pa3*, our *Rh4* as their *Ng4*; transferred move g8c8. Synthesis: no retrieved analog transferred to Rh1+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



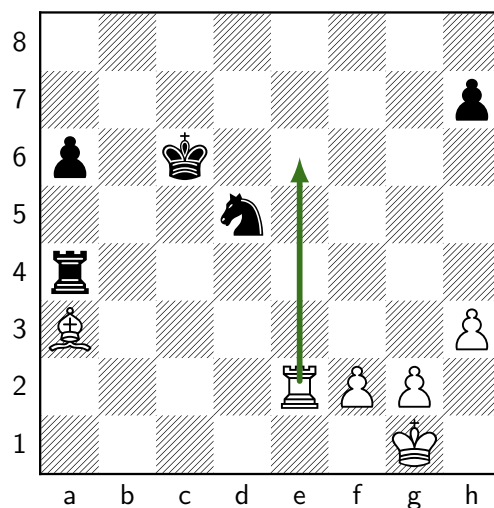
(a)



(b)



(c)



(d)

Figure 43: **Move 50, Black: g5** (g6g5). (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0024#2834` (ply 82) [structural distance $d=35$, QAP obj=15]: the master played **Re1+**; under μ our *Pg3* plays the role of that game’s *Rg1* (a **cross-type** correspondence); cross-type roles: our *Nf5* as their *Pb2*, our *Rh1* as their *Ng4*; transferred move **g6g5 (endorses this move)**. (c) Retrieved from `lichess_elite_2024-09_p0028#2198` (ply 107) [structural distance $d=43$, QAP obj=11]: the master played **e5**; under μ our *Pg3* plays the role of that game’s *Pe3*; cross-type roles: our *ra7* as their *Pg6*, our *Nf5* as their *Pd5*; transferred move **g6g5 (endorses this move)**. (d) Retrieved from `lichess_elite_2025-11_p0000#3879` (ply 98) [structural distance $d=37$, QAP obj=10]: the master played **Re6+**; under μ our *Pg3* plays the role of that game’s *Re2* (a **cross-type** correspondence); cross-type roles: our *Nf5* as their *Pf2*, our *Pb3* as their *Ba3*; transferred move **g6g5 (endorses this move)**. Synthesis: 3 retrieved analogs transferred to g5 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

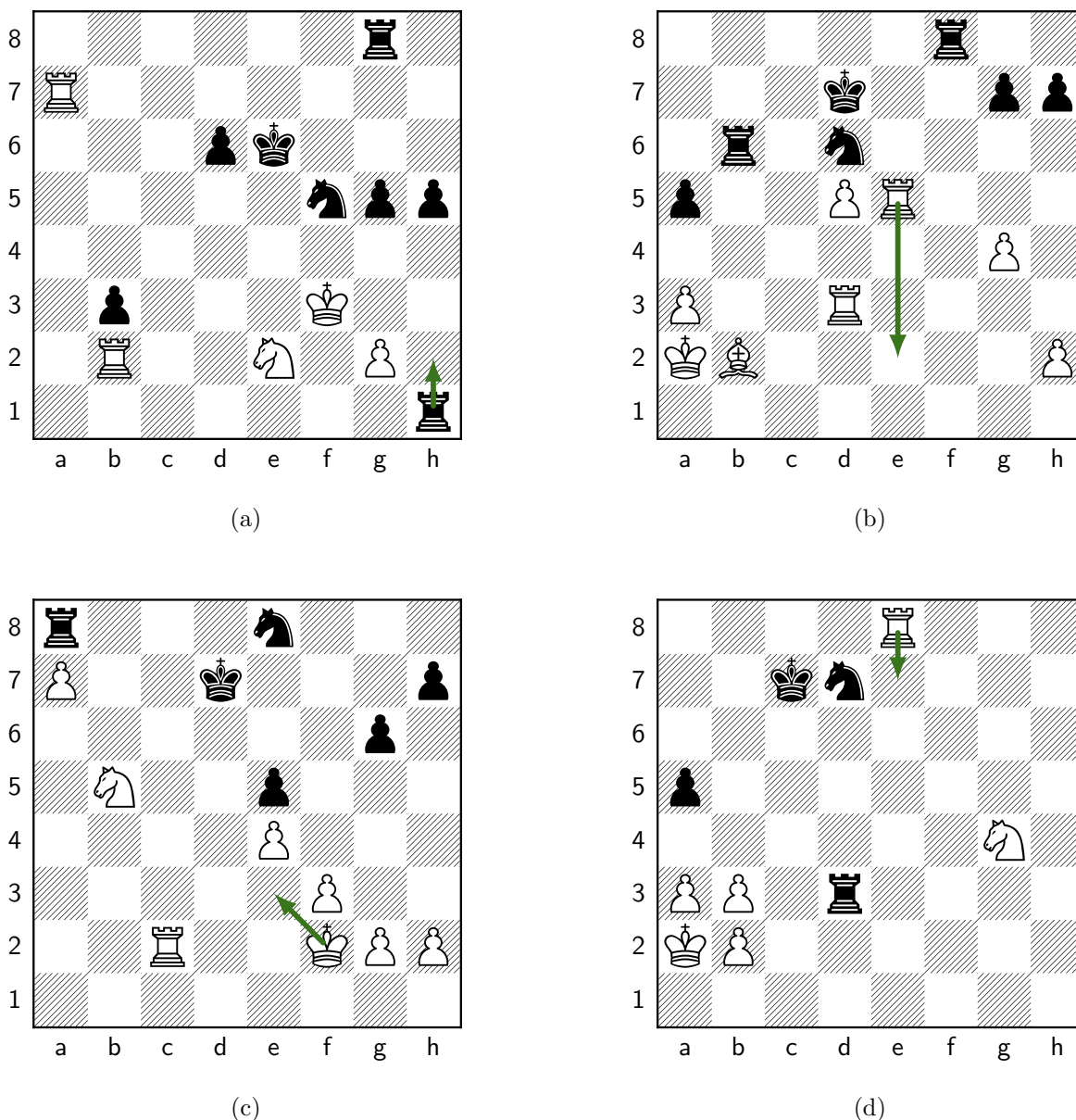


Figure 44: **Move 51, Black: Rh2.** (a) Our position; Melanie plays Rh2 (h1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0018#2850 (ply 84) [structural distance $d=43$, QAP obj=17]: the master played Re2; under μ our Rh8 plays the role of that game's Re5; cross-type roles: our Pb3 as their Bb2, our ra7 as their Ph2; transferred move h1h2 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0030#3715 (ply 88) [structural distance $d=33$, QAP obj=21]: the master played Ke3; under μ our Ke3 plays the role of that game's Kf2; cross-type roles: our Nf5 as their Rc2, our ra7 as their Ph2; transferred move e6d5. (d) Retrieved from lichess_elite_2024-09_p0024#2834 (ply 86) [structural distance $d=43$, QAP obj=13]: the master played Re7; under μ our Pg4 plays the role of that game's Re8 (a **cross-type** correspondence); cross-type roles: our Nf5 as their Pb2, our Rh1 as their Ng4; transferred move g5g4. Synthesis: 1 retrieved analog transferred to Rh2 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

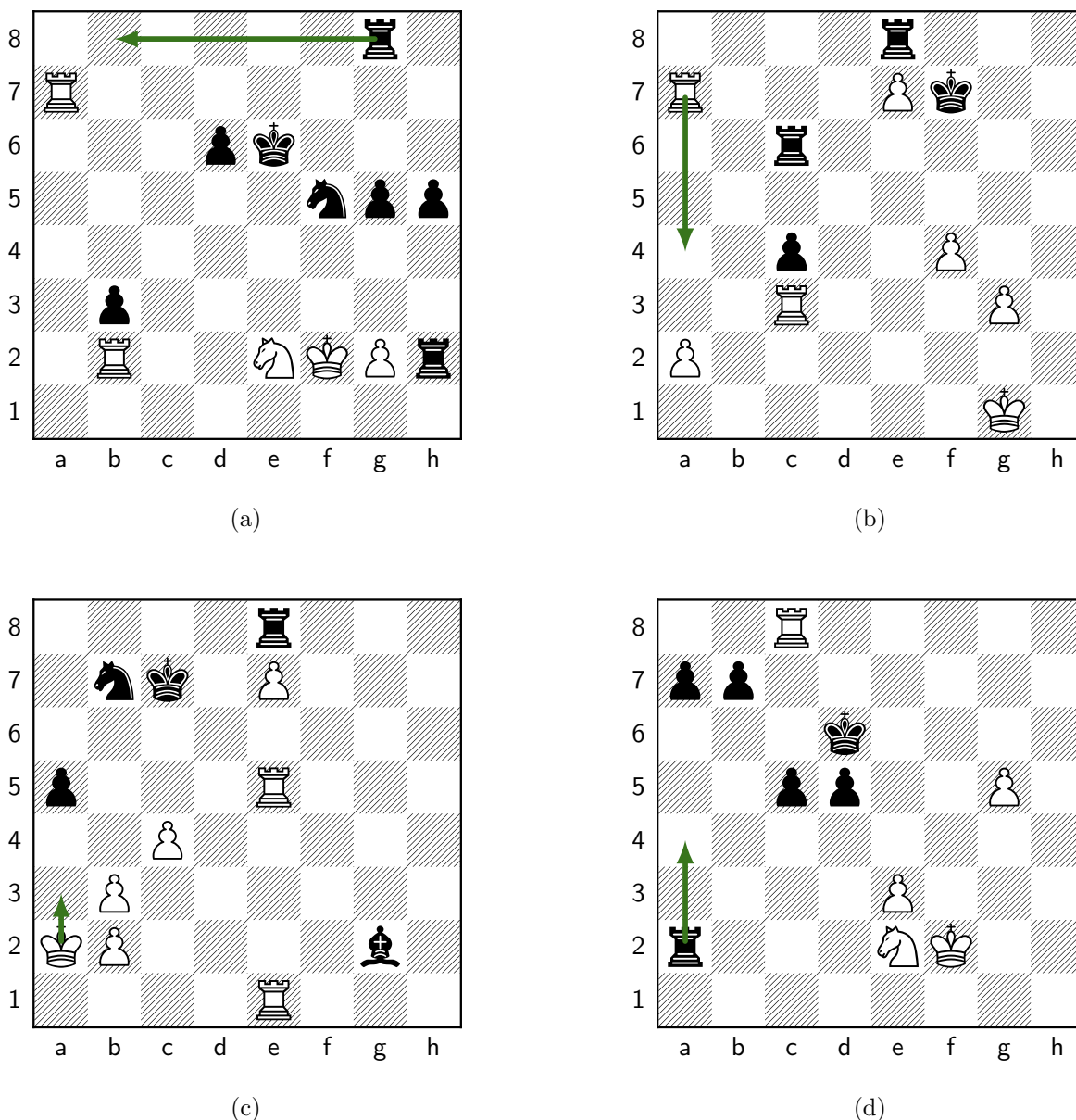
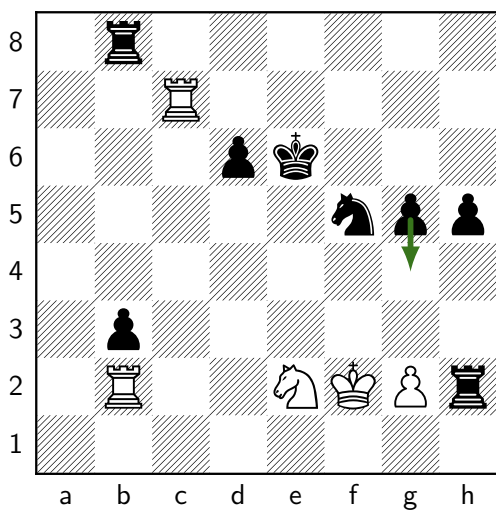
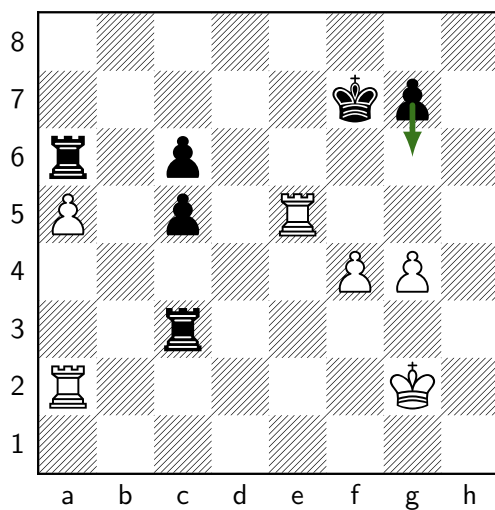


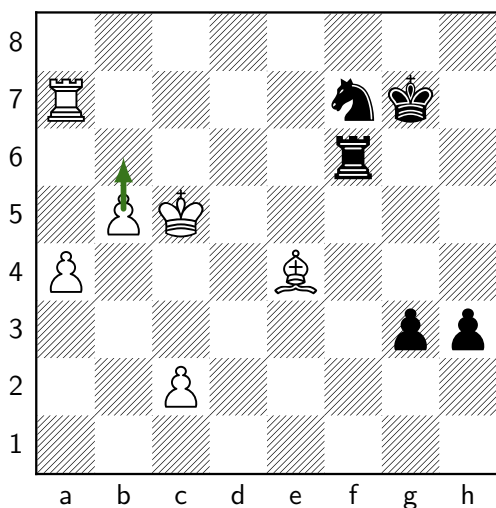
Figure 45: **Move 52, Black: Rb8.** (a) Our position; Melanie plays Rb8 (g8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2025-11_p0000#3680 (ply 96) [structural distance $d=44$, QAP obj=58]: the master played Ra4; under μ our Rg1 plays the role of that game's Ra7; cross-type roles: our Nf5 as their Pa2, our ra7 as their Pf4; transferred move g8g6. (c) Retrieved from lichess_elite_2024-09_p0019#2836 (ply 106) [structural distance $d=45$, QAP obj=46]: the master played Ka3; under μ our Ke3 plays the role of that game's Ka2; cross-type roles: our ne2 as their Re1, our Pg5 as their bg2; transferred move e6e5. (d) Retrieved from lichess_elite_2025-11_p0000#1967 (ply 89) [structural distance $d=41$, QAP obj=39]: the master played Ra4; under μ our Rh7 plays the role of that game's Ra7; cross-type roles: our Rg8 as their Pb7, our Nf5 as their Pd5; transferred move h2h1. Synthesis: no retrieved analog transferred to Rb8 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



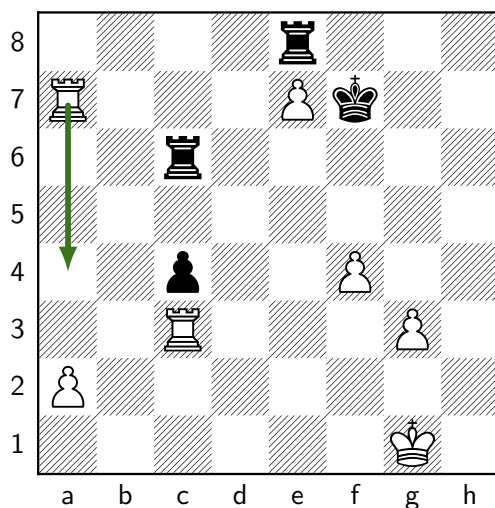
(a)



(b)

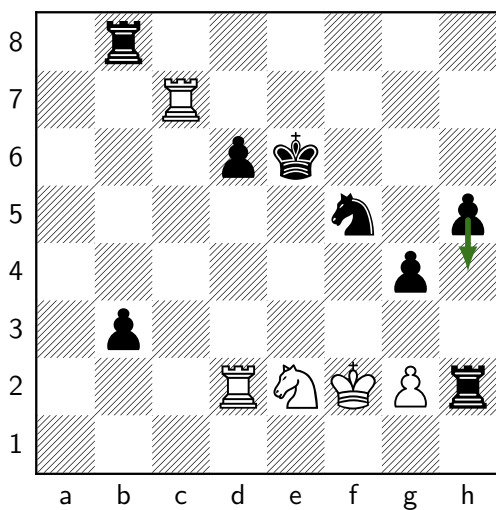


(c)

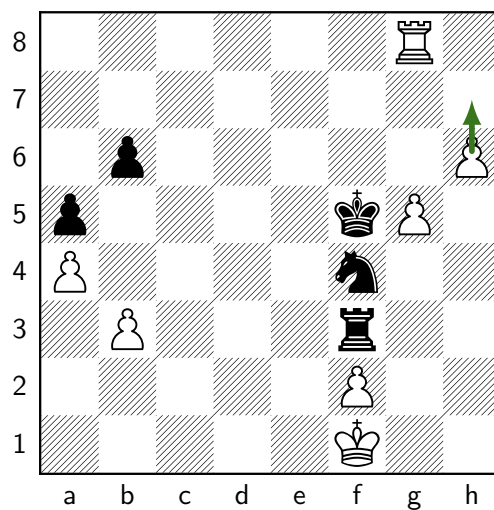


(d)

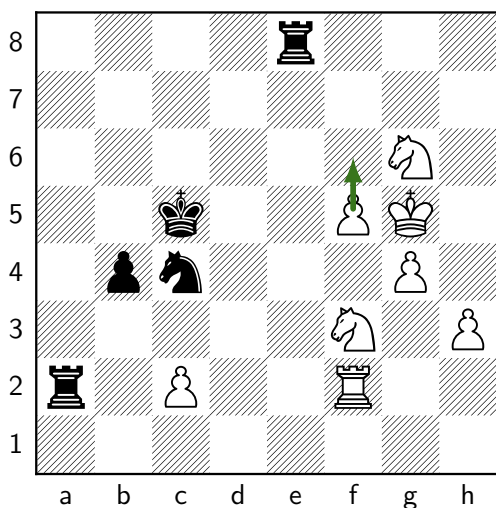
Figure 46: **Move 53, Black: g4** (g5g4). (a) Our position; Melanie plays g4 (g5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0019#1130` (ply 99) [structural distance $d=45$, QAP obj=51]: the master played g6; under μ our $Pg4$ plays the role of that game’s $Pg2$; cross-type roles: our $ne2$ as their $Pc6$, our $rc7$ as their $pf4$; transferred move g5g4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0031#3117` (ply 102) [structural distance $d=44$, QAP obj=27]: the master played b6; under μ our $Pg4$ plays the role of that game’s $Pb5$; cross-type roles: our $rc7$ as their $pg3$, our $Rb8$ as their $ph3$; transferred move g5g4 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3680` (ply 96) [structural distance $d=46$, QAP obj=59]: the master played Ra4; under μ our $Pd3$ plays the role of that game’s $Ra7$ (a **cross-type** correspondence); cross-type roles: our $rc7$ as their $Pa2$, our $ne2$ as their $Pg3$; transferred move d6d5. Synthesis: 2 retrieved analogs transferred to g4 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



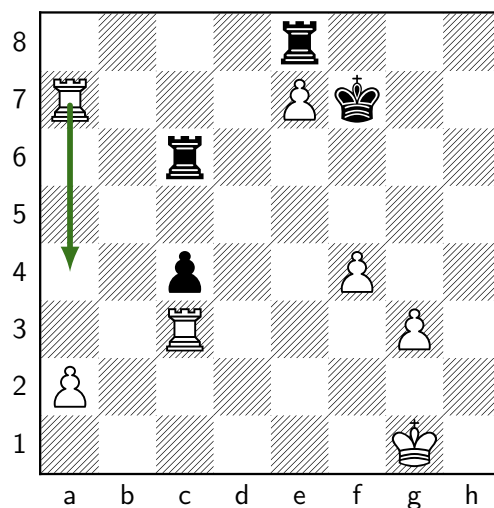
(a)



(b)

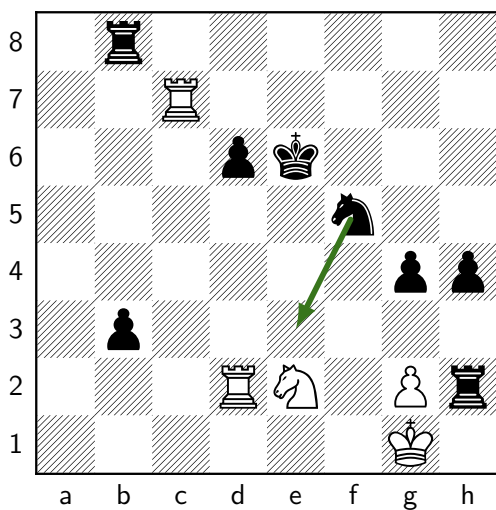


(c)

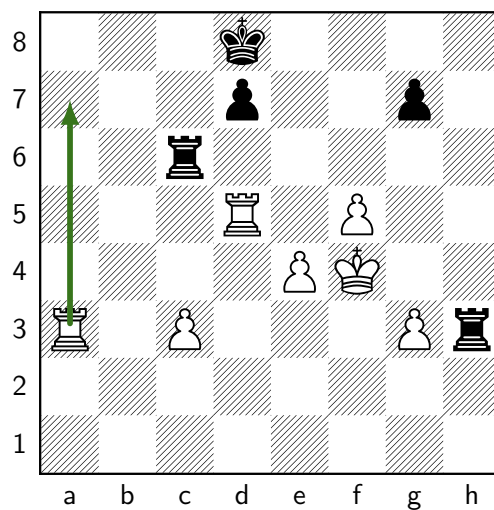


(d)

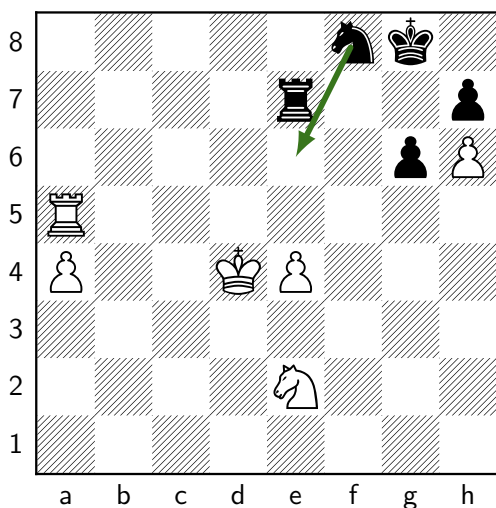
Figure 47: **Move 54, Black: h4**. (a) Our position; Melanie plays h4 (h5h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#1201` (ply 120) [structural distance $d=48$, QAP obj=41]: the master played h7; under μ our Ph_4 plays the role of that game’s Ph_6 ; cross-type roles: our $rd2$ as their Pa_4 , our $rc7$ as their pa_5 ; transferred move h5h4 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0010#3490` (ply 108) [structural distance $d=48$, QAP obj=20]: the master played f6; under μ our Ph_4 plays the role of that game’s Pf_5 ; cross-type roles: our Pb_3 as their Nf_3 , our $ne2$ as their Ph_3 ; transferred move h5h4 (**endorses this move**). (d) Retrieved from `lichess_elite_2025-11_p0000#3680` (ply 96) [structural distance $d=47$, QAP obj=49]: the master played Ra4; under μ our Nf_4 plays the role of that game’s Ra_7 (a **cross-type** correspondence); cross-type roles: our Rb_8 as their Pf_4 ; transferred move f5g7. Synthesis: 2 retrieved analogs transferred to h4 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



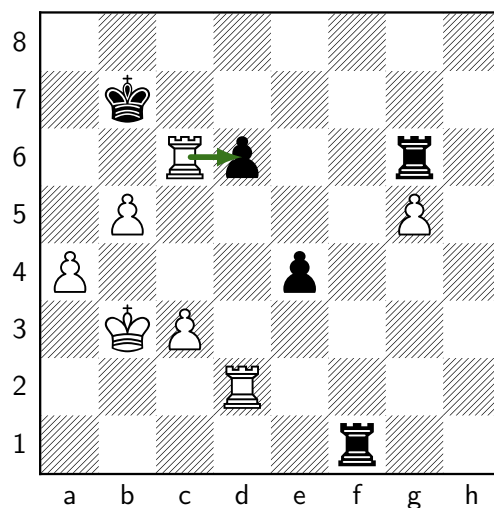
(a)



(b)

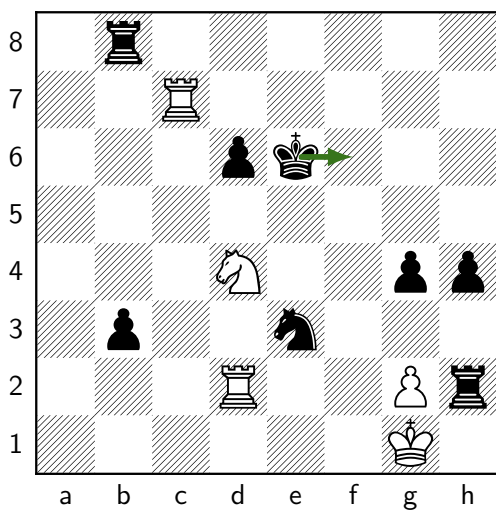


(c)

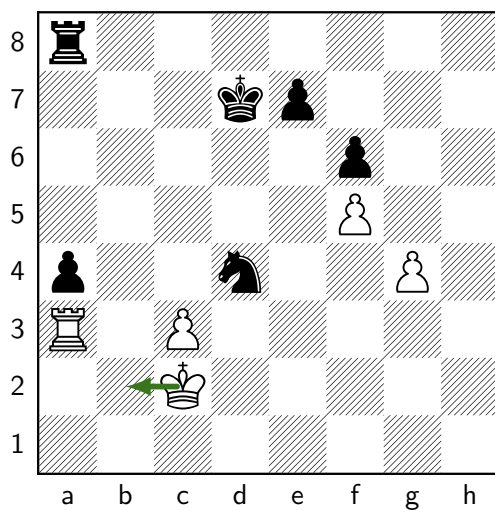


(d)

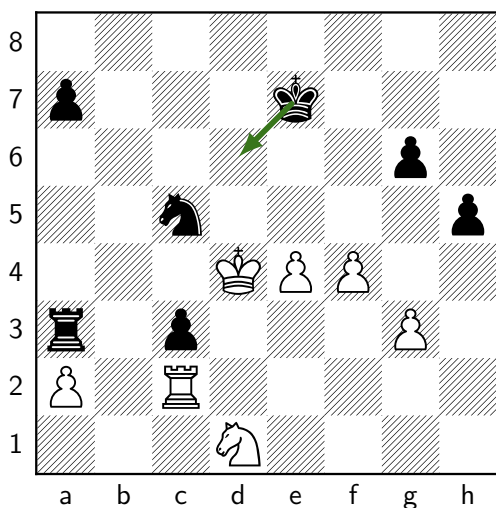
Figure 48: **Move 55, Black: Ne3**. (a) Our position; Melanie plays Ne3 (f5e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#2540` (ply 96) [structural distance $d=49$, QAP obj=16]: the master played Ra7; under μ our Nf4 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pe4, our ne2 as their pg7; transferred move f5e3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0020#1853` (ply 95) [structural distance $d=51$, QAP obj=13]: the master played Ne6+; under μ our Nf4 plays the role of that game’s Nf1; cross-type roles: our rc7 as their ph6; transferred move f5e3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0031#2881` (ply 120) [structural distance $d=48$, QAP obj=60]: the master played Rcx d6; under μ our Rh7 plays the role of that game’s Rc6; cross-type roles: our rc7 as their Pb5, our ne2 as their Pg5; transferred move h2h3. Synthesis: 2 retrieved analogs transferred to Ne3 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



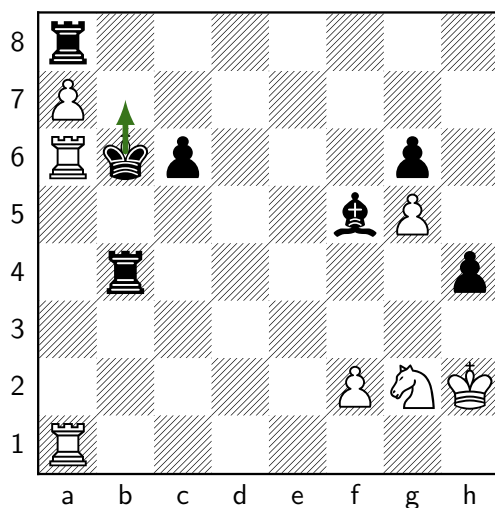
(a)



(b)

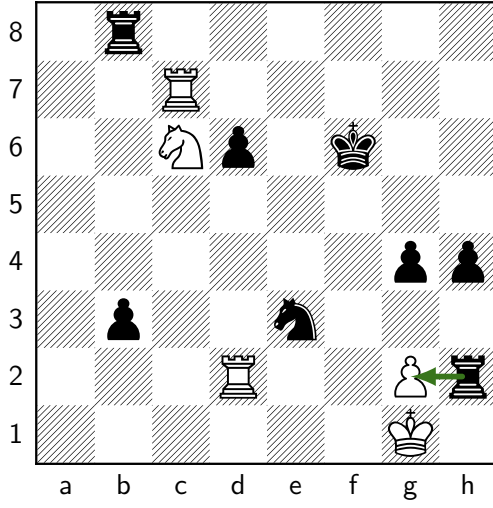


(c)

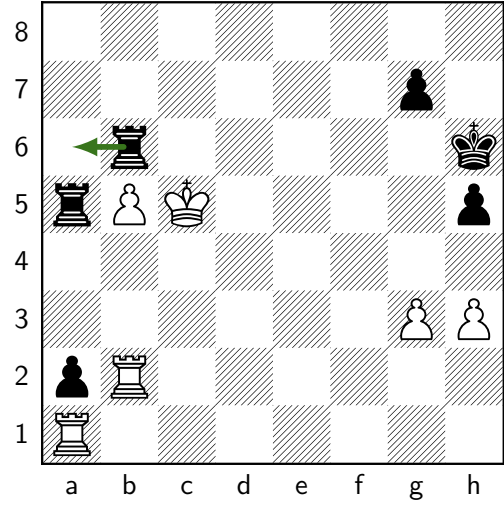


(d)

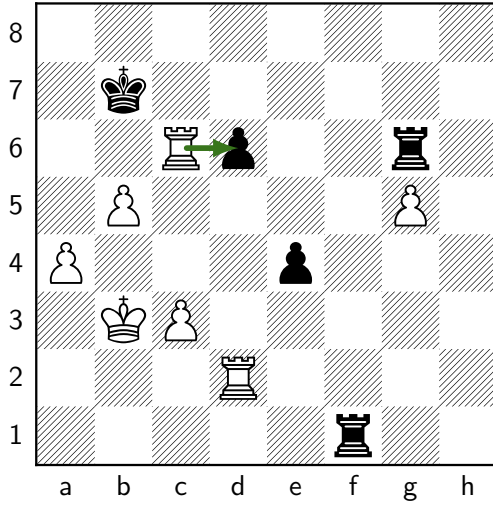
Figure 49: **Move 56, Black: Kf6**. (a) Our position; Melanie plays Kf6 (e6f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0023#1325` (ply 104) [structural distance $d=57$, QAP obj=90]: the master played Kb2; under μ our Ke3 plays the role of that game’s Kc2; cross-type roles: our Ne3 as their Pf5, our rc7 as their pf6; transferred move e6f6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#1989` (ply 95) [structural distance $d=57$, QAP obj=39]: the master played Kd6; under μ our Ke3 plays the role of that game’s Ke2; cross-type roles: our Rb8 as their Pa7, our Pd6 as their Nc5; transferred move e6f6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#647` (ply 121) [structural distance $d=51$, QAP obj=55]: the master played Kb7; under μ our Ke3 plays the role of that game’s Kb3; cross-type roles: our nd4 as their ra6, our rc7 as their Pg6; transferred move e6e5. Synthesis: 2 retrieved analogs transferred to Kf6 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



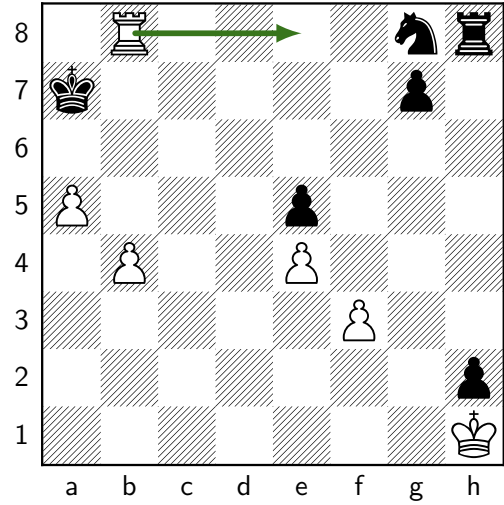
(a)



(b)

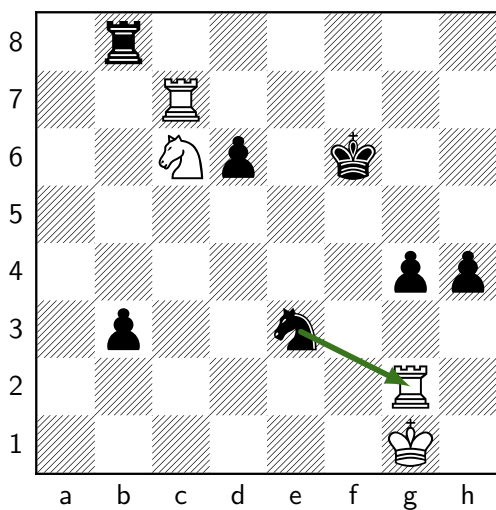


(c)

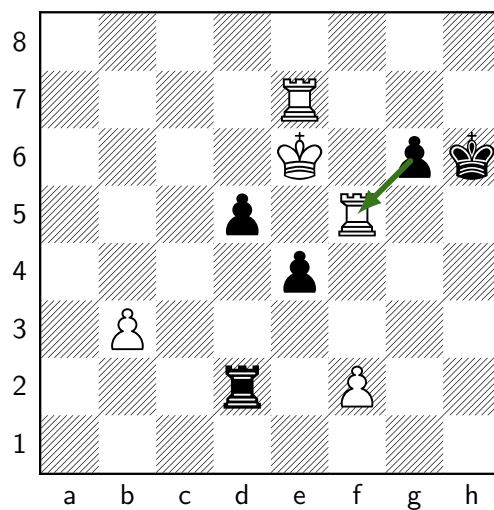


(d)

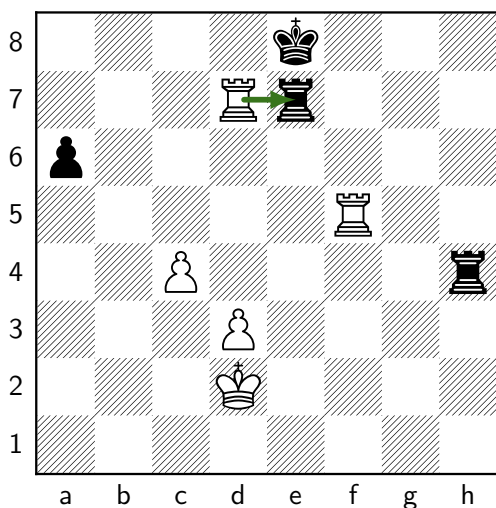
Figure 50: **Move 57, Black: Rxc2+.** (a) Our position; Melanie plays Rxc2+ (h2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0010#3890` (ply 117) [structural distance $d=44$, QAP obj=104]: the master played Rba6; under μ our Rh7 plays the role of that game’s Rb3; cross-type roles: our rc7 as their Pg7, our nc6 as their Ph5; transferred move h2h3. (c) Retrieved from `lichess_elite_2024-09_p0031#2881` (ply 120) [structural distance $d=41$, QAP obj=64]: the master played Rcx d6; under μ our Rh7 plays the role of that game’s Rc6; cross-type roles: our Pg4 as their rf1, our Ne3 as their Rd2; transferred move h2h3. (d) Retrieved from `lichess_elite_2024-09_p0030#184` (ply 100) [structural distance $d=48$, QAP obj=52]: the master played Re8; under μ our Rh7 plays the role of that game’s Rb8; cross-type roles: our rc7 as their ph2, our nc6 as their Pe4; transferred move h2h3. Synthesis: no retrieved analog transferred to Rxc2+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



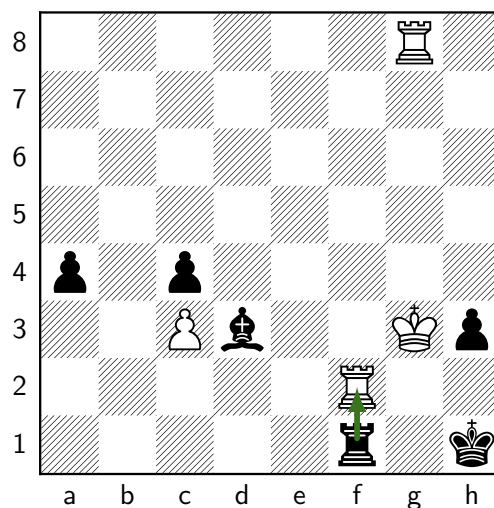
(a)



(b)

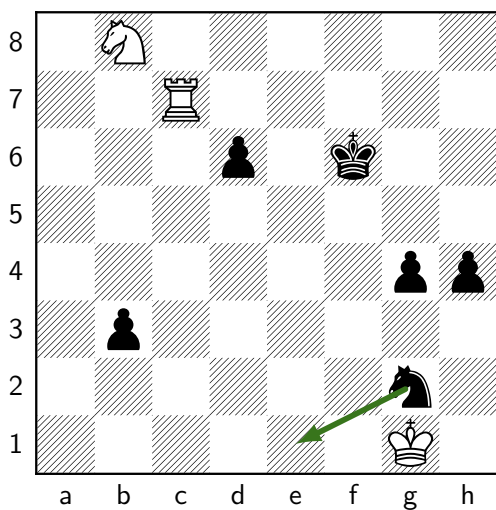


(c)

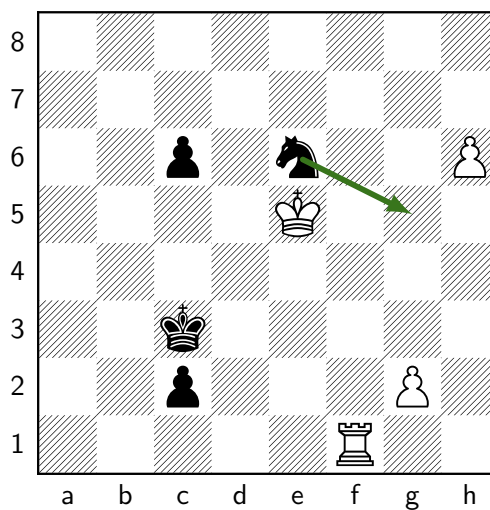


(d)

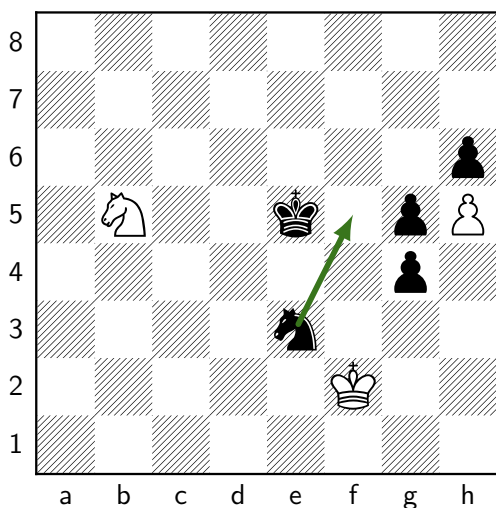
Figure 51: **Move 58, Black: Nxg2**. (a) Our position; Melanie plays Nxg2 (e3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0016#2760 (ply 109) [structural distance $d=37$, QAP obj=21]: the master played gxf5; under μ our Ne6 plays the role of that game’s Pg3 (a **cross-type** correspondence); cross-type roles: our Rb8 as their pb3, our Ph4 as their Rd2; transferred move e3g2 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0016#2466 (ply 110) [structural distance $d=40$, QAP obj=14]: the master played Rxe7+; under μ our Ne6 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our nc6 as their pa6; transferred move e3g2 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0028#1775 (ply 129) [structural distance $d=40$, QAP obj=11]: the master played Rxf2; under μ our Ne6 plays the role of that game’s Rf8 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pa4, our nc6 as their Pc4; transferred move e3g2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nxg2 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



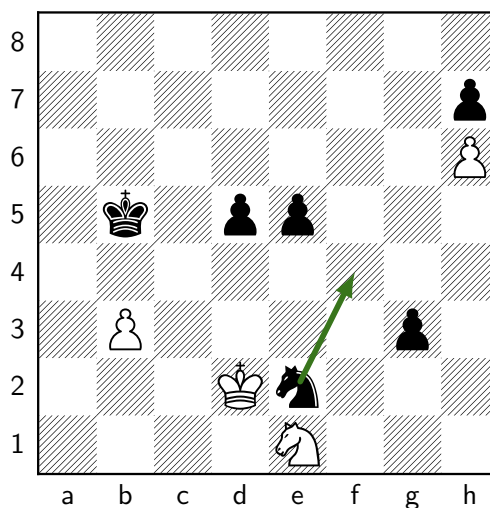
(a)



(b)



(c)



(d)

Figure 52: **Move 59, Black: Ne1**. (a) Our position; Melanie plays Ne1 (g2e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2522` (ply 121) [structural distance $d=16$, QAP obj=18]: the master played Ng5; under μ our Ng7 plays the role of that game’s Ne3; cross-type roles: our nb8 as their Pc6; transferred move g2e1 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#3775` (ply 133) [structural distance $d=14$, QAP obj=20]: the master played Nf5; under μ our Ng7 plays the role of that game’s Ne6; cross-type roles: our nb8 as their Ph6, our rc7 as their nb5; transferred move g2e3. (d) Retrieved from `lichess_elite_2024-09_p0019#3134` (ply 123) [structural distance $d=18$, QAP obj=20]: the master played Nf4; under μ our Ng7 plays the role of that game’s Ne7; cross-type roles: our rc7 as their Ph7, our nb8 as their pb3; transferred move g2e3. Synthesis: 1 retrieved analog transferred to Ne1 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

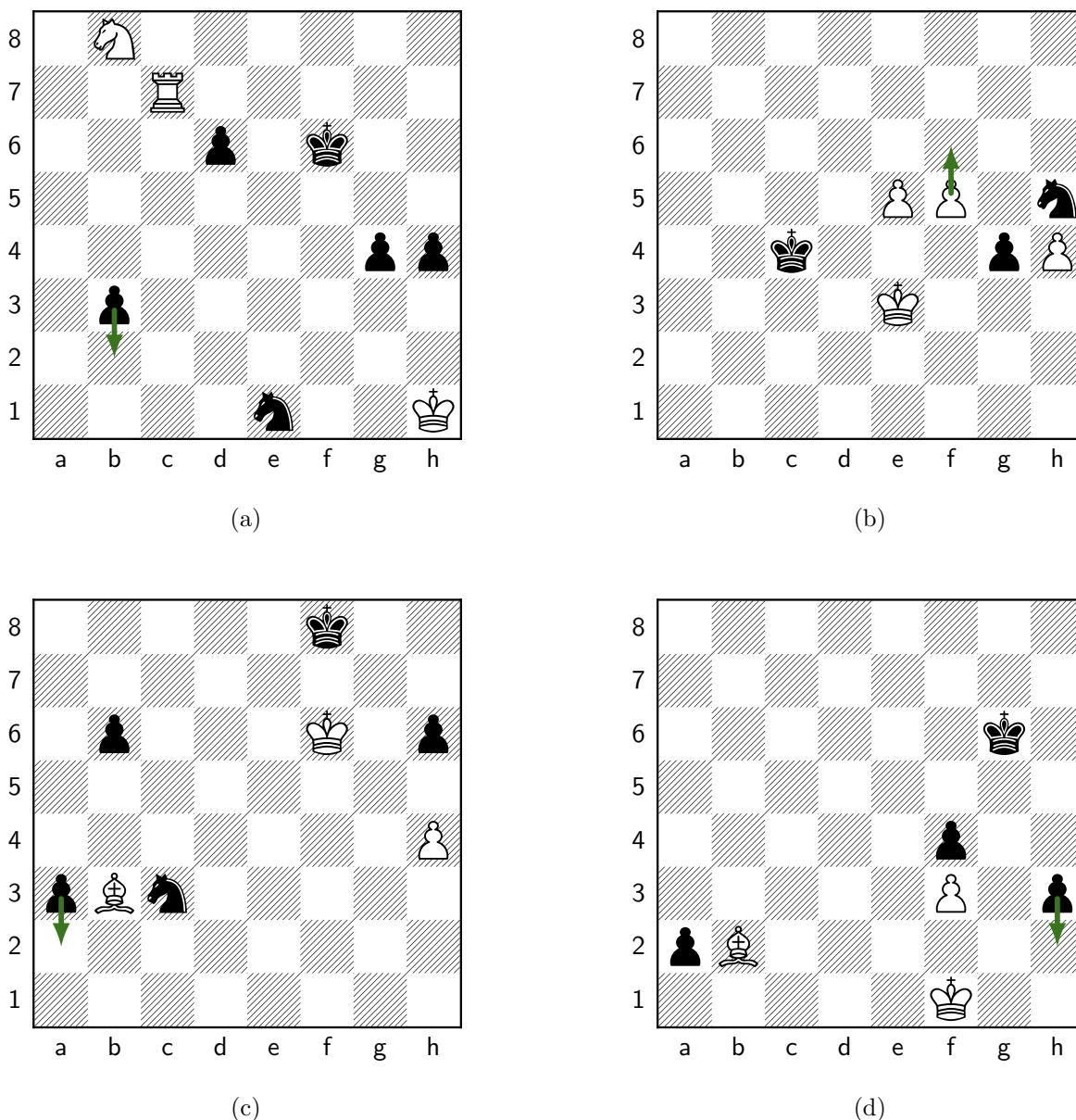
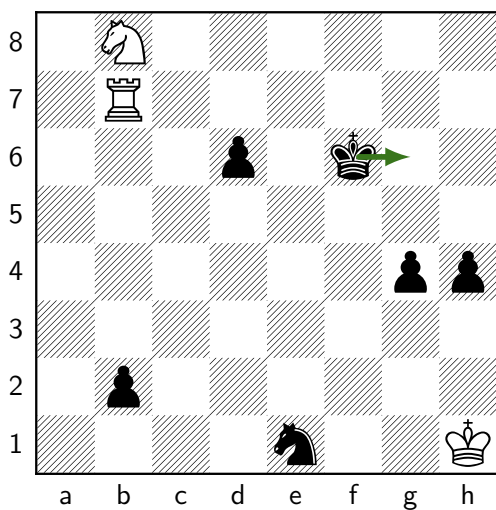
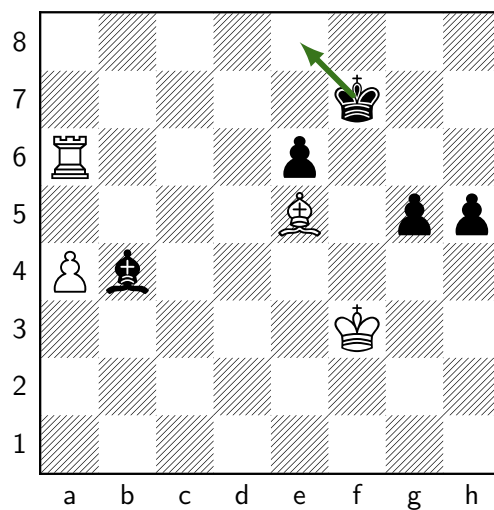


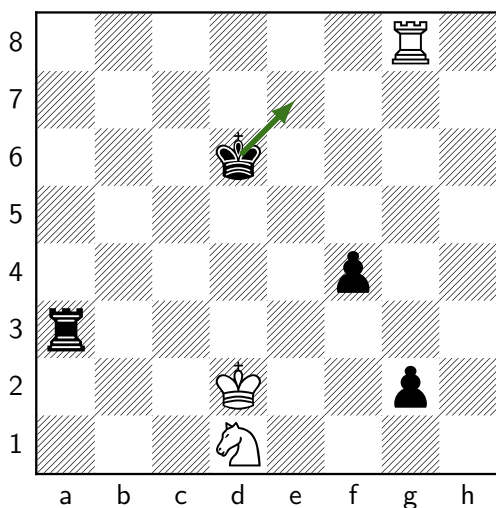
Figure 53: **Move 60, Black: b2**. (a) Our position; Melanie plays b2 (b3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2135` (ply 126) [structural distance $d=13$, QAP obj=1]: the master played f6; under μ our *Pb6* plays the role of that game’s *Pf5*; cross-type roles: our *nb8* as their *pg4*, our *Ne1* as their *Ph4*; transferred move b3b2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0025#2809` (ply 119) [structural distance $d=13$, QAP obj=1]: the master played a2; under μ our *Pb6* plays the role of that game’s *Pa6*; cross-type roles: our *Ne1* as their *Ph6*, our *nb8* as their *ph4*; transferred move b3b2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0022#257` (ply 131) [structural distance $d=14$, QAP obj=1]: the master played h2; under μ our *Ne8* plays the role of that game’s *Ph6* (a **cross-type** correspondence); cross-type roles: our *nb8* as their *Pf4*, our *rc7* as their *bb2*; transferred move e1d3. Synthesis: 2 retrieved analogs transferred to b2 (summed QAP objective 2, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



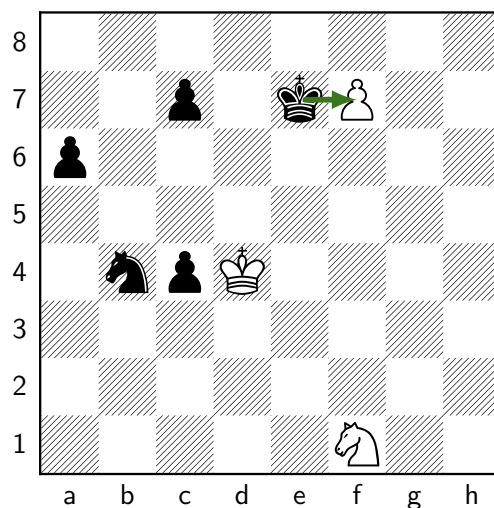
(a)



(b)



(c)



(d)

Figure 54: **Move 61, Black: Kg6**. (a) Our position; Melanie plays Kg6 (f6g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#440` (ply 109) [structural distance $d=19$, QAP obj=16]: the master played Ke8; under μ our Kf3 plays the role of that game’s Kf2; cross-type roles: our Pd6 as their be5, our Ne1 as their Ph5; transferred move f6g6 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#903` (ply 133) [structural distance $d=18$, QAP obj=15]: the master played Ke7; under μ our Kf3 plays the role of that game’s Kd3; cross-type roles: our nb8 as their Pf4, our Ne1 as their Ra3; transferred move f6g6 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#886` (ply 135) [structural distance $d=19$, QAP obj=1]: the master played Kxf7; under μ our Kf3 plays the role of that game’s Ke2; cross-type roles: our nb8 as their Pc7, our rb7 as their pf7; transferred move f6g6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg6 (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

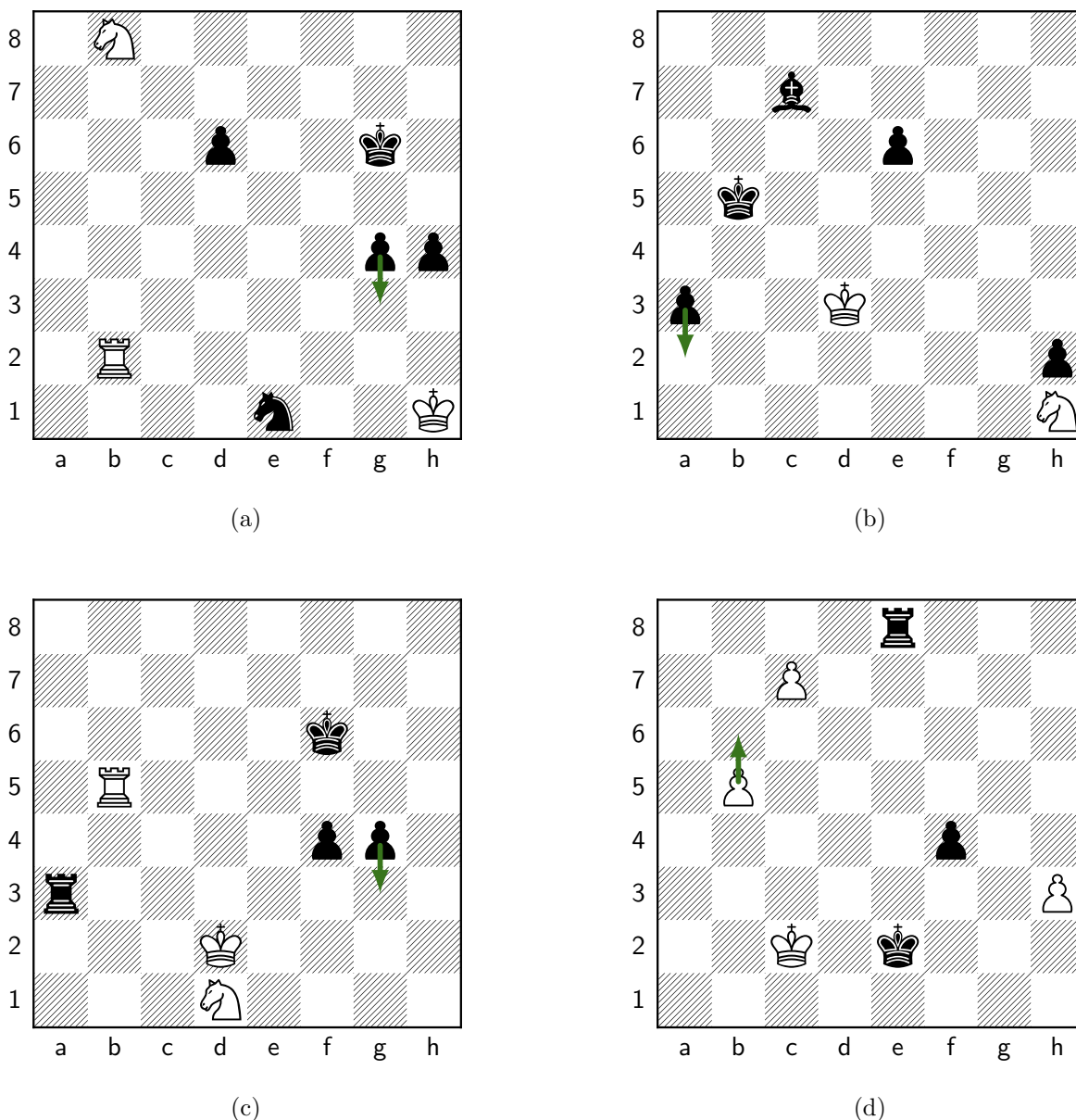
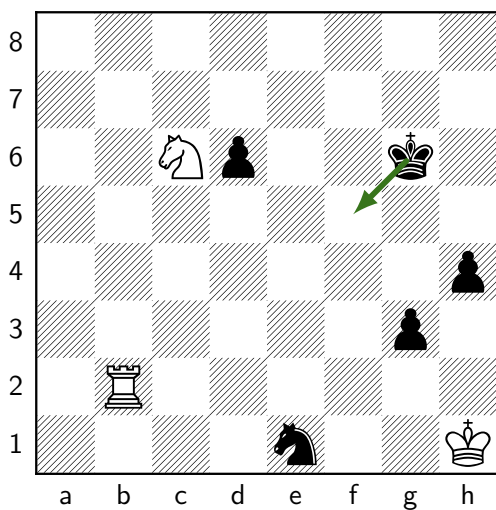
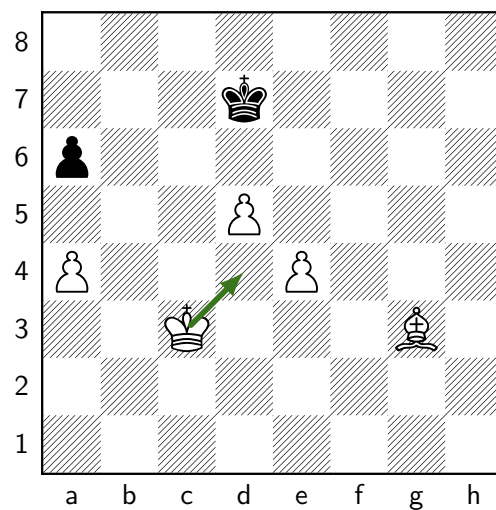


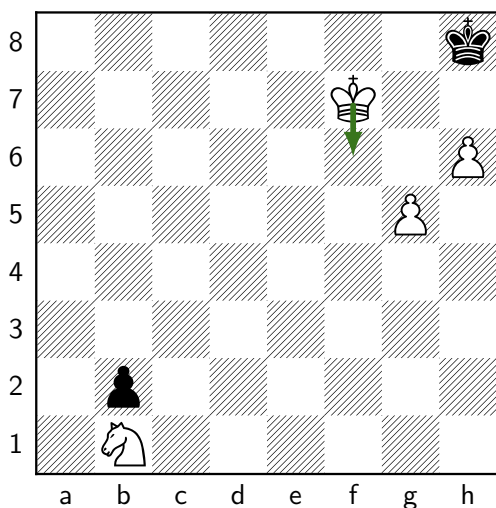
Figure 55: **Move 62, Black: g3**. (a) Our position; Melanie plays **g3** (**g4g3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0025#26` (ply 111) [structural distance $d=14$, QAP obj=1]: the master played **a2**; under μ our *Pg5* plays the role of that game’s *Pa6*; cross-type roles: our *rb2* as their *Bc7*, our *nb8* as their *Ph2*; transferred move **g4g3** (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0021#903` (ply 125) [structural distance $d=12$, QAP obj=1]: the master played **g3**; under μ our *Pg5* plays the role of that game’s *Pg5*; cross-type roles: our *Ne1* as their *Ra3*; transferred move **g4g3** (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0023#221` (ply 116) [structural distance $d=10$, QAP obj=0]: the master played **b6**; under μ our *Pg5* plays the role of that game’s *Pb5*; cross-type roles: our *nb8* as their *Ph3*, our *Ne1* as their *Pc7*; transferred move **g4g3** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **g3** (summed QAP objective 3, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



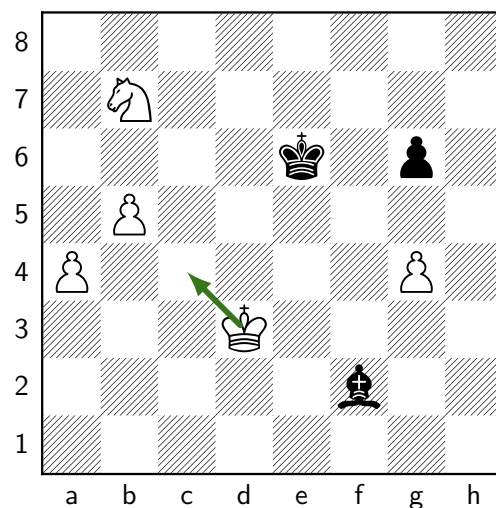
(a)



(b)



(c)



(d)

Figure 56: **Move 63, Black: Kf5.** (a) Our position; Melanie plays Kf5 (g6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0005#2226` (ply 114) [structural distance $d=10$, QAP obj=2]: the master played Kd4; under μ our Kg3 plays the role of that game’s Kc3; cross-type roles: our Ne1 as their Bg3, our nc6 as their Pa4; transferred move g6f5 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#3330` (ply 114) [structural distance $d=13$, QAP obj=2]: the master played Kf6; under μ our Kg3 plays the role of that game’s Kf7; transferred move g6f5 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0030#795` (ply 138) [structural distance $d=9$, QAP obj=2]: the master played Kc4; under μ our Kg3 plays the role of that game’s Kd3; cross-type roles: our rb2 as their bf2, our nc6 as their Pg4; transferred move g6f5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf5 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

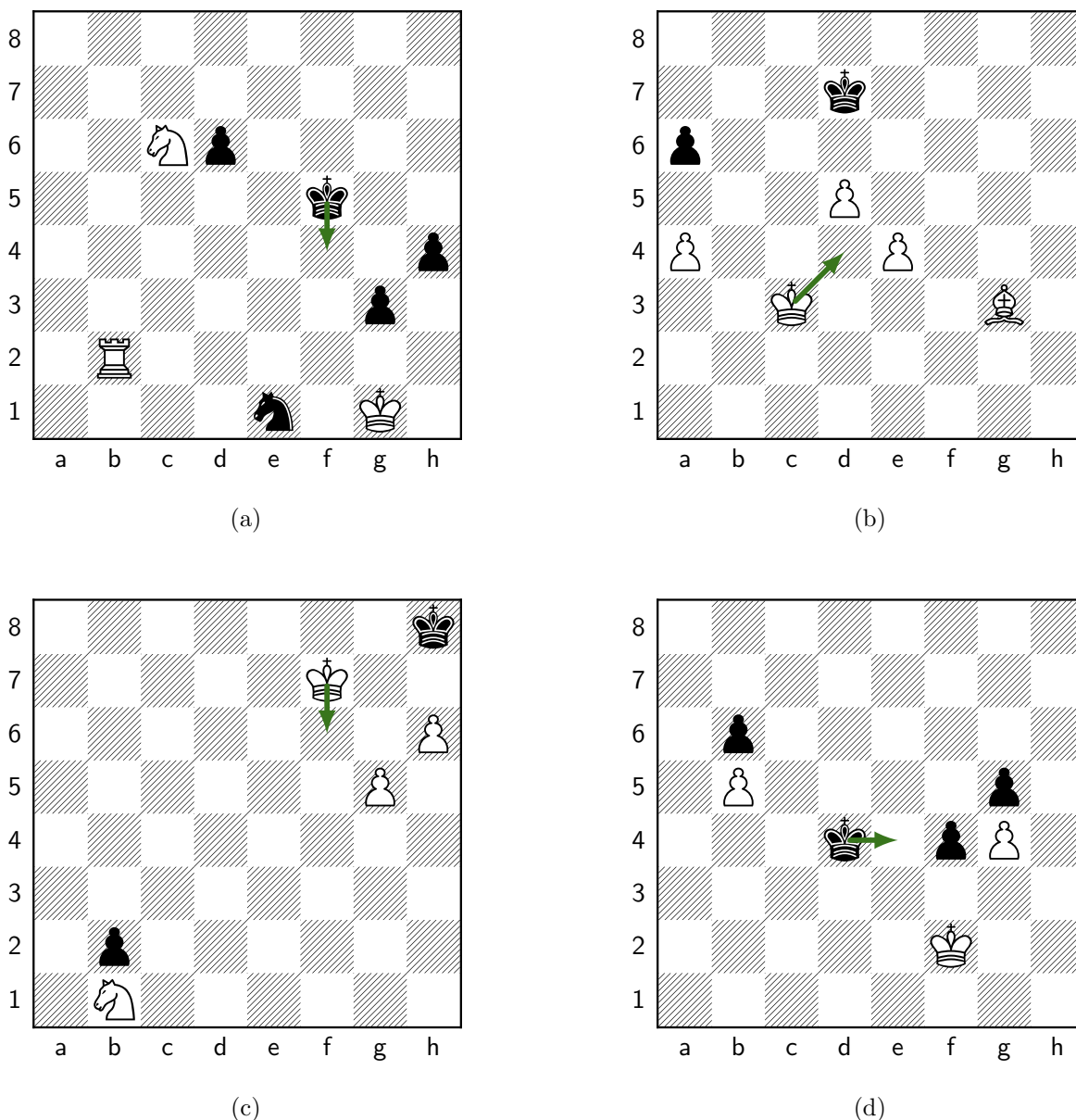
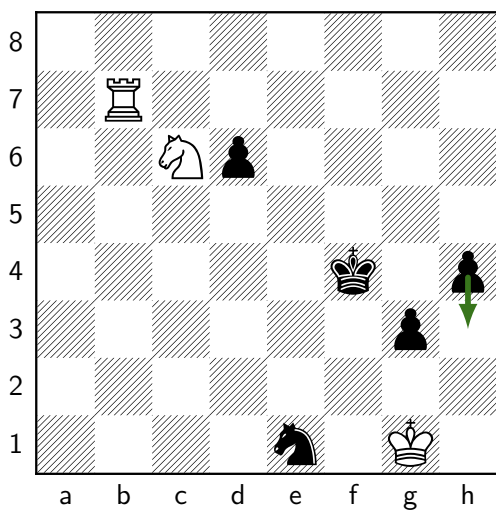
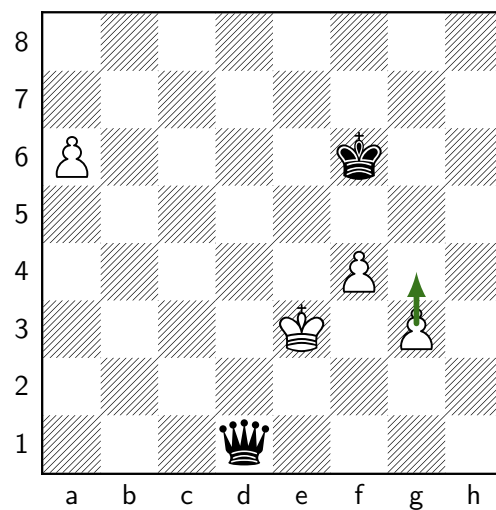


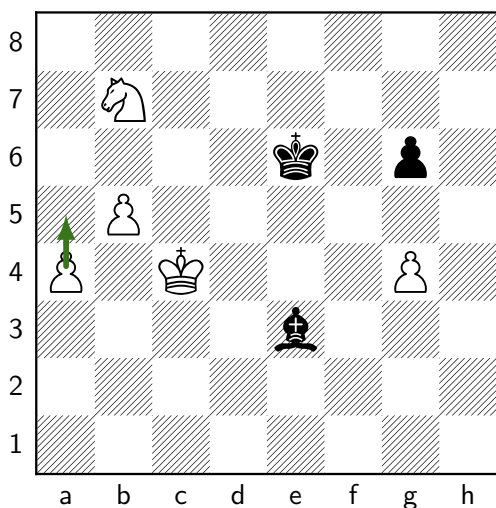
Figure 57: **Move 64, Black: Kf4**. (a) Our position; Melanie plays Kf4 (f5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from lichess_elite_2024-09_p0005#2226 (ply 114) [structural distance $d=11$, QAP obj=2]: the master played Kd4; under μ our Kf4 plays the role of that game’s Kc3; cross-type roles: our Ne1 as their Bg3, our nc6 as their Pa4; transferred move f5f4 (**endorses this move**). (c) Retrieved from lichess_elite_2024-09_p0026#3330 (ply 114) [structural distance $d=14$, QAP obj=2]: the master played Kf6; under μ our Kf4 plays the role of that game’s Kf7; transferred move f5f4 (**endorses this move**). (d) Retrieved from lichess_elite_2024-09_p0017#2651 (ply 139) [structural distance $d=13$, QAP obj=2]: the master played Ke4; under μ our Kf4 plays the role of that game’s Kd5; cross-type roles: our Ne1 as their Pb6, our rb2 as their pb5; transferred move f5f4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf4 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



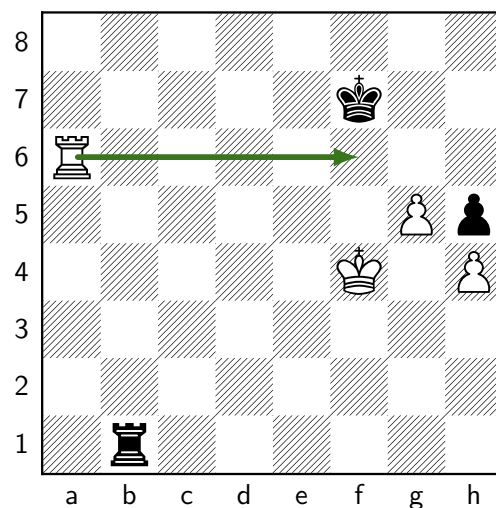
(a)



(b)

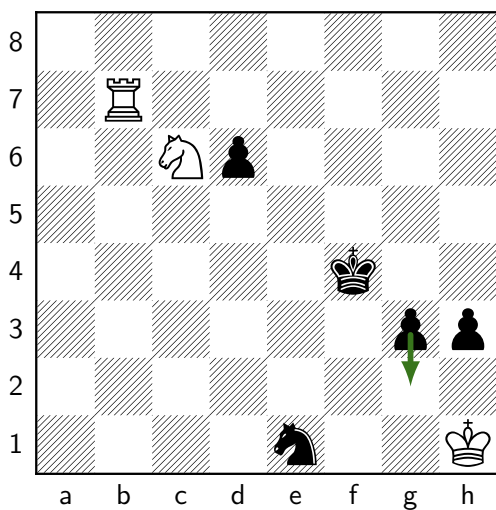


(c)

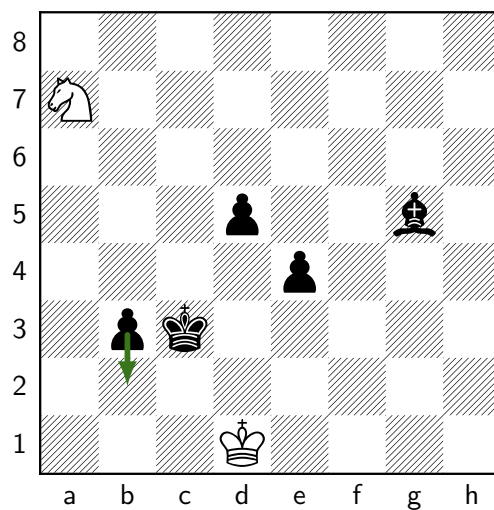


(d)

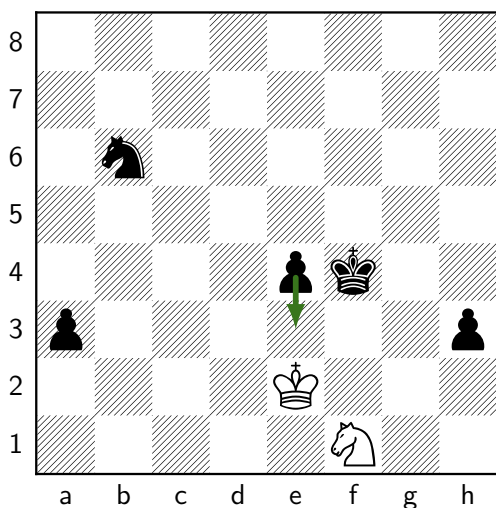
Figure 58: **Move 65, Black: h3**. (a) Our position; Melanie plays h3 (h4h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#805` (ply 140) [structural distance $d=10$, QAP obj=4]: the master played g4; under μ our *Ph5* plays the role of that game’s *Pg3*; cross-type roles: our *nc6* as their *qd1*, our *rb7* as their *Pa6*; transferred move h4h3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0030#795` (ply 140) [structural distance $d=7$, QAP obj=3]: the master played a5; under μ our *Ph5* plays the role of that game’s *Pa4*; cross-type roles: our *nc6* as their *be3*, our *rb7* as their *pg6*; transferred move h4h3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0021#4000` (ply 142) [structural distance $d=9$, QAP obj=3]: the master played Rf6+; under μ our *Ne8* plays the role of that game’s *Ra6* (a **cross-type** correspondence); cross-type roles: our *nc6* as their *ph5*; transferred move e1g2. Synthesis: 2 retrieved analogs transferred to h3 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



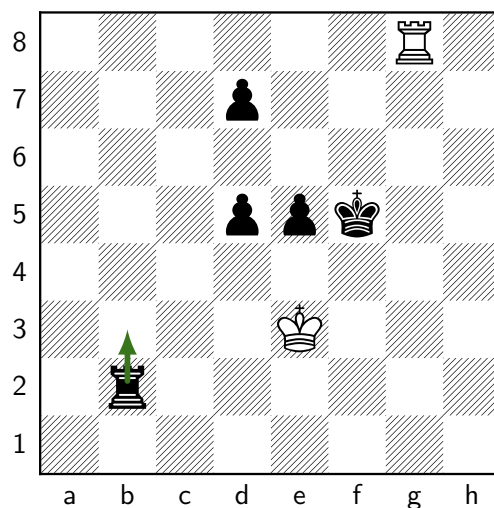
(a)



(b)

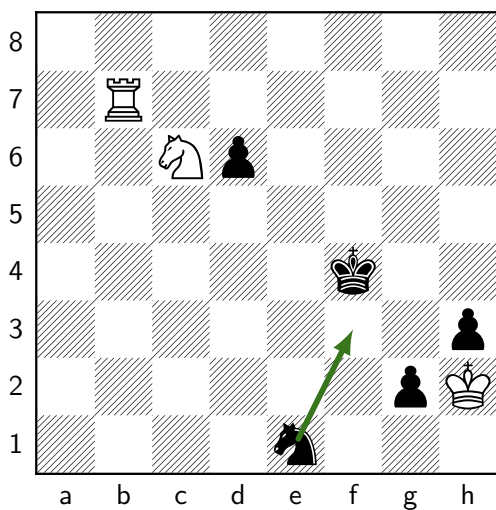


(c)

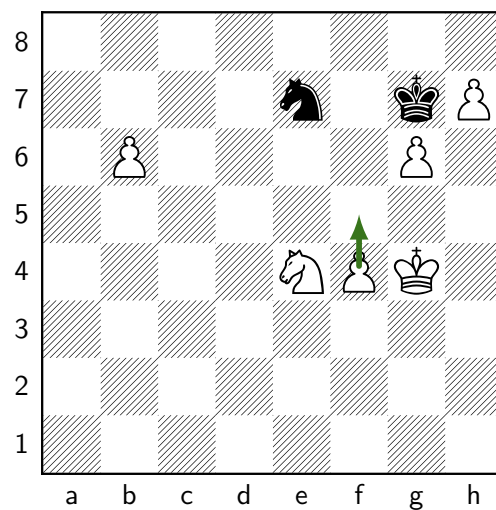


(d)

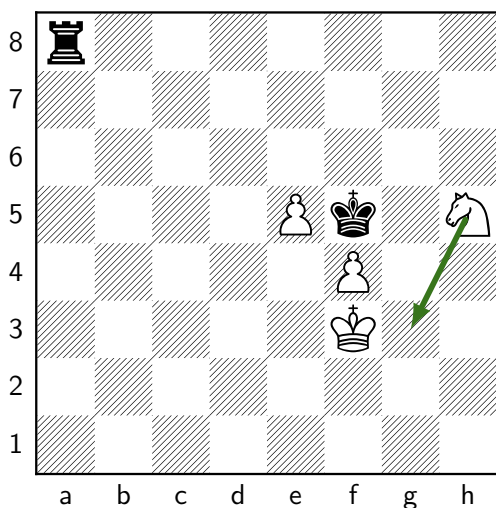
Figure 59: **Move 66, Black: g2+.** (a) Our position; Melanie plays g2+ (g3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0026#2601` (ply 123) [structural distance $d=10$, QAP obj=2]: the master played b2; under μ our *Pg6* plays the role of that game’s *Pb6*; cross-type roles: our *rb7* as their *na7*, our *Ph3* as their *Bg5*; transferred move g3g2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0005#2101` (ply 135) [structural distance $d=8$, QAP obj=1]: the master played e3; under μ our *Pg6* plays the role of that game’s *Pe5*; cross-type roles: our *rb7* as their *Nb6*; transferred move g3g2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0025#3565` (ply 119) [structural distance $d=8$, QAP obj=3]: the master played Rb3+; under μ our *Ne8* plays the role of that game’s *Rb7* (a **cross-type** correspondence); cross-type roles: our *nc6* as their *rg8*, our *rb7* as their *Pd7*; transferred move e1c2. Synthesis: 2 retrieved analogs transferred to g2+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



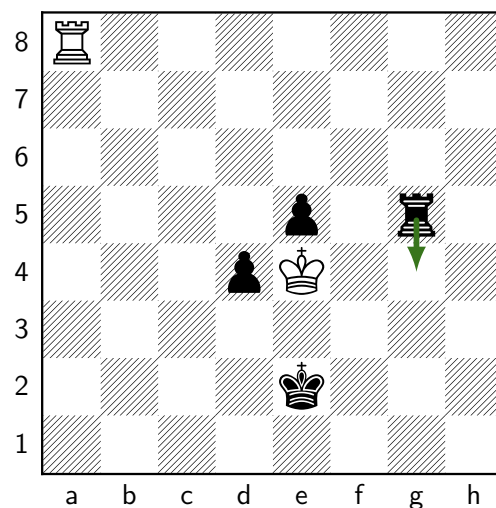
(a)



(b)

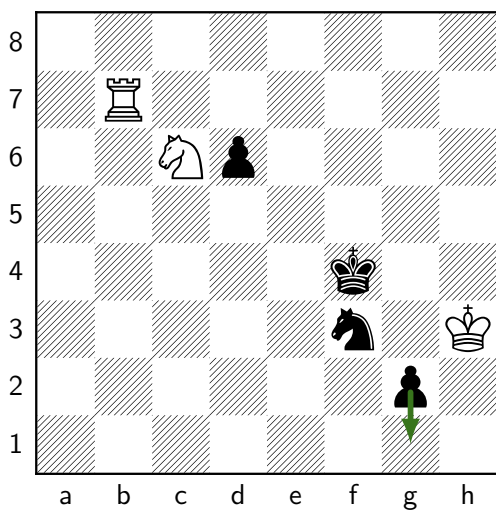


(c)

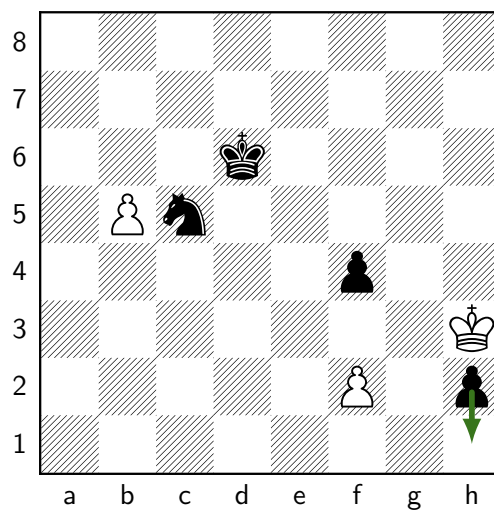


(d)

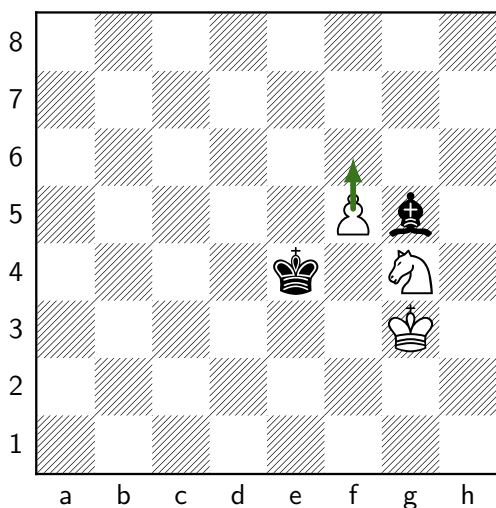
Figure 60: **Move 67, Black: Nf3+.** (a) Our position; Melanie plays Nf3+ (e1f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0031#2212` (ply 142) [structural distance $d=10$, QAP obj=62]: the master played f5; under μ our Ne8 plays the role of that game’s Pf4 (a **cross-type** correspondence); cross-type roles: our rb7 as their Ne4, our nc6 as their Pb6; transferred move e1f3 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0016#1018` (ply 132) [structural distance $d=9$, QAP obj=57]: the master played Ng3+; under μ our Ne8 plays the role of that game’s Nh5; transferred move e1f3 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0012#968` (ply 129) [structural distance $d=14$, QAP obj=58]: the master played Rg4+; under μ our Ne8 plays the role of that game’s Rg4 (a **cross-type** correspondence); transferred move e1c2. Synthesis: 2 retrieved analogs transferred to Nf3+ (summed QAP objective 119, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



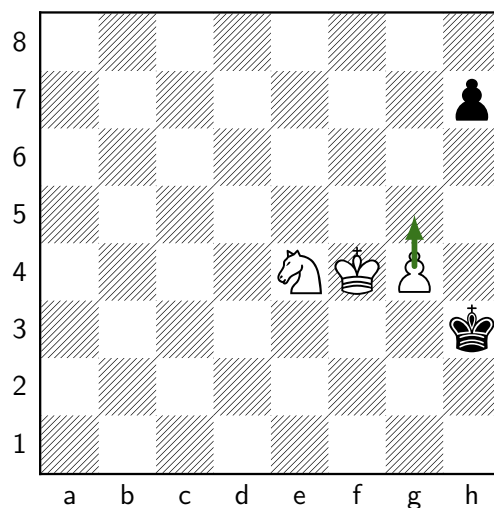
(a)



(b)



(c)



(d)

Figure 61: **Move 68, Black: g1=Q.** (a) Our position; Melanie plays g1=Q (g2g1q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0018#976` (ply 131) [structural distance $d=11$, QAP obj=20]: the master played h1=Q+; under μ our Pg7 plays the role of that game’s Ph7; cross-type roles: our rb7 as their pb5, our nc6 as their pf2; transferred move g2g1q (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0024#3222` (ply 120) [structural distance $d=11$, QAP obj=23]: the master played f6; under μ our Pg7 plays the role of that game’s Pf5; cross-type roles: our nc6 as their bg5; transferred move g2g1r. (d) Retrieved from `lichess_elite_2024-09_p0017#985` (ply 124) [structural distance $d=11$, QAP obj=22]: the master played g5; under μ our Pg7 plays the role of that game’s Pg4; cross-type roles: our rb7 as their ph7; transferred move g2g1r. Synthesis: 1 retrieved analog transferred to g1=Q (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

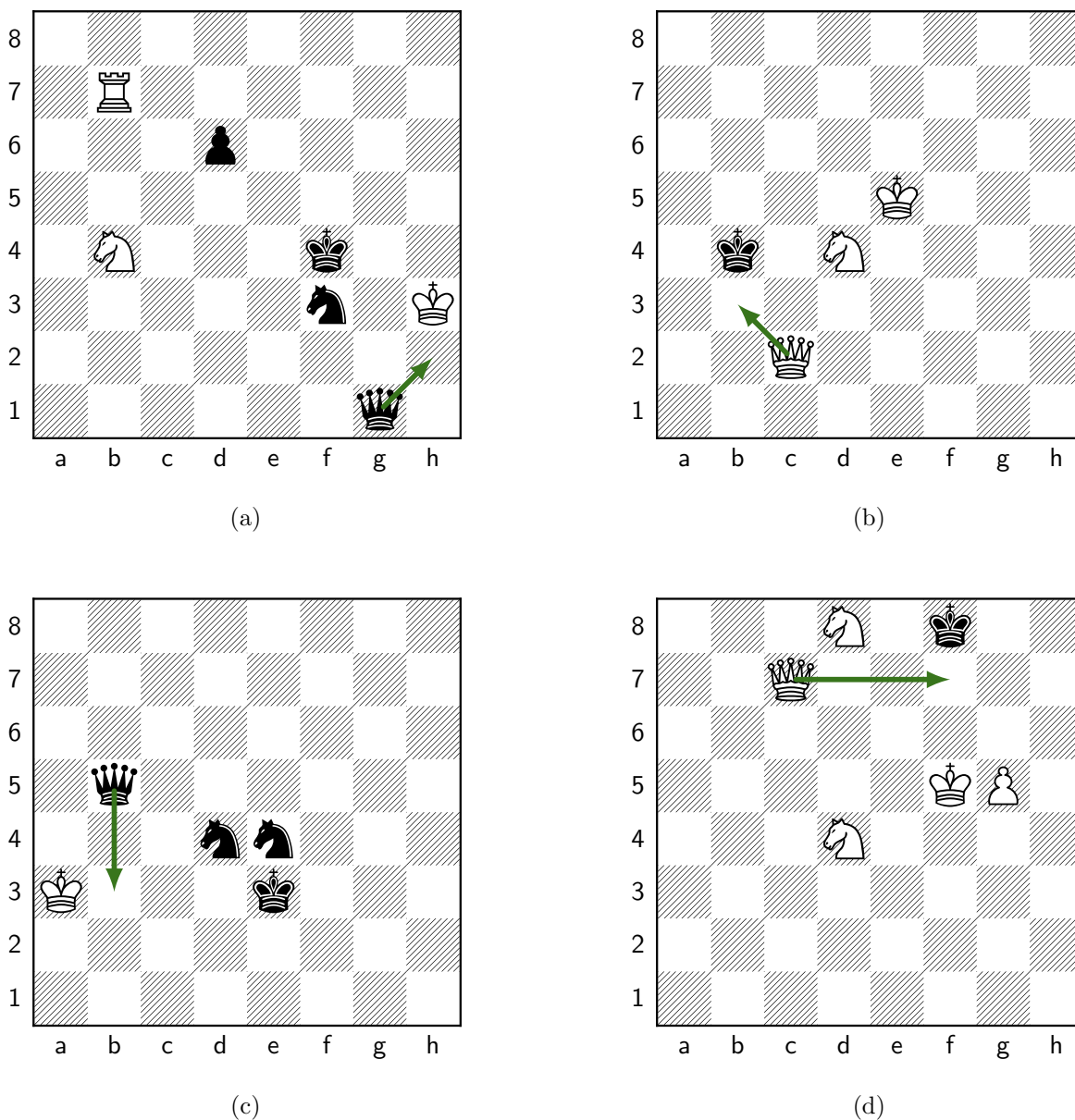


Figure 62: **Move 69, Black: Qh2#**. (a) Our position; Melanie plays Qh2# (g1h2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `lichess_elite_2024-09_p0022#2996` (ply 134) [structural distance $d=17$, QAP obj=4]: the master played Qb3+; under μ our Qg8 plays the role of that game’s Qc2; transferred move g1h2 (**endorses this move**). (c) Retrieved from `lichess_elite_2024-09_p0026#183` (ply 129) [structural distance $d=17$, QAP obj=4]: the master played Qb3#; under μ our Qg8 plays the role of that game’s Qb4; cross-type roles: our rb7 as their Ne4; transferred move g1h2 (**endorses this move**). (d) Retrieved from `lichess_elite_2024-09_p0028#2631` (ply 144) [structural distance $d=18$, QAP obj=1]: the master played Qf7#; under μ our Qg8 plays the role of that game’s Qc7; cross-type roles: our rb7 as their Nd4; transferred move g1h2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qh2# (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.